Polynomial Parametrizations, Birational Involutions, and Genus-1 Fibrations on the Diagonal Quartic Surface $na^4 + b^4 + c^4 = d^4$

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Abstract

We study primitive polynomial parametrizations $(a(p,q), b(p,q), c(p,q), d(p,q)) \in \mathbb{Z}[p,q]^4$ and explicit genus-1 fibrations for the Diophantine equation

$$S_n : \quad na^4 + b^4 + c^4 = d^4, \qquad \gcd(a,b,c,d) = 1,$$

both when $n = n(p,q)$ is a sparse binomial or trinomial expression of arbitrary degree and when $n \in \mathbb{Q}^\times$ is a fixed integer or rational constant. In the classical literature surveyed here, primitive polynomial parametrizations (genus-0 curves on S_n) have been constructed for three fixed constants n up to fourth powers $\mathbb{Q}^{\times 4}$: $n = -1$ (Euler, 1772; Swinnerton-Dyer, 1943, 2000; Bremner, 2025), $n = 4$ (Gerardin, 1911; Carmichael, 1913), and $n = -12$ (Elkies, 2000). By constructing explicit birational involutions defined over $\mathbb{Q}$ from Galois-conjugate skew lines on S_4 and crossed rational skew lines on S_{-1} (defined generically as eigenspace transversal-line maps on the open complements of the defining lines and extended across their codimension-1 base loci to regular K3 automorphisms), we derive four new primitive polynomial families for $n = 4$ (at degrees 8 and 16, representing distinct smooth arithmetic-genus-0 divisor classes in $\mathrm{Pic}(S_4/\mathbb{Q})$) and three compact single-generator involutive polynomial families for $n = -1$ (at degrees 13, 19, and 31). Using Galois descent on the rank-20 geometric Picard lattice spanned by the 48 lines on S_n over $\overline{\mathbb{Q}}$ (certified by an explicit 20-line integral basis $\mathcal{B}_{20}$ of determinant -64 and the full 20×48 coordinate matrix C_{48}), together with the splitting of the Hochschild–Serre exact sequence via $\mathbb{Q}$-rational points on S_n, we prove a sharp geometric trichotomy between $n \in \{-1, 4\}$ and $n = -12$: for $n = 4$ ($\rho(S_4/\mathbb{Q}) = 8$) and $n = -1$ ($\rho(S_{-1}/\mathbb{Q}) = 9$), the birational automorphism group $\mathrm{Bir}(S_n/\mathbb{Q}) = \mathrm{Aut}(S_n/\mathbb{Q})$ is infinite and generates infinitely many projectively inequivalent primitive polynomial families of unbounded degrees via exact quadratic degree towers ($\deg(\Psi_4^k(C^{(1)})) = 8k^2 - 4k + 4$ for $n = 4$, and $\deg(\Psi_{-1}^k(C_{\mathrm{eul}}^{(r)})) = 9k^2 - 3k + 7$, $12k^2 + 7$, and $18k^2 + 6k + 7$ for $n = -1$); whereas for $n = -12$, we prove that $\mathrm{Pic}(S_{-12}/\mathbb{Q}) = \mathrm{Pic}(S_{-12,\overline{\mathbb{Q}}})^{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})} = \mathbb{Z}H$ has rank 1, forcing $\mathrm{Bir}(S_{-12}/\mathbb{Q}) = \mathrm{Aut}(S_{-12}/\mathbb{Q}) \cong (\mathbb{Z}/2\mathbb{Z})^3 \rtimes S_2$ to be finite of order 16 and obstructing any infinite tower of skew-line, conic-bundle, or base-point-free elliptic-fibration polynomial families over $\mathbb{Q}$. Furthermore, applying the factored degree-17 asymptotic inflection-tangent self-maps $\mathcal{A}_\pm$ on S_{-1} to Euler's degree-7 curve yields primitive polynomial families of degrees 43 and 45 whose image curves have positive arithmetic genus ($p_a = 66$ and $p_a = 58$), lying strictly outside the Swinnerton-Dyer and Bremner classification of arithmetic-genus-0 ($C^2 = -2$) curves. Finally, we establish with complete symbolic derivations, parameter-domain classifications, and executable verification certificates the Galois-Field Skew-Line Classification Theorem, the Parabolic-Cone Cardano Classification Theorem, the Factored Degree-17 Asymptotic Tangent Self-Map $\mathcal{A}_\pm$ on S_{-s^2}, the Master Corner Identity, the Master u_4-Eckardt Identity, the 3-Parameter Quadratic-Square Syzygy, the Symmetric Conic-Square Reduction Theorem, the Tacnodal Tangent-Plane Theorem for Coordinate-Zero Points, and a comprehensive prior-literature comparison.

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1 Introduction and the Landscape of Known Parametrizations

Let $S_n \subset \mathbb{P}^3$ denote the smooth diagonal quartic K3 surface defined over $\mathbb{Q}$ by

$$S_n : \quad na^4 + b^4 + c^4 = d^4, \qquad \gcd(a,b,c,d) = 1, \quad a \nmid b, c, d. \tag{1}$$

The arithmetic of S_n splits into two distinct regimes depending on whether n is treated as a sparse polynomial $n(p,q) \in \mathbb{Z}[p,q]$ (monomial, binomial, or trinomial) or as a fixed non-zero constant $n \in \mathbb{Q}^\times$ (considered modulo fourth powers $\mathbb{Q}^{\times 4}$, since replacing $a \mapsto s_0 a$ rescales $n \mapsto s_0^{-4} n$).

Convention on Picard Groups over $\mathbb{Q}$ vs. $\overline{\mathbb{Q}}$. Throughout this paper, $\mathrm{Pic}(S_{n,\overline{\mathbb{Q}}}) = \mathrm{Pic}(S_n \times_{\mathbb{Q}} \overline{\mathbb{Q}})$ denotes the geometric Picard lattice over $\overline{\mathbb{Q}}$, which for every $n \in \mathbb{Q}^\times$ is a free abelian group of rank $\rho(S_{n,\overline{\mathbb{Q}}}) = 20$ and discriminant $\det(\mathrm{Pic}(S_{n,\overline{\mathbb{Q}}})) = -64$ spanned over $\mathbb{Z}$ by the classes of the 48 straight lines on $S_{n,\overline{\mathbb{Q}}}$ (Pinch and Swinnerton-Dyer [13], Bright [12, Proposition 2.15]; see Lemma 5.3). We denote by $\mathrm{Pic}(S_n/\mathbb{Q})$ the arithmetic Picard group of divisor classes defined over $\mathbb{Q}$, and by $\mathrm{Pic}(S_{n,\overline{\mathbb{Q}}})^{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})}$ the Galois-invariant geometric Picard lattice. By the Hochschild–Serre spectral sequence for $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acting on $\mathbb{G}_m$, there is an exact sequence

$$0 \longrightarrow \mathrm{Pic}(S_n/\mathbb{Q}) \longrightarrow \mathrm{Pic}(S_{n,\overline{\mathbb{Q}}})^{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})} \xrightarrow{\delta} \mathrm{Br}(\mathbb{Q}). \tag{2}$$

Whenever $S_n(\mathbb{Q}) \neq \emptyset$, evaluating Brauer classes at any rational point $P_0 \in S_n(\mathbb{Q})$ splits the map $\mathrm{Br}(\mathbb{Q}) \to \mathrm{Br}(S_n)$, forcing the obstruction map δ in (2) to vanish identically [12, Section 2.2]. For all three surfaces S_{-1} (containing $(1:1:1:1)$), S_4 (containing $Q_1 = (0:1:0:1)$), and S_{-12} (containing $Q_1 = (0:1:0:1)$), we have $S_n(\mathbb{Q}) \neq \emptyset$, and hence a canonical isomorphism of lattices

$$\mathrm{Pic}(S_n/\mathbb{Q}) \xrightarrow{\sim} \mathrm{Pic}(S_{n,\overline{\mathbb{Q}}})^{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})}, \qquad \rho(S_n/\mathbb{Q}) = \mathrm{rank}\,\mathrm{Pic}(S_n/\mathbb{Q}) = \mathrm{rank}\,\mathrm{Pic}(S_{n,\overline{\mathbb{Q}}})^{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})}. \tag{3}$$

As surveyed in Dickson [3], Pinch and Swinnerton-Dyer [13], Swinnerton-Dyer [14, 8, 9], Bright [12], Elkies [4, 5], Bremner, Choudhry, and Ulas [1], Logan, McKinnon, and van Luijk [6], Bremner [2], Tomita [10], Andrašek, Kazalicki, and Vlah [15], and the October 2026 Mathematics Stack Exchange online communication of Vepir [11], the known infinite families of solutions for fixed constants $n \in \mathbb{Q}^\times$ fall into two geometric categories summarized in Table 1.

Table 1: Summary of Previously Known Fixed-Constant n Families on $S_n : na^4 + b^4 + c^4 = d^4$.

Class	**Constant** n (mod $\mathbb{Q}^{\times 4}$)	**Geometric Mechanism & Curve Genus**	**Primary Reference**
Genus 0	$n = -1$	Rational skew lines; base curve of degree 7 (and 86 smooth $p_a = 0$ families of degree < 50)	Euler (1772) [3], Swinnerton-Dyer (1943, 2000) [14, 8], Bremner (2025) [2]
Genus 0	$n = 4$	Sophie Germain splitting; base curve of degree 4	Gerardin (1911), Carmichael (1913) [3, 11]
Genus 0	$n = -12$	Parabolic quadric cone $(d-b)^2 = 6ac$; degree 8 (9-nodal curve in $\lvert 2H \rvert$)	Elkies (2000) [5]
Genus 1	$n = -s^2$	Asymptotic inflection tangent ($ABCD = \square$, degree-17 endomorphism)	Euler (1772), Richmond (1944) [7], Logan–McKinnon–van Luijk (2010) [6]
Genus 1	$n = +s^2$	Sphere-quadric / conic-bundle fibration (deg 4)	Elkies (1988) [4], Hahn–Oh (1992)
Genus 1	$n = \pm 2s^2$	Simultaneous quadric diagonalization (deg 4)	Bremner–Choudhry–Ulas (2014) [1]
Genus 1	$n = p_0^4 + q_0^4$	$\mathbb{Q}(\omega)$ bitangent plane $na + p_0^3(b+c) = q_0^3 d$	Vepir (2026, MSE online communication) [11]

In addition, as synthesized in Vepir's October 2026 online communication [11], five fundamental polynomial identities on S_n derive from the classical generating identity

$$(X+Y)^4 - (X-Y)^4 - Z^4 = 8XY(X^2+Y^2) - Z^4 = na^4, \tag{4}$$

Table 2: Five Polynomial Specializations Arising from $(X+Y)^4-(X-Y)^4-Z^4=na^4$ [3, 5, 11].

ID	Author	$X(p,q)$	$Y(p,q)$	$Z(p,q)=b(p,q)$ & Resulting $na^4=8XY(X^2+Y^2)-Z^4$
(0)	Euler (1772)	$p^7+p^5q^2-2p^3q^4+pq^6$	$3p^2q^5$	$e_2(p,q),\quad -1\cdot e_3(p,q)^4\quad (n=-1)$
(1)	Carmichael (1913)	$8p^4$	q^4	$8p^3q,\quad 4\cdot(2pq^3)^4\quad (n=4)$
(2)	Elkies (2000)	$192p^8-q^8$	$24p^4q^4$	$192p^7q,\quad -12\cdot(2pq^7)^4\quad (n=-12)$
(3)	Norrie (1911)	p^4	q^4	$2p^3q,\quad \frac{q^8-p^8}{2}\cdot(2pq)^4$
(4)	Vepir (2026) [11]	$p(64p^8-3q^8)$	$8p^5q^4$	$2q(32p^8-q^8),\quad (20p^8-q^8)\cdot(2q^7)^4$

by setting $d=X+Y$, $c=X-Y$, and $b=Z$, as displayed in Table 2. Here $e_2(p,q)=p^6q+3p^5q^2-2p^4q^3+p^2q^5+q^7$ and $e_3(p,q)=e_2(p,-q)=p^6q-3p^5q^2-2p^4q^3+p^2q^5+q^7$.

Overview of Main Results: Every theorem, proposition, and corollary in this paper is accompanied by a complete, self-contained mathematical proof and an executable symbolic verification suite (Appendix A). We construct four new primitive polynomial families for $n=4$ (degrees 8 and 16, representing distinct smooth rational curve classes in $\mathrm{Pic}(S_4/\mathbb{Q})$) and derive single-generator involutive polynomial representations for $n=-1$ at degrees $13,19,31$, together with higher-degree iterates at degrees $37,55,91$ and singular rational curve families of degrees 43 ($p_a=66$) and 45 ($p_a=58$) via the factored degree-17 asymptotic tangent self-map $\mathcal{A}_{\pm}$. Using the explicit 20-line integral basis $\mathcal{B}_{20}$ of $\mathrm{Pic}(S_{n,\overline{\mathbb{Q}}})$ of determinant -64 and the full 20×48 coordinate matrix C_{48}, we establish exact quadratic degree towers for $n=4$ ($\deg=8k^2-4k+4$) and $n=-1$ ($\deg=9k^2-3k+7,\ 12k^2+7,\ 18k^2+6k+7$), and prove via Galois descent and Hessian plane rigidity why $\mathrm{Pic}(S_{-12}/\mathbb{Q})=\mathbb{Z}H$ has rank 1 and obstructs any infinite tower of birational-automorphism or skew-line families for $n=-12$. Section 11 provides a complete curve-by-curve and identity-by-identity comparison against the existing literature.

2 Binomial and Trinomial $n(p,q)$ Engines Across Arbitrary Degrees

Before restricting to constant n, we examine the algebraic mechanisms that collapse $d(p,q)^4-b(p,q)^4-c(p,q)^4$ into $n(p,q)a(p,q)^4$ where $n(p,q)$ has at most 2 or 3 terms (monomial, binomial, or trinomial) for arbitrary maximum degree $m=\max\deg(a,b,c,d)$.

2.1 The Difference-of-Two-Cubes Collapse Engine

Theorem 2.1 (Generalized Difference-of-Two-Cubes Collapse)**.** *Let X,Y,L be elements of any commutative ring satisfying the quadratic constraint*

$$L^2=3XY. \tag{5}$$

Then the quartic difference $(X-Y+L)^4-(X-Y-L)^4$ collapses identically to a difference of two cubes:

$$(X-Y+L)^4-(X-Y-L)^4=8L(X^3-Y^3). \tag{6}$$

In particular, parametrizing $L^2=3XY$ over $\mathbb{Q}$ by setting $X=3U^2$, $Y=4V^2$, and $L=6UV$ yields the universal two-variable binomial collapse valid for all (U,V):

$$(3U^2+6UV-4V^2)^4-(3U^2-6UV-4V^2)^4=48UV(27U^6-64V^6)=1296U^7V-3072UV^7. \tag{7}$$

Proof. Set $Q=X-Y$. By the binomial expansion $(Q+L)^4-(Q-L)^4=8Q^3L+8QL^3=8QL(Q^2+L^2)$, substituting the hypothesis $L^2=3XY$ gives

$$Q^2+L^2=(X-Y)^2+3XY=X^2-2XY+Y^2+3XY=X^2+XY+Y^2.$$

Multiplying by $Q = X - Y$ yields the classical difference-of-two-cubes factorization $Q(Q^2 + L^2) = (X - Y)(X^2 + XY + Y^2) = X^3 - Y^3$, establishing (6). Next, over any commutative ring, setting $X = 3U^2$, $Y = 4V^2$, and $L = 6UV$ satisfies $L^2 = 36U^2V^2 = 3(3U^2)(4V^2) = 3XY$. Substituting these into (6) gives

$$8(6UV)((3U^2)^3 - (4V^2)^3) = 48UV(27U^6 - 64V^6) = 1296U^7V - 3072UV^7,$$

which completes the proof. □

Corollary 2.2 (Arbitrary Even-Degree Binomial/Monomial-n Hierarchy)**.** *For any integer $k \geq 1$, setting $U = p^k$ and $V = q^k$ in (7) gives degree-$2k$ polynomials $d(p,q) = 3p^{2k} + 6p^kq^k - 4q^{2k}$ and $b(p,q) = 3p^{2k} - 6p^kq^k - 4q^{2k}$ whose fourth-power difference is the pure 2-term binomial $6^4p^{7k}q^k - 12 \cdot 4^4p^kq^{7k}$:*

1. *When $k = 4m$ is a multiple of 4 ($\deg = 8, 16, 24, 32, \dots$), $7k = 28m$ and $k = 4m$ are both multiples of 4, so setting $c = 6p^{7m}q^m$ cancels the first term identically and setting $a = 4p^mq^{7m}$ yields the constant $n = -12$ (recovering Elkies' degree-8 family [5] at $m = 1$ and its higher-power pullbacks at $m \geq 2$).*
2. *For any $k \geq 1$ and any choice of $c = 6p^rq^s$, $a = 2p^uq^v$ with nonnegative integers u, v, r, s satisfying $4u \leq k$, $4v \leq k$, $4r \geq 4u$, $4s \geq 4v$, dividing $1296p^{7k}q^k - 3072p^kq^{7k} - c^4$ by a^4 produces an exact polynomial trinomial or binomial $n(p,q) \in \mathbb{Z}[p,q]$ at degree $\deg = 2k$ ($\deg = 4, 6, 8, 10, 12, 14, 16, \dots$).*

Proof. Substituting $U = p^k$ and $V = q^k$ directly into (7) yields

$$d(p,q)^4 - b(p,q)^4 = 1296p^{7k}q^k - 3072p^kq^{7k} = 6^4p^{7k}q^k - 12 \cdot 4^4p^kq^{7k}.$$

(1) If $k = 4m$, then $6^4p^{7k}q^k = (6p^{7m}q^m)^4 = c^4$ and $-12 \cdot 4^4p^kq^{7k} = -12(4p^mq^{7m})^4 = -12a^4$, so $d^4 - b^4 - c^4 = -12a^4$. (2) For general $k \geq 1$, since $1296 = 16 \cdot 81$, $3072 = 16 \cdot 192$, and $6^4 = 16 \cdot 81$ are all divisible by $2^4 = 16$, and the exponents satisfy $\min(7k, k, 4r) \geq 4u$ in p and $\min(k, 7k, 4s) \geq 4v$ in q, the monomial $a^4 = 16p^{4u}q^{4v}$ divides every term of $d^4 - b^4 - c^4$ in $\mathbb{Z}[p,q]$, giving

$$n(p,q) = 81p^{7k-4u}q^{k-4v} - 192p^{k-4u}q^{7k-4v} - 81p^{4r-4u}q^{4s-4v} \in \mathbb{Z}[p,q],$$

which collapses to the pure 2-term binomial $-192p^{k-4u}q^{7k-4v}$ whenever $(4r, 4s) = (7k, k)$, or to a 2-term binomial whenever $c = 0$. □

2.2 The Arbitrary-Degree (h,t)-Binomial Engine and Degree-15 Constructions

Proposition 2.3 (Arbitrary-Degree Binomial/Trinomial Specialization Engine)**.** *For any integer $k \geq 1$ and non-zero parameters $h, t \in \mathbb{Q}^\times$ (with $h, t \in \mathbb{Z} \setminus \{0\}$ when integer-coefficient polynomials $a, b, c, d \in \mathbb{Z}[p,q]$ are required):*

1. ***Weighted Even-$4k$ Binomial-n Family*** *(generalizing the $k = 1$ family of Bremner–Choudhry–Ulas [1], Piezas, and Vepir [11, Remark III]): Setting $X = hp^{4k}$, $Y = q^{4k}$, $b = tp^{3k}q^k$, $a = 2p^kq^k$, $c = hp^{4k} - q^{4k}$, and $d = hp^{4k} + q^{4k}$ yields the exact 2-term binomial parametrization of degree $4k$ in $\mathbb{Q}[p,q]$ (and in $\mathbb{Z}[p,q]$ whenever $h, t \in \mathbb{Z}$ satisfy $2 \mid h$ and $4 \mid t$, or after absorbing 2 into $a = p^kq^k$):*

$$n(p,q) = \frac{8h^3 - t^4}{16}\, p^{8k} + \frac{h}{2}\, q^{8k}. \tag{8}$$

2. ***Homogeneous Degree-15 Trinomial-n Family*** *($\max\deg = 15$, $\gcd(a,b,c,d) = 1$): Setting*

$$a = 2p^3q^3, \qquad b = 2p^3q^{12}, \qquad c = p^{15} - 2q^{15}, \qquad d = p^{15} + 2q^{15} \tag{9}$$

satisfies $n(p,q)a^4 + b^4 + c^4 = d^4$ in $\mathbb{Z}[p,q]$ with $\max\deg(a,b,c,d) = 15$, $\gcd(a,b,c,d) = 1$, and the 3-term integer trinomial

$$n(p,q) = q^3(p^{33} + 4p^3q^{30} - q^{33}) \in \mathbb{Z}[p,q]. \tag{10}$$

3. ***Homogeneous Degree-15 Leading-Term Cancellation Family*** *(*$\max\deg = 15$*): Setting*

$$b = 2p^{12}q^3, \qquad c = p^{15} - 2p^3q^{12} + q^{15}, \qquad d = p^{15} + 2p^3q^{12} + q^{15} \tag{11}$$

cancels the leading $16p^{48}q^{12}$ *term of* $d^4 - c^4$ *against* b^4*, leaving the factored remainder*

$$d^4 - b^4 - c^4 = 16p^3q^{27}(3p^{30} + 4p^{21}q^9 + 3p^{15}q^{15} + 4p^6q^{24} + q^{30}).$$

4. ***Inhomogeneous Degree-15 Binomial-*** n ***Family*** *(*$\max\deg = 15$*): Setting* $d = hp^{12} + q^{15}$*,* $c = hp^{12} - q^{15}$*,* $b = tp^9q^3$*, and* $a = 2p^3q^3$ *gives* $\max\deg(a,b,c,d) = 15$*,* $\gcd(a,b,c,d) = 1$ *(for* $h = 1$ *or* $\gcd(h,t) = 1$*), and*

$$d^4 - c^4 - b^4 = (8h^3q^3 - t^4)p^{36}q^{12} + 8hp^{12}q^{45} = \left[\frac{1}{2}h^3p^{24}q^3 - \frac{t^4}{16}p^{24} + \frac{1}{2}hq^{33}\right]a^4,$$

which collapses to a 2-term binomial $n(p,q) = \frac{1}{2}hq^3(h^2p^{24} + q^{30})$ *at* $t = 0$ *(or in* $\mathbb{Z}[p,q]$ *whenever* $2 \mid h$ *and* $4 \mid t$*).*

Proof. (1) By (4), with $X = hp^{4k}$ and $Y = q^{4k}$, we have

$$d^4 - c^4 - b^4 = 8(hp^{4k})(q^{4k})(h^2p^{8k} + q^{8k}) - t^4p^{12k}q^{4k} = (8h^3 - t^4)p^{12k}q^{4k} + 8hp^{4k}q^{12k}.$$

Dividing by $a^4 = (2p^kq^k)^4 = 16p^{4k}q^{4k}$ in $\mathbb{Q}[p,q]$ yields (8).

(2) For (9), setting $X = p^{15}$ and $Y = 2q^{15}$ gives $d^4 - c^4 = 8XY(X^2 + Y^2) = 16p^{15}q^{15}(p^{30} + 4q^{30})$. Subtracting $b^4 = (2p^3q^{12})^4 = 16p^{12}q^{48}$ and dividing by $a^4 = (2p^3q^3)^4 = 16p^{12}q^{12}$ gives

$$n(p,q) = \frac{16p^{45}q^{15} + 64p^{15}q^{45} - 16p^{12}q^{48}}{16p^{12}q^{12}} = p^{33}q^3 + 4p^3q^{33} - q^{36} = q^3(p^{33} + 4p^3q^{30} - q^{33}).$$

Since $d(p,q) + c(p,q) = 2p^{15}$ and $d(p,q) - c(p,q) = 4q^{15}$, any irreducible polynomial factor dividing both $c(p,q)$ and $d(p,q)$ in $\mathbb{Q}[p,q]$ must divide both p^{15} and q^{15}, which are coprime; and since $\gcd(2p^3q^3, p^{15} - 2q^{15}, p^{15} + 2q^{15}) = 1$ in $\mathbb{Z}[p,q]$, we have $\gcd(a,b,c,d) = 1$.

(3) For (11), setting $X = p^{15} + q^{15}$ and $Y = 2p^3q^{12}$ gives

$$d^4 - c^4 = 8XY(X^2 + Y^2) = 16p^3q^{12}(p^{15} + q^{15})((p^{15} + q^{15})^2 + 4p^6q^{24}).$$

Expanding the product gives $16p^{48}q^{12} + 48p^{33}q^{27} + 64p^{24}q^{36} + 48p^{18}q^{42} + 64p^9q^{51} + 16p^3q^{57}$. Subtracting $b^4 = (2p^{12}q^3)^4 = 16p^{48}q^{12}$ cancels the leading term identically and factors out $16p^3q^{27}$.

(4) Setting $X = hp^{12}$ and $Y = q^{15}$ gives $d^4 - c^4 - b^4 = 8(hp^{12})(q^{15})(h^2p^{24} + q^{30}) - t^4p^{36}q^{12} = (8h^3q^3 - t^4)p^{36}q^{12} + 8hp^{12}q^{45}$. Dividing by $a^4 = 16p^{12}q^{12}$ yields the stated formula. □

3 New Primitive Polynomial Families and Infinite Towers for $n = 4$

In the classical and modern literature surveyed in Table 1, only a single polynomial parametrization was previously recorded for $4a^4 + b^4 + c^4 = d^4$: the degree-4 curve of Gerardin (1911) and Carmichael (1913) [3, 11], which in symmetrized coordinates ($2p \mapsto r$, $q \mapsto s$) is

$$C_0(r,s): \quad a_0 = 2rs^3, \qquad b_0 = 2r^3s, \qquad c_0 = r^4 - 2s^4, \qquad d_0 = r^4 + 2s^4. \tag{12}$$

To construct higher-degree primitive families on S_4 and S_{-1} with complete geometric rigor, we first recall the fundamental extension theorem for birational self-maps of smooth projective K3 surfaces and explain how global polynomial representations on $\mathbb{P}^3$ relate to regular K3 automorphisms.

Lemma 3.1 (Biregular Extension of Birational Self-Maps on Smooth Projective K3 Surfaces)**.** *Let* $S \subset \mathbb{P}^3$ *be a smooth quartic K3 surface defined over a field* k *of characteristic* 0*, and let* $f : S \dashrightarrow S$ *be a birational self-map defined over* k *(for instance, satisfying* $f \circ f = \mathrm{id}_S$ *on a Zariski-open dense subset* $U \subset S$*).*

1. ***Unique Biregular Extension:*** *Because S has trivial canonical bundle $\omega_S \cong \mathcal{O}_S$ and contains no (-1)-curves (as the adjunction formula $2p_a(E) - 2 = E^2 + K_S \cdot E = E^2$ is even for every irreducible curve $E \subset S$), S is its own unique minimal model. By the minimal-model factorization theorem for birational maps of smooth projective surfaces (Huybrechts [16, Ch. 2, Proposition 1.5 and Corollary 1.6]), f extends uniquely across its locus of indeterminacy to a regular biregular isomorphism $f \in \mathrm{Aut}(S/k)$, inducing an isometry $f^* \in \mathrm{O}(\mathrm{Pic}(S_{\overline{\mathbb{Q}}}))$ that commutes with $\mathrm{Gal}(\overline{\mathbb{Q}}/k)$.*

2. ***Removal of Divisorial Base Components in Projective Coordinates:*** *Suppose $f : S \dashrightarrow S \subset \mathbb{P}^3$ is written in homogeneous coordinates as $P \mapsto (F_0(P) : F_1(P) : F_2(P) : F_3(P))$ for homogeneous polynomials $F_i \in k[a, b, c, d]$ of degree d_0, and suppose the scheme-theoretic vanishing locus $\{F_0 = F_1 = F_2 = F_3 = 0\} \cap S$ has divisorial (codimension-1) part $\sum_r m_r E_r$ on S. Then in $\mathrm{Pic}(S/k)$, the hyperplane class H pulls back under the regular automorphism $f \in \mathrm{Aut}(S/k)$ to the ample divisor class*

$$f^*(H) = d_0 H - \sum_r m_r [E_r]. \tag{13}$$

Locally on S, dividing the sections $F_0, F_1, F_2, F_3 \in H^0(S, \mathcal{O}_S(d_0 H))$ by a local defining equation of the effective Cartier divisor $D_{\mathrm{base}} = \sum_r m_r E_r$ yields a base-point-free linear system in $H^0(S, \mathcal{O}_S(d_0 H - D_{\mathrm{base}})) = H^0(S, \mathcal{O}_S(f^*(H)))$ that defines the regular morphism $f : S \to S$ everywhere on S.

Proof. (1) By Zariski's theorem on the factorization of birational maps between smooth projective surfaces, any birational map $f : S \dashrightarrow S$ factors as $f = \pi_2 \circ \pi_1^{-1}$, where $\pi_1 : \widetilde{S} \to S$ and $\pi_2 : \widetilde{S} \to S$ are finite sequences of blow-ups at closed points. If π_1 is non-trivial, the final blow-up in π_2 must contract a (-1)-curve $E \subset \widetilde{S}$ (since $\mathrm{rank}\,\mathrm{Pic}(\widetilde{S}) = \mathrm{rank}\,\mathrm{Pic}(S) + r$ and $K_{\widetilde{S}} = \pi_1^* K_S + \sum E_i = \sum E_i = \pi_2^* K_S + \sum E'_j$). By symmetry of the minimal model condition when $K_S = 0$, the curves contracted by π_1 and π_2 match, so $\pi_2 \circ \pi_1^{-1}$ contains no uncancelled blow-ups and is a regular isomorphism everywhere on S [16, Ch. 2, Proposition 1.5]. (2) On S, the rational map $P \mapsto (F_0(P) : \cdots : F_3(P))$ pulls back the hyperplane bundle $\mathcal{O}_{\mathbb{P}^3}(1)|_S = \mathcal{O}_S(H)$ outside the base locus. Twisting by $\mathcal{O}_S(-D_{\mathrm{base}})$ removes all codimension-1 base components on the normal (smooth) surface S, leaving at most a codimension-2 (finite point) base locus; since f is a regular morphism by (1) and $f^*\mathcal{O}_S(H)$ is globally generated on S, this codimension-2 base locus is empty and (13) holds in $\mathrm{Pic}(S/k)$. □

Theorem 3.2 (The $\mathbb{Q}(i)$-Conjugate Skew-Line Involution and Regular Extension on S_4)**.** *On the smooth quartic K3 surface $S_4 : 4a^4 + b^4 + c^4 = d^4$, consider the pair of disjoint (skew) lines conjugate over $\mathbb{Q}(i)$:*

$$L_1 : \quad d - ib = 0, \quad c - (1+i)a = 0, \qquad\qquad \overline{L}_1 : \quad d + ib = 0, \quad c - (1-i)a = 0. \tag{14}$$

1. ***Eigenspace Decomposition and Generic Transversal Construction on $S_4 \setminus (L_1 \cup \overline{L}_1)$:*** *Define the $\mathbb{Q}$-rational linear operator $M : \mathbb{P}^3 \to \mathbb{P}^3$ by*

$$M(a, b, c, d) = (c - a,\ d,\ c - 2a,\ -b).$$

Then $M^2 = -I_4$ identically on $\mathbb{Q}^4$, and the $\pm i$-eigenspaces of M over $\mathbb{Q}(i)$ are precisely the skew lines $L_1 = \ker(M - iI_4)$ and $\overline{L}_1 = \ker(M + iI_4)$. Consequently, for any point $P = (a : b : c : d) \in S_4 \setminus (L_1 \cup \overline{L}_1)$ (and in particular for every rational point $P \in S_4(\mathbb{Q})$, since $(L_1 \cup \overline{L}_1)(\mathbb{Q}) = \emptyset$), P and $M(P)$ are linearly independent and span the unique transversal line $T_P \subset \mathbb{P}^3$ passing through P and intersecting both L_1 (at $t = -i$) and $\overline{L}_1$ (at $t = +i$):

$$P(t) = P + t\,M(P) = \big(a + t(c - a),\ b + td,\ c + t(c - 2a),\ d - tb\big), \qquad t \in \mathbb{P}^1. \tag{15}$$

(When $P \in L_1$ or $P \in \overline{L}_1$, $M(P) = \pm iP$ is proportional to P, so $P(t) = (1 \pm it)P$ degenerates to the constant point $P \in \mathbb{P}^3$ for $t \neq \pm i$ and vanishes at $t = \mp i$.)

2. ***Rational Involution and Regular Extension to*** $\mathrm{Aut}(S_4/\mathbb{Q})$***:*** *Along* $P(t)$ *in* (15)*, the surface polynomial decomposes identically in* $\mathbb{Z}[a,b,c,d,t]$ *as*

$$4a(t)^4 + b(t)^4 + c(t)^4 - d(t)^4 = (1-t^4)(4a^4+b^4+c^4-d^4) + 2t(t^2+1)(U_4t+V_4), \tag{16}$$

where $t=0$ *corresponds to* P*,* $t=\mp i$ *correspond to* $T_P\cap L_1$ *and* $T_P\cap\overline{L}_1$*, and the fourth intersection point* $t=-V_4/U_4$ *defines the degree-5 rational self-map* $\Phi_{L,\overline{L}}: S_4 \dashrightarrow S_4$ *over* $\mathbb{Q}$*:*

$$\Phi_{L,\overline{L}}(P) = U_4(P)P - V_4(P)M(P) = \big(U_4a - V_4(c-a),\, U_4b - V_4d,\, U_4c - V_4(c-2a),\, U_4d + V_4b\big), \tag{17}$$

where the homogeneous degree-4 polynomials $U_4, V_4 \in \mathbb{Z}[a,b,c,d]$ *are given by*

$$U_4 = 3(c^2-2ac+2a^2)^2, \qquad V_4 = -2\big(4a^4 - 4a^3c + 2ac^3 - c^4 - bd(b^2+d^2)\big). \tag{18}$$

On the Zariski-open dense set $U = S_4\setminus(L_1\cup\overline{L}_1\cup L_2\cup\overline{L}_2)$ *where* $L_2\cup\overline{L}_2 = \{d\pm ib=0,\, c-(1\pm i)a=0\}$*,* $\Phi_{L,\overline{L}}$ *is a well-defined morphism satisfying* $\Phi^2_{L,\overline{L}} = \mathrm{id}_U$*. On* S_4*, the divisorial base locus of the degree-5 polynomial tuple* (17) *is* $3D_1+D_2$ *(where* $D_1 = [L_1]+[\overline{L}_1]$ *has multiplicity* 3 *and* $D_2 = [L_2]+[\overline{L}_2]$ *has multiplicity* 1*). By Lemma 3.1, removing these divisorial base components extends* $\Phi_{L,\overline{L}}$ *uniquely to a regular biregular involution* $\Phi_{L,\overline{L}} \in \mathrm{Aut}(S_4/\mathbb{Q})$ *satisfying* $\Phi^2_{L,\overline{L}} = \mathrm{id}_{S_4}$*,* $\Phi^*_{L,\overline{L}}(H) = 5H - 3D_1 - D_2$*,* $\Phi^*_{L,\overline{L}}(D_2) = D_2$*, and* $\Phi^*_{L,\overline{L}}(D_1) = 8H - 5D_1 - 2D_2$*.*

Proof. (1) First, on L_1, we have $d=ib$ and $c=(1+i)a$; since $i^4=1$, $(1+i)^2=2i$, and $(1+i)^4 = (2i)^2 = -4$, we obtain $4a^4+b^4+c^4-d^4 = 4a^4+b^4-4a^4-b^4=0$, so $L_1\subset S_4$ and by conjugation $\overline{L}_1\subset S_4$. At $L_1\cap\overline{L}_1$, we have $ib=-ib \implies b=d=0$ and $(1+i)a=(1-i)a \implies a=c=0$, which is empty in $\mathbb{P}^3$; hence L_1 and $\overline{L}_1$ are disjoint skew lines, and $(L_1\cup\overline{L}_1)(\mathbb{Q}) = L_1\cap\overline{L}_1 = \emptyset$.

Next, applying $M(a,b,c,d) = (c-a, d, c-2a, -b)$ twice gives

$$M\big(M(a,b,c,d)\big) = \big((c-2a)-(c-a),\, -b,\, (c-2a)-2(c-a),\, -d\big) = (-a,-b,-c,-d) = -P,$$

so $M^2 = -I_4$. Furthermore, $M(P) = iP \iff c-a = ia$ and $d=ib \iff P\in L_1$, while $M(P) = -iP \iff P\in\overline{L}_1$. Because M has only eigenvalues $\pm i$, a point $P\in\mathbb{P}^3$ is linearly dependent with $M(P)$ if and only if P is an eigenvector of M, i.e., $P\in L_1\cup\overline{L}_1$. For $P\notin L_1\cup\overline{L}_1$, $P+tM(P)$ at $t=-i$ satisfies $M(P-iM(P)) = M(P)+iP = i(P-iM(P))$, so $P(-i)\in L_1$ and $P(+i)\in\overline{L}_1$. Thus $T_P = \{P+tM(P) : t\in\mathbb{P}^1\}$ is the unique transversal line through P meeting both L_1 and $\overline{L}_1$.

(2) Expanding $(b+td)^4-(d-tb)^4$ and $4(a+t(c-a))^4+(c+t(c-2a))^4$ in $\mathbb{Z}[a,b,c,d,t]$ gives

$$\begin{aligned}(b+td)^4-(d-tb)^4 &= (1-t^4)(b^4-d^4) + 4t(t^2+1)bd(b^2+d^2),\\ 4\big(a+t(c-a)\big)^4+\big(c+t(c-2a)\big)^4 &= (1-t^4)(4a^4+c^4)\\ &\quad + 2t(t^2+1)\Big[3(c^2-2ac+2a^2)^2t - 2(4a^4-4a^3c+2ac^3-c^4)\Big].\end{aligned}$$

Adding these two identities establishes (16). For $P\in S_4\setminus(L_1\cup\overline{L}_1)$, P and $M(P)$ are linearly independent, so $\Phi_{L,\overline{L}}(P) = U_4(P)P - V_4(P)M(P)$ in (17) vanishes if and only if $U_4(P)=V_4(P)=0$. On S_4, $U_4=0 \iff c^2-2ac+2a^2=0$, which forces $c=(1\pm i)a$, hence $4a^4+c^4=0$ and $b^4=d^4$. Since $4a^4-4a^3c+2ac^3-c^4 = -(c^2-2ac+2a^2)(c^2-2a^2)$, $U_4=0$ reduces $V_4=0$ to $bd(b^2+d^2)=0$. Since $b^4=d^4$, $bd=0$ forces $b=d=0$ (where $4a^4+c^4=0$ has no points on $L_1\cap\overline{L}_1$), while $b^2+d^2=0$ forces $d=\pm ib$, which cuts out precisely the four lines $L_1\cup\overline{L}_1\cup L_2\cup\overline{L}_2$.

To prove involutivity $\Phi^2_{L,\overline{L}} = \mathrm{id}_U$ on the dense open set $U = S_4\setminus(L_1\cup\overline{L}_1\cup L_2\cup\overline{L}_2)$, let $P\in U$ and $P' = \Phi_{L,\overline{L}}(P) = U_4(P)P - V_4(P)M(P)$. Since $M^2=-I_4$, we have

$$M(P') = U_4(P)M(P) - V_4(P)M^2(P) = V_4(P)P + U_4(P)M(P).$$

Because $U_4(P)^2+V_4(P)^2\neq 0$ on a Zariski-open dense subset of U (for instance at $P=(0:1:0:1)\in U$, $U_4=0$ and $V_4=4\neq 0$, giving $U_4^2+V_4^2=16\neq 0$), $\mathrm{span}(P',M(P')) = \mathrm{span}(P,M(P)) = T_P$. Thus

$T_{P'} = T_P$, and the two non-$L_1, \overline{L}_1$ intersection points of T_P with S_4 (counted with multiplicity) are $\{P, P'\}$, so $\Phi_{L,\overline{L}}(P') = P$. Since U is irreducible and $\Phi^2_{L,\overline{L}} = \mathrm{id}$ on a dense open subset, $\Phi_{L,\overline{L}}$ is a birational involution of S_4. By Lemma 3.1(1), $\Phi_{L,\overline{L}}$ extends uniquely to a regular biregular involution $\Phi_{L,\overline{L}} \in \mathrm{Aut}(S_4/\mathbb{Q})$.

Finally, we determine the exact base-locus multiplicities and divisor pullbacks (incorporated into the proof of Theorem 3.4 below): along $L_2 \cup \overline{L}_2$, $c^2 - 2ac + 2a^2$ vanishes to order 1 (so U_4 vanishes to order 2) and $b^2 + d^2$ vanishes to order 1 (so V_4 vanishes to order 1), giving base multiplicity 1 along D_2. Along L_1, introduce local surface coordinates by setting $a = 1$, $c = (1 + i) + u$, $b = b_0$, and $d = ib_0 + v$; the surface equation $4 + b_0^4 + (1 + i + u)^4 - (ib_0 + v)^4 = 0$ gives $-8(1 - i)u + 4ib_0^3 v + O(2) = 0$, so $v = -2(1 + i)b_0^{-3}u + O(u^2)$. Substituting this into (17) shows that $U_4 = O(u^2)$, $V_4 = O(u^2)$, and $P - iM(P) = O(u)$, so every component of $U_4P - V_4M(P)$ vanishes to exact order 3 along L_1 (and by conjugation along $\overline{L}_1$). Hence $D_{\text{base}} = 3D_1 + D_2$, and Lemma 3.1(2) gives $\Phi^*_{L,\overline{L}}(H) = 5H - 3D_1 - D_2$. □

Corollary 3.3 (Four New Primitive Polynomial Parametrizations for $n = 4$)**.** *Applying $\Phi_{L,\overline{L}}$ to the symmetry images of $C_0(r, s)$ in (12) and dividing out the common polynomial GCD in $\mathbb{Z}[r, s]$ yields four new primitive polynomial parametrizations of $4a^4 + b^4 + c^4 = d^4$ satisfying $\gcd(a, b, c, d) = 1$ in $\mathbb{Z}[r, s]$:*

1. ***Family 4-I (Even-Odd Sparse Degree-8 Family):***

$$\begin{aligned} a_{\mathrm{I}}(r, s) &= 2r^7 s + 10r^5 s^3 + 8r^3 s^5 + 4rs^7, \\ b_{\mathrm{I}}(r, s) &= r^8 + 6r^6 s^2 - 12r^2 s^6 - 4s^8, \\ c_{\mathrm{I}}(r, s) &= 2r^7 s + 8r^5 s^3 + 20r^3 s^5 + 8rs^7, \\ d_{\mathrm{I}}(r, s) &= r^8 + 6r^6 s^2 + 20r^4 s^4 + 12r^2 s^6 + 4s^8. \end{aligned} \tag{19}$$

2. ***Family 4-II (Alternating-Sign Sparse Degree-8 Family):***

$$a_{\mathrm{II}}(r, s) = i\, a_{\mathrm{I}}(r, is), \quad b_{\mathrm{II}}(r, s) = b_{\mathrm{I}}(r, is), \quad c_{\mathrm{II}}(r, s) = i\, c_{\mathrm{I}}(r, is), \quad d_{\mathrm{II}}(r, s) = d_{\mathrm{I}}(r, is), \tag{20}$$

explicitly in $\mathbb{Z}[r, s]$:

$$\begin{aligned} a_{\mathrm{II}}(r, s) &= -2r^7 s + 10r^5 s^3 - 8r^3 s^5 + 4rs^7, \quad & b_{\mathrm{II}}(r, s) &= r^8 - 6r^6 s^2 + 12r^2 s^6 - 4s^8, \\ c_{\mathrm{II}}(r, s) &= -2r^7 s + 8r^5 s^3 - 20r^3 s^5 + 8rs^7, \quad & d_{\mathrm{II}}(r, s) &= r^8 - 6r^6 s^2 + 20r^4 s^4 - 12r^2 s^6 + 4s^8. \end{aligned}$$

3. ***Family 4-III (Full 9-Term Degree-8 Family):***

$$\begin{aligned} a_{\mathrm{III}}(r, s) &= 2r^8 + 16r^7 s + 48r^6 s^2 + 74r^5 s^3 + 68r^4 s^4 + 40r^3 s^5 + 8r^2 s^6 - 12rs^7 - 8s^8, \\ b_{\mathrm{III}}(r, s) &= -2r^8 - 10r^7 s - 12r^6 s^2 + 8r^5 s^3 + 16r^4 s^4 - 20r^3 s^5 - 48r^2 s^6 - 32rs^7 - 8s^8, \\ c_{\mathrm{III}}(r, s) &= -r^8 - 2r^7 s + 12r^6 s^2 + 60r^5 s^3 + 112r^4 s^4 + 100r^3 s^5 + 40r^2 s^6 + 8rs^7 + 4s^8, \\ d_{\mathrm{III}}(r, s) &= 3r^8 + 22r^7 s + 68r^6 s^2 + 108r^5 s^3 + 92r^4 s^4 + 44r^3 s^5 + 24r^2 s^6 + 24rs^7 + 12s^8. \end{aligned} \tag{21}$$

4. ***Family 4-IV (Even-Odd Sparse Degree-16 Family):***

$$\begin{aligned} a_{\mathrm{IV}}(r, s) &= -2r^{15} s - 2r^{13} s^3 - 16r^{11} s^5 + 44r^9 s^7 - 8r^7 s^9 + 24r^5 s^{11} - 16rs^{15}, \\ b_{\mathrm{IV}}(r, s) &= -r^{16} - 2r^{14} s^2 - 28r^{10} s^6 + 56r^6 s^{10} + 16r^2 s^{14} + 16s^{16}, \\ c_{\mathrm{IV}}(r, s) &= -2r^{15} s + 12r^{11} s^5 - 8r^9 s^7 + 88r^7 s^9 - 64r^5 s^{11} - 16r^3 s^{13} - 32rs^{15}, \\ d_{\mathrm{IV}}(r, s) &= r^{16} + 2r^{14} s^2 + 20r^{12} s^4 - 28r^{10} s^6 + 8r^8 s^8 - 56r^6 s^{10} + 80r^4 s^{12} + 16r^2 s^{14} + 16s^{16}. \end{aligned} \tag{22}$$

Proof. Since $C_0(r, s) = (2rs^3, 2r^3 s, r^4 - 2s^4, r^4 + 2s^4)$ has degree 4, substituting any signed coordinate permutation of $C_0(r, s)$ into the degree-5 homogeneous map $\Phi_{L,\overline{L}}$ in (17) produces four homogeneous polynomials of raw degree $5 \times 4 = 20$ satisfying $4A^4 + B^4 + C^4 = D^4$:

1. For the branch $C^{(1)}(r,s) = (2rs^3, -(r^4-2s^4), 2r^3s, r^4+2s^4)$, we have $c^2-2ac+2a^2 = 4r^2s^2(r^4-2r^2s^2+2s^4)$. Because $C^{(1)}$ intersects $L_1 \cup \overline{L}_1$ at the four roots of $r^4-2r^2s^2+2s^4=0$ (where $\Phi_{L,\overline{L}}$ vanishes to order 3), the exact polynomial greatest common divisor in $\mathbb{Z}[r,s]$ of the four raw degree-20 outputs is $G_1(r,s) = 4(r^4-2r^2s^2+2s^4)^3$ of degree 12. Dividing by $G_1(r,s)$ yields the degree-8 polynomials (19).

2. For the branch $C^{(2)}(r,s) = (2rs^3, -(r^4-2s^4), -2r^3s, r^4+2s^4)$, the exact polynomial GCD in $\mathbb{Z}[r,s]$ is $G_2(r,s) = 4(r^4+2r^2s^2+2s^4)^3$ of degree 12, and dividing by $G_2(r,s)$ yields (20).

3. For the branch $C^{(3)}(r,s) = (2rs^3, 2r^3s, -(r^4-2s^4), r^4+2s^4)$, we have $c^2-2ac+2a^2 = (r^4-2r^3s+4r^2s^2-4rs^3+2s^4)(r^4+2r^3s-4r^2s^2-4rs^3+2s^4)$, and the exact polynomial GCD in $\mathbb{Z}[r,s]$ of the raw degree-20 outputs is $G_3(r,s) = (r^4-2r^3s+4r^2s^2-4rs^3+2s^4)^3$ of degree 12, leaving the degree-8 family (21).

4. For the branch $C^{(4)}(r,s) = (2rs^3, r^4-2s^4, 2r^3s, r^4+2s^4)$, the curve is disjoint from $L_1 \cup \overline{L}_1$ ($C^{(4)} \cdot D_1 = 0$) and instead meets the bisecant line pair $L_2 \cup \overline{L}_2 = \{d = \mp ib, c = (1 \pm i)a\}$ at the four roots of $r^4-2r^2s^2+2s^4=0$, where both U_4 and V_4 vanish to order 1. Thus only a single copy $G_4(r,s) = 4(r^4-2r^2s^2+2s^4)$ of degree 4 divides the four raw degree-20 outputs, leaving the degree-16 family (22).

In all four families, direct Euclidean polynomial GCD computation in $\mathbb{Q}[r,s]$ (dehomogenizing at $s=1$, where $\gcd_{\mathbb{Q}[r]}(a(r,1), b(r,1), c(r,1), d(r,1)) = 1$, and checking at $s=0, r=1$ that $(b(1,0), d(1,0)) = (\pm 1, 1)$ or $(-2, 3)$ has no factor of s and integer content 1 in $\mathbb{Z}$; see Appendix A) confirms $\gcd_{\mathbb{Z}[r,s]}(a,b,c,d) = 1$, and symbolic expansion confirms $4a^4+b^4+c^4-d^4 = 0$ identically in $\mathbb{Z}[r,s]$. □

Theorem 3.4 (Existence of Infinitely Many Primitive Polynomial Families for $n=4$)**.** *Let $\tau_b : S_4 \to S_4$ denote the coordinate sign-change involution $\tau_b(a,b,c,d) = (a,-b,c,d)$, and let $\Psi_4 = \tau_b \circ \Phi_{L,\overline{L}} \in \mathrm{Aut}(S_4/\mathbb{Q})$. Then Ψ_4 has infinite order in $\mathrm{Aut}(S_4/\mathbb{Q})$, and the iterated sequence of smooth rational curves $C_k = \Psi_4^k(C^{(1)})$ ($k \in \mathbb{Z}$, each with $C_k^2 = -2$ and arithmetic genus $p_a(C_k) = 0$) consists of infinitely many projectively inequivalent primitive polynomial parametrizations of $4a^4+b^4+c^4 = d^4$ whose degrees grow strictly quadratically according to the exact formula*

$$\deg(\Psi_4^k(C^{(1)})) = 8k^2 - 4k + 4 \qquad (k \in \mathbb{Z}), \tag{23}$$

yielding $\deg = 4, 8, 28, 64, 116, 184, \ldots$ *for* $k = 0, 1, 2, 3, 4, 5, \ldots$ *and* $\deg = 4, 16, 44, 88, 148, 224, \ldots$ *for* $k = 0, -1, -2, -3, -4, -5, \ldots$ *(while combining $\Phi_{L,\overline{L}}$ with the coordinate swap $\sigma(a,b,c,d) = (a,c,b,d)$ and $\tau_c(a,b,c,d) = (a,b,-c,d)$ on Family 4-I produces additional primitive families of degrees* 32 *and* 36 *at the second step).*

Proof. Let $H \in \mathrm{Pic}(S_4/\mathbb{Q})$ denote the hyperplane section class ($H^2 = 4$), and let $D_1 = [L_1] + [\overline{L}_1] \in \mathrm{Pic}(S_4/\mathbb{Q})$ be the rational divisor class of the disjoint conjugate line pair $L_1 \cup \overline{L}_1$ in (14). Under the sign-change involution $\tau_b(a,b,c,d) = (a,-b,c,d)$, the image of D_1 is the disjoint conjugate line pair $D_2 = \tau_b^*(D_1) = [L_2] + [\overline{L}_2] \in \mathrm{Pic}(S_4/\mathbb{Q})$, where

$$L_2 : \quad d+ib = 0, \quad c-(1+i)a = 0, \qquad\qquad \overline{L}_2 : \quad d-ib = 0, \quad c-(1-i)a = 0.$$

On the K3 surface S_4, each line has self-intersection -2, and since $L_1 \cap \overline{L}_1 = \emptyset$ and $L_2 \cap \overline{L}_2 = \emptyset$, we have $D_1^2 = D_2^2 = -4$ and $H \cdot D_1 = H \cdot D_2 = 2$. Moreover, checking the four cross-pair intersections in $\mathbb{P}^3$ gives

$$\begin{aligned} L_1 \cap L_2 &= \{(1:0:1+i:0)\}, & L_1 \cap \overline{L}_2 &= \{(0:1:0:i)\}, \\ \overline{L}_1 \cap L_2 &= \{(0:1:0:-i)\}, & \overline{L}_1 \cap \overline{L}_2 &= \{(1:0:1-i:0)\}, \end{aligned}$$

so every line in D_2 is a bisecant meeting both L_1 and $\overline{L}_1$, and $D_1 \cdot D_2 = 1+1+1+1 = 4$. Thus the rank-3 Galois-invariant sublattice $\Lambda = \mathbb{Z}H \oplus \mathbb{Z}D_1 \oplus \mathbb{Z}D_2 \subset \mathrm{Pic}(S_4/\mathbb{Q})$ has non-degenerate intersection matrix

$$G_\Lambda = \begin{pmatrix} 4 & 2 & 2 \\ 2 & -4 & 4 \\ 2 & 4 & -4 \end{pmatrix}, \qquad \det(G_\Lambda) = 64 \neq 0.$$

Now we determine the exact action of $\Phi^*_{L,\overline{L}}$ on Λ (and on all of $\mathrm{Pic}(S_4, \overline{\mathbb{Q}})$). By Theorem 3.2(2) and Lemma 3.1(2), subtracting the base-locus multiplicities (3 along D_1 and 1 along D_2) from the raw degree 5 gives $\Phi^*_{L,\overline{L}}(H) = 5H - 3D_1 - D_2$. Next, because L_2 meets L_1 at $P_1 = (1:0:1+i:0)$ and meets $\overline{L}_1$ at $P_2 = (0:1:0:-i)$, for every point $P \in L_2 \setminus \{P_1, P_2\}$ the unique line through P meeting both L_1 and $\overline{L}_1$ is the line L_2 itself ($T_P = L_2$). Since $L_2 \subset S_4$, the residual intersection of T_P with S_4 lies on L_2; indeed, evaluating (17) on a perturbation $P_\varepsilon \to P \in L_2$ shows that $\Phi_{L,\overline{L}}$ restricts to a fractional linear involution on L_2 (and by conjugation on $\overline{L}_2$). Hence $\Phi_{L,\overline{L}}(L_2) = L_2$ and $\Phi_{L,\overline{L}}(\overline{L}_2) = \overline{L}_2$, so $\Phi^*_{L,\overline{L}}(D_2) = D_2$. Finally, since $\Phi_{L,\overline{L}} \in \mathrm{Aut}(S_4/\mathbb{Q})$ is a regular involution satisfying $(\Phi^*_{L,\overline{L}})^2 = \mathrm{id}_{\mathrm{Pic}(S_4/\mathbb{Q})}$, applying $\Phi^*_{L,\overline{L}}$ to $\Phi^*_{L,\overline{L}}(H) = 5H - 3D_1 - D_2$ yields

$$H = 5(5H - 3D_1 - D_2) - 3\Phi^*_{L,\overline{L}}(D_1) - D_2 \implies 3\Phi^*_{L,\overline{L}}(D_1) = 24H - 15D_1 - 6D_2.$$

Since $\mathrm{Pic}(S_4/\mathbb{Q})$ is torsion-free, dividing by 3 proves $\Phi^*_{L,\overline{L}}(D_1) = 8H - 5D_1 - 2D_2$ (which is the reflection $\Phi^*_{L,\overline{L}}(X) = X + \frac{X\cdot R}{8}R$ across the root class $R = 4H - 3D_1 - D_2$ of self-intersection $R^2 = -16$, mapping L_1 and $\overline{L}_1$ to smooth degree-7 rational curves of self-intersection -2).

Consequently, in the ordered basis (H, D_1, D_2) of Λ, the automorphisms $\Phi^*_{L,\overline{L}}$, τ_b^*, and $\Psi_4^* = \Phi^*_{L,\overline{L}} \circ \tau_b^*$ act by the exact unimodular integer matrices

$$M_\Phi = \begin{pmatrix} 5 & 8 & 0 \\ -3 & -5 & 0 \\ -1 & -2 & 1 \end{pmatrix}, \qquad M_{\tau_b} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad M_{\Psi_4} = M_\Phi M_{\tau_b} = \begin{pmatrix} 5 & 0 & 8 \\ -3 & 0 & -5 \\ -1 & 1 & -2 \end{pmatrix},$$

satisfying $\det(M_\Phi) = \det(M_{\tau_b}) = -1$, $M_\Phi^2 = M_{\tau_b}^2 = I_3$, $M_\Phi^T G_\Lambda M_\Phi = G_\Lambda$, and $M_{\Psi_4}^T G_\Lambda M_{\Psi_4} = G_\Lambda$. The matrix M_{Ψ_4} has $\det(M_{\Psi_4}) = +1$, $\mathrm{Tr}(M_{\Psi_4}) = 3$, and characteristic polynomial $\det(\lambda I - M_{\Psi_4}) = (\lambda-1)^3$, with nilpotent Jordan blocks

$$N = M_{\Psi_4} - I_3 = \begin{pmatrix} 4 & 0 & 8 \\ -3 & -1 & -5 \\ -1 & 1 & -3 \end{pmatrix}, \qquad N^2 = \begin{pmatrix} 8 & 8 & 8 \\ -4 & -4 & -4 \\ -4 & -4 & -4 \end{pmatrix} \neq 0, \qquad N^3 = 0.$$

Therefore M_{Ψ_4} is a unipotent parabolic transformation of Jordan block size 3, and for every integer $k \in \mathbb{Z}$, the k-th power $M_{\Psi_4}^k = I_3 + kN + \frac{k(k-1)}{2}N^2$ has first column

$$(\Psi_4^*)^k(H) = (4k^2+1)H - (2k^2+k)D_1 - (2k^2-k)D_2.$$

For the initial curve $C^{(1)}$ (which has $C^{(1)} \cdot H = 4$, $C^{(1)} \cdot D_1 = 4$, and $C^{(1)} \cdot D_2 = 0$), the degree of the k-th iterate $\Psi_4^k(C^{(1)})$ is

$$\deg\big(\Psi_4^k(C^{(1)})\big) = C^{(1)} \cdot (\Psi_4^*)^k(H) = 4(4k^2+1) - 4(2k^2+k) - 0(2k^2-k) = 8k^2 - 4k + 4,$$

establishing (23). Since two projectively equivalent curves in $\mathbb{P}^3$ must have the same degree, the curves $\{\Psi_4^k(C^{(1)})\}_{k\in\mathbb{Z}}$ represent infinitely many projectively inequivalent primitive polynomial families on S_4. □

Remark 3.5 (Hyperbolic Action of $\Phi_{L,\overline{L}} \circ \sigma$ Across Coordinate Pairings). If we also include the $b \leftrightarrow c$ coordinate swap $\sigma(a,b,c,d) = (a,c,b,d)$, let $D_1' = \sigma^*(D_1) = [L_1'] + [\overline{L}_1']$ where $L_1' = \{d = ic, b = (1+i)a\}$. Direct solution of the four linear equations shows that $L_1 \cap L_1' = \{(1:1+i:1+i:-1+i)\} \subset S_4$

and $L_1 \cap \overline{L}_1' = \emptyset$, while $D_2 \cdot D_1' = 0$, so $D_1 \cdot D_1' = 2$. Moreover, L_1 and L_1' span the plane $\Pi_{L_1,L_1'} : (-1+i)a - ib - ic + d = 0$, whose intersection with S_4 splits into $L_1 + L_1' + \Phi_{L,\overline{L}}(L_1') \sim H$, giving $\Phi^*_{L,\overline{L}}(D_1') = 2H - D_1 - D_1'$. Applying $\Phi_{L,\overline{L}} \circ \sigma$ and $\Phi_{L,\overline{L}} \circ \tau_c$ to Family 4-I ($k = 1, \deg = 8$) divides out GCDs of degrees 8 and 4 from raw degree $5 \times 8 = 40$, producing verified primitive polynomial families of degrees 32 and 36.

4 Primitive Polynomial Families and Infinite Towers for $n = -1$

On Euler's surface $S_{-1} : -a^4 + b^4 + c^4 = d^4$ (equivalently $b^4 + c^4 = a^4 + d^4$), there are 8 rational lines (four in Pairing I: $\{d = \pm b, c = \pm a\}$ and four in Pairing II: $\{d = \pm c, b = \pm a\}$). Choosing two skew rational lines from the same pairing induces a trivial coordinate swap, whereas choosing one rational line from each of the two distinct pairings yields a non-trivial birational involution of degree 5:

Theorem 4.1 (The Crossed-Pairing Skew-Line Involution and Regular Extension on S_{-1})**.** *Let $L_1, L_2 \subset S_{-1}$ be the two disjoint rational lines from opposite coordinate pairings:*

$$L_1 : \quad d - b = 0, \quad c - a = 0, \qquad\qquad L_2 : \quad d + c = 0, \quad b - a = 0. \tag{24}$$

1. ***Eigenspace Decomposition and Generic Transversal Construction on*** $S_{-1} \setminus (L_1 \cup L_2)$***:*** *Define the linear involution $M_{-1} : \mathbb{P}^3 \to \mathbb{P}^3$ by*

$$M_{-1}(a, b, c, d) = (-b + c + d,\ -a + c + d,\ a - b + d,\ -a + b + c).$$

Then $M_{-1}^2 = +I_4$ identically on $\mathbb{Q}^4$, with $+1$-eigenspace $\ker(M_{-1} - I_4) = L_1$ and -1-eigenspace $\ker(M_{-1} + I_4) = L_2$. For any point $P = (a : b : c : d) \in S_{-1} \setminus (L_1 \cup L_2)$, the eigenspace projections $Q_1 = P + M_{-1}(P) = (u_1, v_1, u_1, v_1) \in L_1$ and $Q_2 = P - M_{-1}(P) = (w_2, w_2, -z_2, z_2) \in L_2$ with

$$u_1 = a - b + c + d, \qquad v_1 = -a + b + c + d, \qquad w_2 = a + b - c - d, \qquad z_2 = a - b - c + d$$

are both non-zero in $\mathbb{P}^3$ and span the unique transversal line $T_P \subset \mathbb{P}^3$ passing through P (at $t = 1$), meeting L_1 (at $t = 0$) and L_2 (at $t = \infty$), given by $P(t) = Q_1 + tQ_2 = (1 + t)P + (1 - t)M_{-1}(P)$.

2. ***Rational Involution and Regular Extension to*** $\mathrm{Aut}(S_{-1}/\mathbb{Q})$***:*** *Along $P(t)$, the surface polynomial decomposes identically in $\mathbb{Z}[a, b, c, d, t]$ as*

$$-a(t)^4 + b(t)^4 + c(t)^4 - d(t)^4 = 8t(t^2+1)(-a^4 + b^4 + c^4 - d^4) - 16t(t-1)(C_+(t+1) + C_-(t-1)), \tag{25}$$

so the fourth intersection point $t = (C_- - C_+)/(C_- + C_+)$ defines the degree-5 rational self-map $\Phi_{L_1,L_2} : S_{-1} \dashrightarrow S_{-1}$:

$$\Phi_{L_1,L_2}(a, b, c, d) = (-C_-)(a, b, c, d) - C_+(-b + c + d,\ -a + c + d,\ a - b + d,\ -a + b + c), \tag{26}$$

where the degree-4 invariants C_- and C_+ are given in factored form by

$$\begin{aligned} -C_- &= 3(a - b)(a - c)(b - d)(c + d), \\ C_+ &= a^3(b - c - d) - a(b^3 - c^3 - d^3) + b^3(c + d) - b(c^3 + d^3) + cd(c^2 - d^2). \end{aligned} \tag{27}$$

On the Zariski-open dense set $U = S_{-1} \setminus (L_1 \cup L_2 \cup M_1 \cup M_2 \cup M_3 \cup M_4)$, where $M_1 = \{d = b, c = -a\}$, $M_2 = \{d = -b, c = a\}$, $M_3 = \{d = c, b = a\}$, and $M_4 = \{d = -c, b = -a\}$ are the four rational bisecant lines meeting both L_1 and L_2, Φ_{L_1,L_2} is a regular morphism satisfying $\Phi^2_{L_1,L_2} = \mathrm{id}_U$. On S_{-1}, the divisorial base locus of (26) consists of $L_1 \cup L_2$ (to order 3) and $M_1 \cup M_2 \cup M_3 \cup M_4$ (to order 1). By Lemma 3.1, removing these codimension-1 base components extends Φ_{L_1,L_2} uniquely to a regular biregular involution $\Phi_{L_1,L_2} \in \mathrm{Aut}(S_{-1}/\mathbb{Q})$.

Proof. (1) On L_1, $d = b$ and $c = a$, so $-a^4 + b^4 + c^4 - d^4 = 0$; on L_2, $d = -c$ and $b = a$, so $-a^4 + b^4 + c^4 - d^4 = 0$. At $L_1 \cap L_2$, $d = b = a = c = -d \implies a = b = c = d = 0$, so $L_1 \cap L_2 = \emptyset$. Direct matrix multiplication of M_{-1} gives $M_{-1}^2 = I_4$, with $M_{-1}(P) = P \iff d = b, c = a$ $(P \in L_1)$ and $M_{-1}(P) = -P \iff d = -c, b = a$ $(P \in L_2)$. Hence for $P \in S_{-1} \setminus (L_1 \cup L_2)$, $Q_1 = P + M_{-1}(P) \in L_1$ and $Q_2 = P - M_{-1}(P) \in L_2$ are both non-zero, and $Q_1 + Q_2 = 2P \sim P$, so $P(t) = Q_1 + tQ_2 = (1+t)P + (1-t)M_{-1}(P)$ is the unique line passing through $Q_1 \in L_1$ at $t = 0$, through P at $t = 1$, and through $Q_2 \in L_2$ at $t = \infty$.

(2) Expanding $-a(t)^4 + b(t)^4 + c(t)^4 - d(t)^4$ along $P(t) = (1+t)P + (1-t)M_{-1}(P)$ yields (25) with C_+, C_- as in (27). Solving $C_+(t+1) + C_-(t-1) = 0$ gives $\frac{1-t}{1+t} = \frac{C_+}{C_-} = -\frac{C_+}{-C_-}$; substituting into $P(t) \propto (1+t)P + (1-t)M_{-1}(P)$ yields (26). For $P \notin L_1 \cup L_2$, P and $M_{-1}(P)$ are linearly independent, so $\Phi_{L_1,L_2}(P) = 0 \iff C_-(P) = C_+(P) = 0$. On S_{-1}, $C_- = 0$ requires $(a-b)(a-c)(b-d)(c+d) = 0$. Checking each of the four planes on S_{-1} shows that $C_- = C_+ = 0$ cuts out precisely $L_1 \cup L_2 \cup M_1 \cup M_2 \cup M_3 \cup M_4$. For $P \in U = S_{-1} \setminus (L_1 \cup L_2 \cup M_1 \cup M_2 \cup M_3 \cup M_4)$, $P' = \Phi_{L_1,L_2}(P) = (-C_-)P - C_+M_{-1}(P)$ satisfies $M_{-1}(P') = (-C_-)M_{-1}(P) - C_+P \in \mathrm{span}(P, M_{-1}(P)) = T_P$, so $T_{P'} = T_P$ and $\Phi_{L_1,L_2}(P') = P$. By Lemma 3.1, Φ_{L_1,L_2} extends uniquely across the divisorial base locus $3([L_1] + [L_2]) + \sum_{r=1}^{4}[M_r]$ to a regular biregular involution $\Phi_{L_1,L_2} \in \mathrm{Aut}(S_{-1}/\mathbb{Q})$. □

Corollary 4.2 (Explicit Degree-13, Degree-19, and Degree-31 Single-Generator Primitive Families for $n = -1$)**.** *Applying Φ_{L_1,L_2} to the eight sign/permutation branches of Euler's degree-7 curve $(a_{\mathrm{eul}}, b_{\mathrm{eul}}, c_{\mathrm{eul}}, d_{\mathrm{eul}})$ from Row (0) of Table 2 produces three primitive polynomial families for $-a^4 + b^4 + c^4 = d^4$ at degrees* $\deg = 13$, $\deg = 19$, *and* $\deg = 31$, *each generated by a single master polynomial via involutive (p,q)-symmetries and satisfying* $\gcd(a,b,c,d) = 1$ *in $\mathbb{Z}[p,q]$:*

1. ***The Degree-13 Single-Generator Primitive Family*** *(from $(a_{\mathrm{eul}}, -c_{\mathrm{eul}}, -b_{\mathrm{eul}}, d_{\mathrm{eul}})$): Let*

$$\begin{aligned} F_{13}(p,q) = {} & p^{13} + p^{12}q + 2p^{11}q^2 - 4p^{10}q^3 - 3p^9q^4 + 3p^8q^5 + 7p^7q^6 \\ & + 4p^6q^7 - 12p^5q^8 - 6p^4q^9 + 5p^3q^{10} - p^2q^{11} + pq^{12} + q^{13}. \end{aligned} \tag{28}$$

Then $-a_{13}^4 + b_{13}^4 + c_{13}^4 = d_{13}^4$ is satisfied identically in $\mathbb{Z}[p,q]$ by

$$a_{13}(p,q) = F_{13}(p,q),\ b_{13}(p,q) = -F_{13}(q,p),\ c_{13}(p,q) = F_{13}(p,-q),\ d_{13}(p,q) = -F_{13}(-q,p). \tag{29}$$

2. ***The Degree-19 Single-Generator Primitive Family*** *(from $(a_{\mathrm{eul}}, c_{\mathrm{eul}}, b_{\mathrm{eul}}, d_{\mathrm{eul}})$): Let*

$$\begin{aligned} F_{19}(p,q) = {} & p^{18}q + 6p^{17}q^2 - 6p^{16}q^3 + 12p^{15}q^4 + 6p^{14}q^5 + 36p^{13}q^6 + 19p^{12}q^7 \\ & - 12p^{11}q^8 - 24p^{10}q^9 + 36p^9q^{10} - 12p^8q^{11} + 12p^7q^{12} - 5p^6q^{13} \\ & + 6p^5q^{14} - 18p^4q^{15} + 6p^2q^{17} + q^{19}. \end{aligned} \tag{30}$$

Then $-a_{19}^4 + b_{19}^4 + c_{19}^4 = d_{19}^4$ is satisfied identically in $\mathbb{Z}[p,q]$ by

$$a_{19}(p,q) = F_{19}(p,q),\ b_{19}(p,q) = F_{19}(q,p),\ c_{19}(p,q) = -F_{19}(p,-q),\ d_{19}(p,q) = F_{19}(q,-p). \tag{31}$$

3. ***The Degree-31 Single-Generator Primitive Family*** *(from $(a_{\mathrm{eul}}, -b_{\mathrm{eul}}, c_{\mathrm{eul}}, d_{\mathrm{eul}})$): Let*

$$\begin{aligned} F_{31}(p,q) = {} & p^{31} + p^{30}q + 11p^{29}q^2 - p^{28}q^3 + 42p^{27}q^4 - 24p^{26}q^5 - 19p^{25}q^6 + 32p^{24}q^7 \\ & - 154p^{23}q^8 + 254p^{22}q^9 + 266p^{21}q^{10} - 718p^{20}q^{11} + 126p^{19}q^{12} + 303p^{18}q^{13} \\ & - 478p^{17}q^{14} + 830p^{16}q^{15} + 770p^{15}q^{16} - 916p^{14}q^{17} - 738p^{13}q^{18} - 21p^{12}q^{19} \\ & + 350p^{11}q^{20} + 434p^{10}q^{21} + 50p^9q^{22} - 142p^8q^{23} - 91p^7q^{24} - 76p^6q^{25} \\ & + 15p^5q^{26} + 3p^4q^{27} + 8p^3q^{28} + 8p^2q^{29} + pq^{30} + q^{31}. \end{aligned} \tag{32}$$

Then $-a_{31}^4 + b_{31}^4 + c_{31}^4 = d_{31}^4$ is satisfied identically in $\mathbb{Z}[p,q]$ by

$$a_{31}(p,q) = -F_{31}(-q,p),\ b_{31}(p,q) = -F_{31}(q,p),\ c_{31}(p,q) = F_{31}(p,-q),\ d_{31}(p,q) = F_{31}(p,q). \tag{33}$$

Proof. On S_{-1}, besides L_1 and L_2 (where Φ_{L_1,L_2} vanishes to order 3), there are four rational bisecant lines meeting both L_1 and L_2:

$$M_1 = \{d = b, c = -a\}, \quad M_2 = \{d = -b, c = a\}, \quad M_3 = \{d = c, b = a\}, \quad M_4 = \{d = -c, b = -a\},$$

along each of which both C_- and C_+ vanish to order 1. Substituting Euler's degree-7 polynomials into the degree-5 homogeneous map Φ_{L_1,L_2} in (26) produces raw homogeneous polynomials of degree $5 \times 7 = 35$:

1. On the branch $C^{(1)}_{\text{eul}} = (a_{\text{eul}}, -c_{\text{eul}}, -b_{\text{eul}}, d_{\text{eul}})$, the curve meets M_1, M_2, M_3, M_4 at $p = 0, q = 0, p + q = 0, p - q = 0$ (contributing $pq(p^2 - q^2)$ of degree 4) and meets L_2 along $(p^2 + q^2)(p^2 - pq - q^2)(p^2 + pq - q^2) = 0$ (degree 6, contributing cube power of degree 18). Thus the four raw degree-35 outputs share the exact polynomial greatest common divisor $G_{13}(p, q) = 8pq(p^2 - q^2)(p^2 + q^2)^3(p^2 - pq - q^2)^3(p^2 + pq - q^2)^3$ of degree 22. Dividing out $G_{13}(p, q)$ leaves degree $35 - 22 = 13$ polynomials given by (28)–(29).

2. On the branch $C^{(2)}_{\text{eul}} = (a_{\text{eul}}, c_{\text{eul}}, b_{\text{eul}}, d_{\text{eul}})$, the curve meets $M_1, \ldots, M_4$ along $pq(p^2 - q^2) = 0$ (degree 4) and meets L_1 along $p^2q^2 = 0$ (contributing $(p^2q^2)^3 = p^6q^6$ of degree 12). Thus the four raw degree-35 outputs share the degree-16 polynomial GCD $G_{19}(p, q) = 216p^7q^7(p^2 - q^2)$, leaving degree $35 - 16 = 19$ polynomials given by (30)–(31).

3. On the branch $C^{(3)}_{\text{eul}} = (a_{\text{eul}}, -b_{\text{eul}}, c_{\text{eul}}, d_{\text{eul}})$, the curve is disjoint from $L_1 \cup L_2$ and meets $M_1, \ldots, M_4$ along $pq(p^2 - q^2) = 0$, so the four raw degree-35 outputs share the degree-4 polynomial GCD $G_{31}(p, q) = 8pq(p^2 - q^2)$, leaving degree $35 - 4 = 31$ polynomials given by (32)–(33).

In all three families, direct symbolic Euclidean GCD computation in $\mathbb{Q}[p, q]$ (dehomogenizing at $q = 1$ gives $\gcd_{\mathbb{Q}[p]}(a_m, b_m, c_m, d_m) = 1$, and evaluating at $(p, q) = (1, 0)$ gives $(\pm1, \pm1, \pm1, \pm1)$ or $(0, 1, 0, 1)$ with integer content 1) proves $\gcd_{\mathbb{Z}[p,q]}(a_m, b_m, c_m, d_m) = 1$, and direct expansion confirms $-a_m^4 + b_m^4 + c_m^4 - d_m^4 = 0$ identically in $\mathbb{Z}[p, q]$. □

Theorem 4.3 (Infinite Quadratic Degree Towers and Positive-Arithmetic-Genus Families for $n = -1$)**.** *Let $\sigma(a, b, c, d) = (a, c, b, d)$, and let $\Psi_{-1} = \sigma \circ \Phi_{L_1,L_2} \in \mathrm{Aut}(S_{-1}/\mathbb{Q})$. Then Ψ_{-1} has infinite order in $\mathrm{Aut}(S_{-1}/\mathbb{Q})$, and iterating Ψ_{-1} on the three branches $C^{(1)}_{\text{eul}}, C^{(2)}_{\text{eul}}, C^{(3)}_{\text{eul}}$ of Euler's degree-7 curve produces three infinite towers of smooth rational curves ($C^2 = -2$, $p_a(C) = 0$) giving projectively inequivalent primitive polynomial families of strictly increasing quadratic degrees ($k \geq 0$):*

$$\begin{aligned}
&\deg(\Psi_{-1}^k(C^{(1)}_{\text{eul}})) = 9k^2 - 3k + 7 \quad &&(\deg = 7, 13, 37, 79, 139, 217, \ldots),\\
&\deg(\Psi_{-1}^k(C^{(2)}_{\text{eul}})) = 12k^2 + 7 \quad &&(\deg = 7, 19, 55, 115, 199, 307, \ldots),\\
&\deg(\Psi_{-1}^k(C^{(3)}_{\text{eul}})) = 18k^2 + 6k + 7 \quad &&(\deg = 7, 31, 91, 187, 319, 487, \ldots).
\end{aligned} \tag{34}$$

In addition, applying the degree-17 asymptotic inflection-tangent self-maps $\mathcal{A}_+$ and $\mathcal{A}_-$ of Theorem 6.1 to C_{eul} on S_{-1} ($s = 1$) divides out base-locus polynomial GCDs of degrees 76 and 74 from raw degree $17 \times 7 = 119$, producing two further primitive polynomial families of degrees 43 and 45 whose image curves $C_{43} = \mathcal{A}_+(C_{\text{eul}})$ and $C_{45} = \mathcal{A}_-(C_{\text{eul}})$ are singular rational curves ($p_g = 0$) of positive self-intersection $C_{43}^2 = 130$ ($p_a(C_{43}) = 66$) and $C_{45}^2 = 114$ ($p_a(C_{45}) = 58$).

Proof. Partition the 8 rational lines on S_{-1} into four crossed skew-line pairs:

$$\begin{aligned}
D_1 &= [L_1] + [L_2] = [\{d = b, c = a\}] + [\{d = -c, b = a\}],\\
D_2 &= \sigma^*(D_1) = [M_3] + [M_2] = [\{d = c, b = a\}] + [\{d = -b, c = a\}],\\
D_3 &= [M_1] + [M_4] = [\{d = b, c = -a\}] + [\{d = -c, b = -a\}],\\
D_4 &= \sigma^*(D_3) = [N_2] + [N_1] = [\{d = c, b = -a\}] + [\{d = -b, c = -a\}].
\end{aligned}$$

Each pair consists of two disjoint lines ($D_r^2 = -4$, $H \cdot D_r = 2$). For D_1 and $D_2 = \sigma^*(D_1)$, all four cross-pair intersections in $\mathbb{P}^3$ are non-empty:

$$L_1 \cap M_3 = \{(1:1:1:1)\}, \qquad L_1 \cap M_2 = \{(1:0:1:0)\},$$
$$L_2 \cap M_3 = \{(1:1:0:0)\}, \qquad L_2 \cap M_2 = \{(1:1:1:-1)\},$$

so $D_1 \cdot D_2 = 4$. By the same direct check across all pairs, $D_1 \cdot D_3 = D_2 \cdot D_4 = D_3 \cdot D_4 = 4$ and $D_1 \cdot D_4 = D_2 \cdot D_3 = 0$, giving the intersection matrix on (H, D_1, D_2, D_3, D_4):

$$G_5 = \begin{pmatrix} 4 & 2 & 2 & 2 & 2 \\ 2 & -4 & 4 & 4 & 0 \\ 2 & 4 & -4 & 0 & 4 \\ 2 & 4 & 0 & -4 & 4 \\ 2 & 0 & 4 & 4 & -4 \end{pmatrix}, \qquad \det(G_5|_{H,D_1,D_2,D_3}) = -192 \neq 0.$$

Under Φ_{L_1,L_2}, the degree-5 map vanishes to order 3 along D_1 and to order 1 along the four bisecant lines M_1, M_2, M_3, M_4 ($D_2 + D_3$), so Lemma 3.1(2) gives $\Phi^*_{L_1,L_2}(H) = 5H - 3D_1 - D_2 - D_3$. Because each bisecant line M_r meets both L_1 and L_2, for every point $P \in M_r \setminus (L_1 \cup L_2)$ the unique transversal line T_P through P meeting L_1 and L_2 is M_r itself, so $\Phi_{L_1,L_2}(M_r) = M_r$ for $r = 1, 2, 3, 4$, proving $\Phi^*_{L_1,L_2}(D_2) = D_2$ and $\Phi^*_{L_1,L_2}(D_3) = D_3$. On $N_1 = \{d = -b, c = -a\}$ (which is disjoint from $L_1 \cup L_2$), evaluating (26) on $(a, b, -a, -b)$ gives $-C_- = 12ab(a+b)^2$ and $C_+ = -4ab(a+b)^2 - 4(a^4 + b^4)$, which maps N_1 isomorphically onto $N_2 = \{d = c, b = -a\}$ and vice versa, so $\Phi^*_{L_1,L_2}(D_4) = D_4$. Since $(\Phi^*_{L_1,L_2})^2 = \mathrm{id}$, solving $5\Phi^*_{L_1,L_2}(H) - 3\Phi^*_{L_1,L_2}(D_1) - D_2 - D_3 = H$ gives $\Phi^*_{L_1,L_2}(D_1) = 8H - 5D_1 - 2D_2 - 2D_3$ (the exact reflection $\Phi^*_{L_1,L_2}(X) = X + \frac{X \cdot R}{6} R$ across the root class $R = 4H - 3D_1 - D_2 - D_3$ of self-intersection $R^2 = -12$). In the basis (H, D_1, D_2, D_3, D_4), $\Phi^*_{L_1,L_2}$, σ^*, and $\Psi^*_{-1} = \Phi^*_{L_1,L_2} \circ \sigma^*$ act by the exact unimodular integer matrices

$$M_\Phi = \begin{pmatrix} 5 & 8 & 0 & 0 & 0 \\ -3 & -5 & 0 & 0 & 0 \\ -1 & -2 & 1 & 0 & 0 \\ -1 & -2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad M_\sigma = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \quad M_{\Psi_{-1}} = M_\Phi M_\sigma = \begin{pmatrix} 5 & 0 & 8 & 0 & 0 \\ -3 & 0 & -5 & 0 & 0 \\ -1 & 1 & -2 & 0 & 0 \\ -1 & 0 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix},$$

satisfying $M_\Phi^2 = M_\sigma^2 = I_5$, $M_\Phi^T G_5 M_\Phi = G_5$, $M_\sigma^T G_5 M_\sigma = G_5$, and $\det(\lambda I - M_{\Psi_{-1}}) = (\lambda - 1)^4(\lambda + 1)$ with $(M^2_{\Psi_{-1}} - I_5)^2 \neq 0$ and $(M^2_{\Psi_{-1}} - I_5)^3 = 0$. Notice that $M_{\Psi_{-1}}$ is block lower-triangular whose top-left 3×3 block on (H, D_1, D_2) is identical to M_{Ψ_4} in Theorem 3.4. By induction on $k \geq 0$, the first column of $M^k_{\Psi_{-1}}$ gives the exact closed-form pullback

$$(\Psi^*_{-1})^k(H) = (4k^2+1)H - k(2k+1)D_1 - k(2k-1)D_2 - \frac{k(k+1)}{2}D_3 - \frac{k(k-1)}{2}D_4. \tag{35}$$

From the intersection counts in the proof of Corollary 4.2, the three branches $C^{(1)}_{\mathrm{eul}}, C^{(2)}_{\mathrm{eul}}, C^{(3)}_{\mathrm{eul}}$ have intersection vectors $v^{(r)} = (C \cdot H, C \cdot D_1, C \cdot D_2, C \cdot D_3, C \cdot D_4)$ equal to

$$v^{(1)} = (7, 6, 2, 2, 4), \qquad v^{(2)} = (7, 4, 2, 2, 6), \qquad v^{(3)} = (7, 0, 2, 2, 10).$$

Taking the dot product of each $v^{(r)}$ with the coefficient vector of $(\Psi^*_{-1})^k(H)$ in (35) gives:

$$\deg\big(\Psi^k_{-1}(C^{(1)}_{\mathrm{eul}})\big) = 7(4k^2+1) - 6k(2k+1) - 2k(2k-1) - 2\frac{k(k+1)}{2} - 4\frac{k(k-1)}{2} = 9k^2 - 3k + 7,$$
$$\deg\big(\Psi^k_{-1}(C^{(2)}_{\mathrm{eul}})\big) = 7(4k^2+1) - 4k(2k+1) - 2k(2k-1) - 2\frac{k(k+1)}{2} - 6\frac{k(k-1)}{2} = 12k^2 + 7,$$
$$\deg\big(\Psi^k_{-1}(C^{(3)}_{\mathrm{eul}})\big) = 7(4k^2+1) - 0 - 2k(2k-1) - 2\frac{k(k+1)}{2} - 10\frac{k(k-1)}{2} = 18k^2 + 6k + 7,$$

establishing (34). Explicitly, at $k = 2$, direct symbolic composition of $\Psi_{-1} = \sigma \circ \Phi_{L_1,L_2}$ on all three $k = 1$ curves (of degrees 13, 19, and 31) confirms (34):

- On Branch 1 ($k=2$), applying Ψ_{-1} to the degree-13 curve (28) divides out the exact degree-28 polynomial GCD $G_{37}(p,q) = 24p^2q^5(p+q)^3(p^2+q^2)(p^2-pq-q^2)(p^2+pq-q^2)(2p^{12}+p^{10}q^2-9p^8q^4+11p^6q^6-9p^4q^8+p^2q^{10}+2q^{12})$ from raw degree $5\times 13=65$ to yield degree $65-28=37=9(2)^2-3(2)+7$.
- On Branch 2 ($k=2$), applying Ψ_{-1} to the degree-19 curve (30) divides out the exact degree-40 polynomial GCD $G_{55}(p,q) = 48p^5q^2(p-q)^3(p^2+q^2)(p^2-pq-q^2)(p^2+pq-q^2)(p^4-p^2q^2+q^4)(p^4-p^3q+2p^2q^2+pq^3+q^4)(p^4+p^3q+2p^2q^2-pq^3+q^4)(p^{12}-4p^{10}q^2-12p^8q^4-2p^6q^6-12p^4q^8-4p^2q^{10}+q^{12})$ from raw degree $5\times 19=95$ to yield degree $95-40=55=12(2)^2+7$.
- On Branch 3 ($k=2$), applying Ψ_{-1} to the degree-31 curve (32) divides out the exact degree-64 polynomial GCD $G_{91}(p,q) = 24p^2q^5(p-q)^3(p^2+q^2)(p^4+q^4)(p^2-pq-q^2)(p^2+pq-q^2)(p^4+3p^2q^2+q^4)(p^{16}+7p^{14}q^2+11p^{12}q^4-84p^{10}q^6+146p^8q^8-84p^6q^{10}+11p^4q^{12}+7p^2q^{14}+q^{16})(2p^{24}+11p^{22}q^2+5p^{20}q^4-95p^{18}q^6+83p^{16}q^8+173p^{14}q^{10}-326p^{12}q^{12}+173p^{10}q^{14}+83p^8q^{16}-95p^6q^{18}+5p^4q^{20}+11p^2q^{22}+2q^{24})$ from raw degree $5\times 31=155$ to yield degree $155-64=91=18(2)^2+6(2)+7$.

Finally, for $\mathcal{A}_+(C_{\text{eul}})$ and $\mathcal{A}_-(C_{\text{eul}})$, symbolic evaluation of (48) on C_{eul} divides out exact polynomial GCDs of degrees 76 and 74 (factored explicitly in Appendix A), leaving primitive families of degrees $119-76=43$ and $119-74=45$ whose divisor classes in $\mathrm{Pic}(S_{-1}/\mathbb{Q})$ have $C_{43}^2=130$ ($p_a=66$) and $C_{45}^2=114$ ($p_a=58$) as proved in Section 11. □

5 Classification of $n\in\{-1,4,-12\}$ and the Picard Rigidity Theorem for $n=-12$

We now establish the two classification theorems that explain why $n\in\{-1,4,-12\}$ are the unique constant-n families arising from Galois-invariant skew-line pairs and rational-discriminant parabolic quadric cones, followed by the Picard Rigidity Theorem explaining why $n=-12$ behaves fundamentally differently from $n=-1$ and $n=4$.

5.1 The Galois-Field Skew-Line Classification Theorem ($n=-1$ and $n=4$)

Theorem 5.1 (Classification of Galois-Invariant Skew-Line Pairs on S_n)**.** *A diagonal quartic surface $S_n: na^4+b^4+c^4=d^4$ ($n\in\mathbb{Q}^\times$) possesses a pair of disjoint lines $\{L_1,L_2\}\subset S_n(\overline{\mathbb{Q}})$ whose union $L_1\cup L_2$ is defined over $\mathbb{Q}$ if and only if $n\equiv -1 \pmod{\mathbb{Q}^{\times 4}}$ or $n\equiv 4 \pmod{\mathbb{Q}^{\times 4}}$.*

Proof. Over the algebraic closure $\overline{\mathbb{Q}}$, S_n contains (for any $n\in\mathbb{Q}^\times$) exactly 48 lines [13, 12], partitioned into three coordinate pairings of 16 lines each:

$$\begin{aligned}
\text{Pairing I:}\quad & L_{\mathrm{I}}(j,k): \quad d=i^jb,\quad c=\eta_0 i^k a \qquad (\eta_0^4=-n,\ 0\le j,k\le 3),\\
\text{Pairing II:}\quad & L_{\mathrm{II}}(j,k): \quad d=i^jc,\quad b=\eta_0 i^k a \qquad (\eta_0^4=-n,\ 0\le j,k\le 3),\\
\text{Pairing III:}\quad & L_{\mathrm{III}}(j,k): \quad d=\theta_0 i^j a,\quad c=\zeta_8 i^k b \qquad (\theta_0^4=n,\ \zeta_8=e^{2\pi i/8},\ 0\le j,k\le 3).
\end{aligned}$$

In each pairing, the two variables on the right-hand sides ((a,b) in Pairings I and III, (a,c) in Pairing II) serve as free homogeneous coordinates on the line: specifically, every line $L_{\mathrm{I}}(j,k)$ contains the points $(1:0:\eta_0 i^k:0)$ and $(0:1:0:i^j)$, so a field automorphism $\sigma\in\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ fixes $L_{\mathrm{I}}(j,k)$ (or $L_{\mathrm{II}}(j,k)$) as a subvariety of $\mathbb{P}^3$ if and only if $\sigma(i^j)=i^j$ and $\sigma(\eta_0 i^k)=\eta_0 i^k$. Similarly, $L_{\mathrm{III}}(j,k)$ contains $(1:0:0:\theta_0 i^j)$ and $(0:1:\zeta_8 i^k:0)$, so $\sigma(L_{\mathrm{III}}(j,k))=L_{\mathrm{III}}(j,k)$ if and only if $\sigma(\theta_0 i^j)=\theta_0 i^j$ and $\sigma(\zeta_8 i^k)=\zeta_8 i^k$. Thus the minimal field of definition $\mathbb{Q}(L)$ of each line L over $\mathbb{Q}$ is generated by its two defining ratios of coefficients:

$$\mathbb{Q}(L_{\mathrm{I}}(j,k))=\mathbb{Q}(L_{\mathrm{II}}(j,k))=\mathbb{Q}(i^j,\eta_0 i^k),\qquad \mathbb{Q}(L_{\mathrm{III}}(j,k))=\mathbb{Q}(\theta_0 i^j,\zeta_8 i^k).$$

For a pair of disjoint lines $\{L_1,L_2\}\subset S_n(\overline{\mathbb{Q}})$ to have $L_1\cup L_2$ defined over $\mathbb{Q}$, the absolute Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ must stabilize the two-element set $\{L_1,L_2\}$. Hence the stabilizer of each line L_r ($r=1,2$)

has index at most 2 in $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, so its field of definition $K_r = \mathbb{Q}(L_r)$ satisfies $[K_r : \mathbb{Q}] \le 2$. Since $\zeta_8 i^k \in \{\zeta_8, \zeta_8^3, \zeta_8^5, \zeta_8^7\}$ is a primitive 8-th root of unity generating the cyclotomic field $\mathbb{Q}(\zeta_8) = \mathbb{Q}(i, \sqrt{2})$ of degree $[\mathbb{Q}(\zeta_8) : \mathbb{Q}] = 4 > 2$, no line of Pairing III can have $[K_r : \mathbb{Q}] \le 2$. Thus L_1 and L_2 must belong to Pairing I or Pairing II, and two exhaustive Galois cases remain:

1. **Case 1 ($[K_1 : \mathbb{Q}] = [K_2 : \mathbb{Q}] = 1$, so $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ fixes L_1 and L_2 individually):** Here L_1 and L_2 are each defined over $\mathbb{Q}$ (whether they lie in the same pairing or in opposite pairings). For L_1 (in Pairing I or II) to be defined over $\mathbb{Q}$, we must have $i^j \in \mathbb{Q}$ (so $i^j = \pm 1$) and $\eta_0 i^k = \alpha \in \mathbb{Q}^\times$, which forces $-n = (\eta_0 i^k)^4 = \alpha^4$, i.e., $n = -\alpha^4 \equiv -1 \pmod{\mathbb{Q}^{\times 4}}$. Conversely, when $n = -\alpha^4$, $L_1 = \{d = b, c = \alpha a\}$ (in Pairing I) and $L_2 = \{d = -c, b = \alpha a\}$ (in Pairing II, or $\{d = -b, c = -\alpha a\}$ in Pairing I) are disjoint rational lines on S_n.

2. **Case 2 ($[K_1 : \mathbb{Q}] = 2$ and $L_2 = \tau(L_1)$ for the non-trivial automorphism $\tau \in \mathrm{Gal}(K_1/\mathbb{Q})$):** Because the defining equations $\{d^4 = b^4, c^4 = -na^4\}$ of the 16-line union of Pairing I and $\{d^4 = c^4, b^4 = -na^4\}$ of the 16-line union of Pairing II have coefficients in $\mathbb{Q}$, $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ preserves Pairing I and Pairing II separately. Thus L_1 and $L_2 = \tau(L_1)$ must belong to the same pairing; up to $b \leftrightarrow c$ symmetry, assume both lie in Pairing I, so $L_1 = \{d = \mu b, c = \beta a\}$ and $L_2 = \{d = \tau(\mu) b, c = \tau(\beta) a\}$ with $\mu = i^j \in K_1^\times$ and $\beta = \eta_0 i^k \in K_1^\times$ ($\beta^4 = -n$). Two lines $\{d = \mu_1 b, c = \beta_1 a\}$ and $\{d = \mu_2 b, c = \beta_2 a\}$ in Pairing I are disjoint in $\mathbb{P}^3$ if and only if $\mu_1 \ne \mu_2$ (otherwise they intersect at $(0 : 1 : 0 : \mu_1)$) and $\beta_1 \ne \beta_2$ (otherwise they intersect at $(1 : 0 : \beta_1 : 0)$). Hence $L_1 \cap L_2 = \emptyset$ requires both $\tau(\mu) \ne \mu$ and $\tau(\beta) \ne \beta$. Since $\mu \in \{\pm 1, \pm i\}$ and $\tau(\mu) \ne \mu$, we must have $\mu = \pm i \notin \mathbb{Q}$, which uniquely forces the quadratic field to be $K_1 = \mathbb{Q}(i)$ and τ to be complex conjugation. Writing $\beta = u + iv \in \mathbb{Q}(i)^\times$ ($v \ne 0$ since $\tau(\beta) \ne \beta$), we have

$$-n = \beta^4 = (u + iv)^4 = (u^4 - 6u^2v^2 + v^4) + 4uv(u^2 - v^2)i \in \mathbb{Q}^\times \iff uv(u^2 - v^2) = 0.$$

Since $v \ne 0$, either $u = 0$ (giving $\beta = iv$ and $-n = v^4 \implies n = -v^4 \equiv -1 \pmod{\mathbb{Q}^{\times 4}}$) or $u = \pm v = s \in \mathbb{Q}^\times$ (giving $\beta = s(1 \pm i)$ and $-n = s^4(1 \pm i)^4 = -4s^4 \implies n = 4s^4 \equiv 4 \pmod{\mathbb{Q}^{\times 4}}$). Conversely, when $n = 4s^4$, $L_1 = \{d = ib, c = s(1 + i)a\}$ and $\overline{L}_1 = \{d = -ib, c = s(1 - i)a\}$ are disjoint and $L_1 \cup \overline{L}_1$ is defined over $\mathbb{Q}$. □

5.2 The Parabolic-Cone Cardano Classification Theorem ($n = -12$)

Theorem 5.2 (Parabolic Quadric Cone Cardano Reduction & Uniqueness of $n \equiv -12$ Within the Cone Mechanism)**.** *Let $S_n : na^4 + b^4 + c^4 = d^4$ ($n \in \mathbb{Q}^\times$) be intersected with the 1-parameter family of parabolic quadric cones $\mathcal{Q}_k : (d - b)^2 = k^2 ac$ ($k \in \mathbb{Q}^\times$), each having its vertex at the Eckardt point $Q_1 = (0 : 1 : 0 : 1) \in S_n$.*

1. ***Cardano Cubic Reduction:*** *On the open complement of the two boundary ruling planes $\{a = 0\} \cup \{c = 0\}$ (which cut the cone $\mathcal{Q}_k$ along its two boundary ruling lines $\{a = 0, d = b\}$ and $\{c = 0, d = b\}$ through Q_1, and whose simultaneous intersection $\mathcal{Q}_k \cap S_n \cap \{ac = 0\}$ reduces strictly to the single vertex point $Q_1 = (0 : 1 : 0 : 1)$), pulling back the rulings of $\mathcal{Q}_k$ via $(a : c : d - b) = (u^2 : v^2 : kuv)$ with $uv \ne 0$ and setting $M = d - b = kuv$ and $P = d + b = Wuv$ reduces $d^4 - b^4 = \frac{1}{2} MP(P^2 + M^2) = c^4 + na^4$ to the depressed cubic in W:*

$$W(W^2 + k^2) = \frac{2}{k}\left(\frac{v^4}{u^4} + n\frac{u^4}{v^4}\right). \tag{36}$$

Under Cardano's substitution $W = T - \frac{k^2}{3T}$ ($T \ne 0$), equation (36) transforms into the quadratic in T^3:

$$(T^3)^2 - \frac{2}{k}\left(\frac{v^4}{u^4} + n\frac{u^4}{v^4}\right) T^3 - \frac{k^6}{27} = 0, \tag{37}$$

whose discriminant with respect to T^3 is

$$\Delta_{T^3}(u, v) = \frac{4}{k^2}\left[\left(\frac{v^4}{u^4} - n\frac{u^4}{v^4}\right)^2 + \left(4n + \frac{k^8}{27}\right)\right]. \tag{38}$$

2. ***Uniqueness of*** $n \equiv -12 \pmod{\mathbb{Q}^{\times 4}}$ ***Within the Parabolic Cone Ansatz:*** *The Cardano discriminant* $\Delta_{T^3}(u,v)$ *in* (38) *is the square of a rational function in* $\mathbb{Q}(u,v)$ *(so that the degree-6 spectral cover* (37) *splits over* $\mathbb{Q}(u,v)$ *into two binomial cubic equations* $T^3 = R_\pm(u,v)$*) if and only if the constant obstruction* $4n + \frac{k^8}{27}$ *vanishes identically:*

$$4n + \frac{k^8}{27} = 0 \iff n = -\frac{k^8}{108} = -12\left(\frac{k^2}{6}\right)^4 \equiv -12 \pmod{\mathbb{Q}^{\times 4}}. \tag{39}$$

At $k = 6$*, we have* $n = -12 \cdot 6^4 = -15552$*, so in terms of the fourth-power rescaled coordinate* $a_0 = 6a = 6u^2$ *on* $S_{-12} : -12a_0^4 + b^4 + c^4 = d^4$ *(where the quadric cone is* $(d-b)^2 = 36u^2v^2 = 6a_0c$*), equation* (37) *factors over* $\mathbb{Q}(u,v)$ *as* $(T^3 - \frac{1}{3}\frac{v^4}{u^4})(T^3 + 4(\frac{6u}{v})^4) = 0$*. Setting* $u = s^3$ *and* $v = 3r^3$ *makes* $T^3 = 27(r/s)^{12} = (3r^4/s^4)^3$*, rationalizing* $T = 3r^4/s^4 \in \mathbb{Q}(r,s)$ *over* $\mathbb{Q}$ *and recovering (after multiplying* (a_0, b, c, d) *by* $\frac{2rs}{3}$*) Elkies' degree-8 curve* $C_0(r,s) = (4rs^7,\, 3r^8 - 6r^4s^4 - 4s^8,\, 6r^7s,\, 3r^8 + 6r^4s^4 - 4s^8)$*.*

3. ***Universal Cardano–Eckardt Birational Form:*** *Without restricting to* $\mathcal{Q}_k$*, projecting* S_n *from* $Q_1 = (0:1:0:1)$ *via* $\frac{d+b}{d-b} = Z - \frac{1}{3Z}$ *gives the universal polynomial identity valid for all* (Z, X, Y, n)*:*

$$(3Z^2+3Z-1)^4 - (3Z^2-3Z-1)^4 - (6ZX)^4 - n(6ZY)^4 = 24Z(27Z^6 - 54(X^4+nY^4)Z^3 - 1), \tag{40}$$

reducing S_n *birationally (for* $Z \neq 0$*) to* $27Z^3 - Z^{-3} = 54(X^4 + nY^4)$*, or equivalently* $(3Z)^3 - 27(3Z)^{-3} = 54(X^4 + nY^4)$*.*

4. ***Odd-Parity Cancellation Law on*** $C_0(r,s) \subset S_{-12}$***:*** *For two points* $P_1 = C_0(r,s)$ *and* $P_2 = C_0(u,v)$ *on the degree-8 curve* $C_0 \subset S_{-12}$*, the tangency polynomial* $\Phi(r,s;u,v) = \nabla F(C_0(r,s)) \cdot C_0(u,v) = -12(rv - su)^2\Psi_6(r,s;u,v)$ *has all even-degree blocks in* (r^4s^4, u^4v^4) *vanish identically, so setting* $u = tr$ *(*$s = v = 1$*) collapses* $\Psi_6(r,1;tr,1)/r^2$ *into a cubic in* $R = r^8$*:*

$$27t^4(3t^2+2t+1)R^3 - 108(t+1)^2(t^2+1)^2R^2 + 144(t+1)^2(t^2+1)^2R - 64(t^2+2t+3) = 0. \tag{41}$$

Proof. (1) Writing $M = d - b$ and $P = d + b$, we have $d = \frac{1}{2}(P + M)$ and $b = \frac{1}{2}(P - M)$, so $d^4 - b^4 = \frac{1}{2}MP(P^2 + M^2)$. On the quadric cone $\mathcal{Q}_k : M^2 = k^2ac$, the boundary locus $ac = 0$ forces $M = d - b = 0$, cutting $\mathcal{Q}_k$ along its two ruling lines $\{a = 0, d = b\}$ and $\{c = 0, d = b\}$ through $Q_1 = (0:1:0:1)$, and reducing the simultaneous intersection $S_n \cap \mathcal{Q}_k \cap \{ac = 0\}$ to $\{a = 0, d = b, c^4 = 0\} = \{Q_1\}$ and $\{c = 0, d = b, na^4 = 0\} = \{Q_1\}$. Away from $ac = 0$, any point on $\mathcal{Q}_k$ has $M/(ka) = kc/M = v/u \in \mathbb{P}^1 \setminus \{0, \infty\}$, and after the quadratic base change $(a : c) = (u^2 : v^2)$ on the ruling parameter $(u : v)$, we have $M = kuv$ and $P = Wuv$. Substituting these into $d^4 - b^4 = c^4 + na^4$ gives $\frac{1}{2}kWu^4v^4(W^2 + k^2) = v^8 + nu^8$; dividing by $\frac{1}{2}ku^4v^4$ yields (36). Next, since $k \neq 0$, the mapping $T \mapsto W = T - \frac{k^2}{3T}$ is a finite 2-to-1 cover $\mathbb{P}^1 \setminus \{0, \infty\} \to \mathbb{A}^1$, and substituting $W = T - \frac{k^2}{3T}$ into $W(W^2 + k^2) = W^3 + k^2W$ cancels the T and T^{-1} terms identically:

$$\left(T - \frac{k^2}{3T}\right)^3 + k^2\left(T - \frac{k^2}{3T}\right) = T^3 - k^2T + \frac{k^4}{3T} - \frac{k^6}{27T^3} + k^2T - \frac{k^4}{3T} = T^3 - \frac{k^6}{27T^3}.$$

Multiplying by T^3 gives (37), whose quadratic discriminant is (38).

(2) Clearing denominators by multiplying $\Delta_{T^3}(u,v)$ in (38) by $\frac{k^2u^8v^8}{4}$ yields the homogeneous degree-16 polynomial in $\mathbb{Q}[u,v]$:

$$P_{16}(u,v) = (v^8 - nu^8)^2 + Cu^8v^8 = v^{16} + (C - 2n)u^8v^8 + n^2u^{16}, \qquad C = 4n + \frac{k^8}{27}.$$

By Gauss's Lemma, $\Delta_{T^3}(u,v)$ is a square in $\mathbb{Q}(u,v)$ if and only if $P_{16}(u,v) = G(u,v)^2$ for some homogeneous degree-8 polynomial $G(u,v) \in \mathbb{Q}[u,v]$. Over $\overline{\mathbb{Q}}$, the roots of $P_{16}(1,v) = (v^8)^2 + (C - 2n)v^8 + n^2 = 0$ are $v^8 = \alpha_1, \alpha_2$, where α_1, α_2 are the roots of $\lambda^2 + (C - 2n)\lambda + n^2 = 0$. Since $n \neq 0$,

$\alpha_1\alpha_2 = n^2 \neq 0$, so each root $v^8 = \alpha_i$ has 8 distinct simple roots in $\overline{\mathbb{Q}}$. Therefore $P_{16}(1, v)$ has only even-multiplicity roots in $\overline{\mathbb{Q}}$ (a necessary condition for $P_{16} = G^2$) if and only if $\alpha_1 = \alpha_2$, which occurs if and only if the quadratic discriminant $(C-2n)^2 - 4n^2 = C(C-4n)$ vanishes. Since $C - 4n = \frac{k^8}{27} \neq 0$ for $k \in \mathbb{Q}^\times$, we must have $C = 4n + \frac{k^8}{27} = 0$, proving (39). Conversely, when $C = 0$, $P_{16}(u, v) = (v^8 - nu^8)^2$ is visibly a square in $\mathbb{Q}[u, v]$. At $k = 6$, we have $n = -12 \cdot 6^4$ and $\sqrt{\Delta_{T^3}} = \frac{1}{3}(v^4/u^4 + 12 \cdot 6^4 u^4/v^4)$, so the roots of (37) are $T^3 = \frac{1}{3}v^4/u^4$ and $T^3 = -4(6u/v)^4$. Setting $u = s^3$ and $v = 3r^3$ gives $T^3 = 27(r/s)^{12} \implies T = 3r^4/s^4$, $W = 3r^4/s^4 - 4s^4/r^4 = \frac{3r^8 - 4s^8}{r^4 s^4}$, $a_0 = 6u^2 = 6s^6$, $c = v^2 = 9r^6$, $M = 18r^3 s^3$, and $P = \frac{3(3r^8 - 4s^8)}{rs}$. Multiplying $(a_0, \frac{P-M}{2}, c, \frac{P+M}{2})$ by $\frac{2rs}{3}$ gives $C_0(r, s)$.

(3) Setting $Q = 3Z^2 - 1$ and $L = 3Z$ in $(Q + L)^4 - (Q - L)^4 = 8QL(Q^2 + L^2)$ gives

$$(3Z^2 + 3Z - 1)^4 - (3Z^2 - 3Z - 1)^4 = 24Z(3Z^2 - 1)(9Z^4 + 3Z^2 + 1) = 24Z(27Z^6 - 1).$$

Subtracting $(6ZX)^4 + n(6ZY)^4 = 1296Z^4(X^4 + nY^4) = 24Z \cdot 54(X^4 + nY^4)Z^3$ yields (40). Setting the right-hand side of (40) to zero for $Z \neq 0$ gives $27Z^6 - 54(X^4 + nY^4)Z^3 - 1 = 0$, and dividing by Z^3 yields $27Z^3 - Z^{-3} = 54(X^4 + nY^4)$, which in terms of $3Z$ is $(3Z)^3 - 27(3Z)^{-3} = 54(X^4 + nY^4)$.

(4) Write $A_1 = 3r^8 - 4s^8$, $B_1 = 6r^4 s^4$ and $A_2 = 3u^8 - 4v^8$, $B_2 = 6u^4 v^4$, so that $b_i = A_i - B_i$ and $d_i = A_i + B_i$. Then

$$b_1^3 b_2 - d_1^3 d_2 = (A_1 - B_1)^3(A_2 - B_2) - (A_1 + B_1)^3(A_2 + B_2) = -2\big[A_1(A_1^2 + 3B_1^2)B_2 + B_1(3A_1^2 + B_1^2)A_2\big].$$

Setting $s = v = 1$ and $u = tr$ makes $A_1 = 3R - 4$, $A_2 = 3t^8 R - 4$, $B_1 = 6r^4$, and $B_2 = 6t^4 r^4$ (where $R = r^8$), while $-12a_1^3 a_2 + c_1^3 c_2 = -3072tr^4 + 1296t^7 r^4 R^3$. Thus every term of $\Phi(r, 1; tr, 1) = -12a_1^3 a_2 + c_1^3 c_2 + b_1^3 b_2 - d_1^3 d_2$ is r^4 times a polynomial in $R = r^8$:

$$\Phi(r, 1; tr, 1) = -12r^4(t-1)^2\Big[27t^4(3t^2 + 2t + 1)R^3 - 108(t+1)^2(t^2+1)^2 R^2 + 144(t+1)^2(t^2+1)^2 R - 64(t^2 + 2t + 3)\Big].$$

Dividing out $-12(rv - su)^2 r^2 = -12r^4(t-1)^2$ yields (41). □

5.3 The Picard Rigidity and Finiteness Theorem for $n = -12$

Why do $n = 4$ and $n = -1$ admit infinite towers of primitive polynomial families generated by birational involutions (Theorems 3.4 and 4.3), whereas only a single primitive polynomial family (Elkies' degree-8 curve C_0) is known for $n = -12$? We resolve this question by exhibiting the explicit 20-line integral basis $\mathcal{B}_{20}$ of $\mathrm{Pic}(S_{n,\overline{\mathbb{Q}}})$, computing the Galois-invariant Picard lattice $\mathrm{Pic}(S_{-12}/\mathbb{Q})$ via Reynolds averaging over the subgroup $H_0 = \mathrm{Gal}(K/\mathbb{Q}(\zeta_8))$, and proving via the Hessian discriminant that S_{-12} is Picard-rigid over $\mathbb{Q}$ with finite automorphism group of order 16.

Lemma 5.3 (Explicit 20-Line Integral Basis of $\mathrm{Pic}(S_{n,\overline{\mathbb{Q}}})$)**.** *For any $n \in \mathbb{Q}^\times$, let $\eta_0 \in \overline{\mathbb{Q}}^\times$ satisfy $\eta_0^4 = -n$ and set $\theta_0 = \zeta_8\eta_0$ ($\theta_0^4 = n$), where $\zeta_8 = (1 + i)/\sqrt{2}$. Index the 48 lines on S_n over $\overline{\mathbb{Q}}$ by $m = 16p + 4j + k \in \{0, \ldots, 47\}$, where $p \in \{0, 1, 2\}$ corresponds to Pairings I, II, III and $0 \leq j, k \leq 3$. Then the 48×48 intersection matrix G_{48} ($L_m^2 = -2$, and $L_m \cdot L_{m'} = 1$ or 0 according as L_m and $L_{m'}$ are coplanar or skew in $\mathbb{P}^3$) is independent of n and has rank 20. Specifically, the 20 pivot lines*

$$\mathcal{B}_{20} = \{L_0, L_1, L_2, L_3, L_4, L_5, L_6, L_8, L_9, L_{10}, L_{16}, L_{17}, L_{18}, L_{20}, L_{21}, L_{22}, L_{32}, L_{33}, L_{34}, L_{36}\} \tag{42}$$

have the explicit 20×20 *intersection matrix* $G_{20} = G_{48}[\mathcal{B}_{20}, \mathcal{B}_{20}]$ *of determinant* $\det(G_{20}) = -64 \neq 0$*:*

$$G_{20} = \begin{pmatrix}
-2 & 1 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
1 & -2 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\
1 & 1 & -2 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\
1 & 1 & 1 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\
1 & 0 & 0 & 0 & -2 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 1 & -2 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\
0 & 0 & 1 & 0 & 1 & 1 & -2 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\
1 & 0 & 0 & 0 & 1 & 0 & 0 & -2 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\
0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & -2 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & -2 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & -2 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & -2 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & -2 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & -2 & 1 & 1 & 0 & 0 & 0 & 1 \\
1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & -2 & 1 & 0 & 0 & 1 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & -2 & 0 & 1 & 0 & 0 \\
0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -2 & 1 & 1 & 1 \\
0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & -2 & 1 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & -2 & 0 \\
0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & -2
\end{pmatrix}, \tag{43}$$

and every one of the 48 *lines is an integer linear combination of* $\mathcal{B}_{20}$*, with coordinate matrix* $C_{48} = G_{20}^{-1} G_{48}[\mathcal{B}_{20}, \cdot] \in \mathbb{Z}^{20\times 48}$ *recorded in full in Table 8 (Appendix A) and satisfying* $G_{48} = C_{48}^T G_{20} C_{48}$*. Since* $\rho(S_{n,\overline{\mathbb{Q}}}) \leq h^{1,1}(S_{n,\mathbb{C}}) = 20$ *and* $\det(\mathrm{Pic}(S_{n,\overline{\mathbb{Q}}})) = -64$ *(Pinch–Swinnerton-Dyer [13], Bright [12, Proposition 2.15]),* $\mathcal{B}_{20}$ *is an integral* $\mathbb{Z}$*-basis of* $\mathrm{Pic}(S_{n,\overline{\mathbb{Q}}}) \cong \mathbb{Z}^{20}$*.*

Proof. Rescaling $\tilde{a} = \eta_0 a$ over $\overline{\mathbb{Q}}$ reduces the defining equations of the 48 lines to a fixed projective configuration in $\mathbb{P}^3$ ($d = i^j b, c = i^k \tilde{a}$ for $p = 0$; $d = i^j c, b = i^k \tilde{a}$ for $p = 1$; $d = \zeta_8 i^j \tilde{a}, c = \zeta_8 i^k b$ for $p = 2$), where two lines $L_m, L_{m'}$ ($m \neq m'$) intersect if and only if the 4×4 matrix formed by their four defining planes has determinant 0. Direct symbolic evaluation of all $\binom{48}{2} = 1128$ pairs (Appendix A) yields G_{48} of rank 20, with row-reduced echelon pivot columns given by (42), G_{20} given by (43), and $\det(G_{20}) = -64$. Multiplying G_{20}^{-1} against the 20×48 submatrix of G_{48} gives the all-integer 20×48 coordinate matrix $C_{48} \in \mathbb{Z}^{20\times 48}$ in Table 8, proving that $\mathcal{B}_{20}$ generates the full 48-line lattice Λ_{48} over $\mathbb{Z}$. Because $\rho(S_{n,\overline{\mathbb{Q}}}) \leq 20$ for any characteristic-0 K3 surface and $\det(\mathrm{Pic}(S_{n,\overline{\mathbb{Q}}})) = -64 = \det(G_{20})$ [12, Proposition 2.15], the lattice index $[\mathrm{Pic}(S_{n,\overline{\mathbb{Q}}}) : \Lambda_{48}] = \sqrt{|\det(G_{20})/\det(\mathrm{Pic}(S_{n,\overline{\mathbb{Q}}}))|} = 1$, so $\mathrm{Pic}(S_{n,\overline{\mathbb{Q}}}) = \bigoplus_{L\in\mathcal{B}_{20}} \mathbb{Z}[L]$. □

Theorem 5.4 (Picard Rigidity & Obstruction to Infinite Automorphism Towers on S_{-12})**.** *Let* $S_{-12} \subset \mathbb{P}^3$ *be the smooth K3 surface* $-12a^4 + b^4 + c^4 = d^4$*. Then:*

1. ***Galois Orbits of the 48 Lines:*** *The splitting field of the* 48 *lines on* S_{-12} *is* $K = \mathbb{Q}(\zeta_8, 3^{1/4})$ *of degree* $[K : \mathbb{Q}] = 16$*. Under the cyclic normal subgroup* $H_0 = \mathrm{Gal}(K/\mathbb{Q}(\zeta_8)) \cong \mathbb{Z}/4\mathbb{Z}$*, the* 48 *lines partition into* 12 *coplanar 4-line orbits* $\mathcal{B}_{\mathrm{I}}(j), \mathcal{B}_{\mathrm{II}}(j), \mathcal{B}_{\mathrm{III}}(k)$ *(*$0 \leq j, k \leq 3$*), each of which forms a complete plane section of class* H *in* $\mathrm{Pic}(S_{-12,\overline{\mathbb{Q}}})$*. Under the full Galois group* $\mathrm{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}) \rtimes \mathbb{Z}/4\mathbb{Z}$*, these* 12 *blocks merge into* 7 *Galois orbits* $\mathcal{O}_1, \ldots, \mathcal{O}_7$ *of sizes* $4, 8, 4, 4, 8, 4, 16$ *(Table 3), whose orbit sums* $D_r = \sum_{L\in\mathcal{O}_r} [L]$ *satisfy*

$$D_1 = D_3 = D_4 = D_6 = H, \qquad D_2 = D_5 = 2H, \qquad D_7 = 4H. \tag{44}$$

2. ***Picard Rank 1 over*** $\mathbb{Q}$ ***(and over*** $\mathbb{Q}(\zeta_8)$***):*** *Whereas* $\rho(S_{-1}/\mathbb{Q}) = 9$ *(spanned by* 24 *Galois orbits of sizes* $1^8 2^{12} 4^4$*) and* $\rho(S_4/\mathbb{Q}) = 8$ *(spanned by* 20 *Galois orbits of sizes* $2^{16} 4^4$*), both the arithmetic Picard group and the Galois-invariant geometric Picard lattice of* S_{-12} *over* $\mathbb{Q}$ *(and even over* $\mathbb{Q}(\zeta_8)$*) have rank* 1 *and are generated by the hyperplane class* H *(*$H^2 = 4$*):*

$$\mathrm{Pic}(S_{-12}/\mathbb{Q}) = \mathrm{Pic}(S_{-12,\overline{\mathbb{Q}}})^{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})} = \mathrm{Pic}(S_{-12,\overline{\mathbb{Q}}})^{H_0} = \mathbb{Z}H. \tag{45}$$

3. ***Finiteness of*** $\mathrm{Bir}(S_{-12}/\mathbb{Q})$ ***and Non-Existence of Infinite Automorphism Towers:*** *Every birational self-map* $f \in \mathrm{Bir}(S_{-12}/\mathbb{Q})$ *is a regular projective linear automorphism preserving* H*, and by Hessian plane rigidity of diagonal quartic surfaces,*

$$\mathrm{Bir}(S_{-12}/\mathbb{Q}) = \mathrm{Aut}(S_{-12}/\mathbb{Q}) \cong (\mathbb{Z}/2\mathbb{Z})^3 \rtimes S_2 \qquad (\textit{of finite order } 16), \tag{46}$$

generated by coordinate sign changes $(a,b,c,d) \mapsto (\pm a, \pm b, \pm c, d)$ and the swap $b \leftrightarrow c$. Consequently, unlike $n = 4$ and $n = -1$, S_{-12} admits no non-trivial skew-line involution, conic-bundle involution, or base-point-free genus-1 fibration over $\mathbb{Q}$, and the entire $\mathrm{Bir}(S_{-12}/\mathbb{Q})$-orbit of Elkies' degree-8 curve C_0 consists of a single projective equivalence class of degree 8.

4. ***9-Node Geometry of Elkies' Curve*** $C_0 \in |2H|$***:*** *Every rational curve $C \subset S_{-12}$ defined over $\mathbb{Q}$ lies in $|mH|$ for some integer $m \geq 1$ ($\deg(C) = 4m$), with arithmetic genus $p_a(C) = 2m^2 + 1$, and can have geometric genus 0 only if it acquires $\sum_{P \in C} \delta_P = 2m^2 + 1$ singular delta-invariants on S_{-12}. For $m = 2$, $C_0 = \mathcal{Q}_6 \cap S_{-12} \in |2H|$ achieves $p_g(C_0) = 9 - 9 = 0$ via 9 ordinary double points: 1 node at the Eckardt vertex $Q_1 = (0:1:0:1)$ (where $(r:s) = (1:0)$ and $(0:1)$ meet) and 8 Galois-conjugate nodes where pairs $(r_1 : s_1)$ and $(\omega r_1 : s_1)$ ($\omega^3 = 1, \omega \neq 1$) satisfy $9r^{16} - 12r^8 s^8 + 16 s^{16} = 0$.*

Table 3: Complete Classification of the 7 $\mathrm{Gal}(K/\mathbb{Q})$-Orbits (and 12 H_0-Orbits) of the 48 Lines on S_{-12}. Here $\eta_0 = \sqrt{2} \cdot 3^{1/4}$ ($\eta_0^4 = 12$) and $\theta_0 = (1+i) \cdot 3^{1/4}$ ($\theta_0^4 = -12$).

Orbit	Pairing & H_0-Blocks	Explicit Line Equations (k or $j \in \{0,1,2,3\}$)	Size	Complete Intersection & Class
$\mathcal{O}_1$	Type I ($j=0$): $\mathcal{B}_{\mathrm{I}}(0)$	$d = b,\ c = \eta_0 i^k a$	4	$S_{-12} \cap \{d - b = 0\} \implies D_1 = H$
$\mathcal{O}_2$	Type I ($j \in \{1,3\}$): $\mathcal{B}_{\mathrm{I}}(1) \cup \mathcal{B}_{\mathrm{I}}(3)$	$d = \pm ib,\ c = \eta_0 i^k a$	8	$S_{-12} \cap \{d^2 + b^2 = 0\} \implies D_2 = 2H$
$\mathcal{O}_3$	Type I ($j=2$): $\mathcal{B}_{\mathrm{I}}(2)$	$d = -b,\ c = \eta_0 i^k a$	4	$S_{-12} \cap \{d + b = 0\} \implies D_3 = H$
$\mathcal{O}_4$	Type II ($j=0$): $\mathcal{B}_{\mathrm{II}}(0)$	$d = c,\ b = \eta_0 i^k a$	4	$S_{-12} \cap \{d - c = 0\} \implies D_4 = H$
$\mathcal{O}_5$	Type II ($j \in \{1,3\}$): $\mathcal{B}_{\mathrm{II}}(1) \cup \mathcal{B}_{\mathrm{II}}(3)$	$d = \pm ic,\ b = \eta_0 i^k a$	8	$S_{-12} \cap \{d^2 + c^2 = 0\} \implies D_5 = 2H$
$\mathcal{O}_6$	Type II ($j=2$): $\mathcal{B}_{\mathrm{II}}(2)$	$d = -c,\ b = \eta_0 i^k a$	4	$S_{-12} \cap \{d + c = 0\} \implies D_6 = H$
$\mathcal{O}_7$	Type III ($k \in \{0,1,2,3\}$): $\bigcup_{k=0}^{3} \mathcal{B}_{\mathrm{III}}(k)$	$d = \theta_0 i^j a,\ c = \zeta_8 i^k b$	16	$S_{-12} \cap \{b^4 + c^4 = 0\} \implies D_7 = 4H$

Proof. (1)–(2) By Lemma 5.3, $\mathrm{Pic}(S_{-12,\overline{\mathbb{Q}}}) \cong \mathbb{Z}^{20}$ is spanned over $\mathbb{Z}$ by the classes of the 48 lines (with explicit 20-line integral basis $\mathcal{B}_{20}$). For $n = -12$, we have $\eta_0 = 12^{1/4} = \sqrt{2} \cdot 3^{1/4}$, $\theta_0 = (1+i) \cdot 3^{1/4} = \zeta_8 \eta_0$, and $\zeta_8 = (1+i)/\sqrt{2}$, so the splitting field of the 48 lines is $K = \mathbb{Q}(\zeta_8, 3^{1/4})$. Since $\mathbb{Q}(\zeta_8)$ has quadratic subfields $\mathbb{Q}(i), \mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{-2})$ (ramified only at 2) while $\mathbb{Q}(3^{1/4})$ has unique quadratic subfield $\mathbb{Q}(\sqrt{3})$ (ramified at 3), we have $\mathbb{Q}(\zeta_8) \cap \mathbb{Q}(3^{1/4}) = \mathbb{Q}$ and $[K : \mathbb{Q}] = 4 \times 4 = 16$. Every automorphism $\sigma_{u,v} \in \mathrm{Gal}(K/\mathbb{Q})$ is uniquely determined by $\sigma_{u,v}(\zeta_8) = \zeta_8^u$ ($u \in \{1,3,5,7\}$) and $\sigma_{u,v}(3^{1/4}) = i^v 3^{1/4}$ ($v \in \{0,1,2,3\}$).

Consider the cyclic subgroup $H_0 = \mathrm{Gal}(K/\mathbb{Q}(\zeta_8)) = \{\sigma_{1,v} : v \in \{0,1,2,3\}\} \cong \mathbb{Z}/4\mathbb{Z}$. Since $\sigma_{1,v}$ fixes ζ_8 (and hence fixes $i, \sqrt{2}$, and $1+i$) pointwise while sending $3^{1/4} \mapsto i^v 3^{1/4}$, it acts on η_0 and θ_0 by $\sigma_{1,v}(\eta_0) = i^v \eta_0$ and $\sigma_{1,v}(\theta_0) = i^v \theta_0$. Consequently:

- In Type I, for each fixed $j \in \{0,1,2,3\}$, H_0 acts transitively on the 4-line block $\mathcal{B}_{\mathrm{I}}(j) = \{d = i^j b,\ c = \eta_0 i^k a\}_{k=0}^{3}$, whose union is the complete intersection of the plane $d - i^j b = 0$ with S_{-12} ($-12a^4 + c^4 = 0$), so $\sum_{L \in \mathcal{B}_{\mathrm{I}}(j)} [L] = H$ (specifically, in $\mathcal{B}_{20}$, $[L_0] + [L_1] + [L_2] + [L_3] = H$).
- In Type II, for each fixed $j \in \{0,1,2,3\}$, H_0 acts transitively on the 4-line block $\mathcal{B}_{\mathrm{II}}(j) = \{d = i^j c,\ b = \eta_0 i^k a\}_{k=0}^{3} = S_{-12} \cap \{d - i^j c = 0\}$, so $\sum_{L \in \mathcal{B}_{\mathrm{II}}(j)} [L] = H$.
- In Type III, for each fixed $k \in \{0,1,2,3\}$, H_0 acts transitively on the 4-line block $\mathcal{B}_{\mathrm{III}}(k) = \{d = \theta_0 i^j a,\ c = \zeta_8 i^k b\}_{j=0}^{3} = S_{-12} \cap \{c - \zeta_8 i^k b = 0\}$, so $\sum_{L \in \mathcal{B}_{\mathrm{III}}(k)} [L] = H$.

Thus every single one of the 48 lines $L \subset S_{-12}$ satisfies $\sum_{\tau \in H_0} [\tau(L)] = H$ in $\mathrm{Pic}(S_{-12,\overline{\mathbb{Q}}})$. Under the quotient $\mathrm{Gal}(\mathbb{Q}(\zeta_8)/\mathbb{Q}) = \{\sigma_{u,0} : u \in \{1,3,5,7\}\}$, $\mathcal{B}_{\mathrm{I}}(1)$ and $\mathcal{B}_{\mathrm{I}}(3)$ merge into $\mathcal{O}_2$ ($D_2 = 2H$), $\mathcal{B}_{\mathrm{II}}(1)$ and $\mathcal{B}_{\mathrm{II}}(3)$ merge into $\mathcal{O}_5$ ($D_5 = 2H$), and $\mathcal{B}_{\mathrm{III}}(0), \ldots, \mathcal{B}_{\mathrm{III}}(3)$ merge into $\mathcal{O}_7$ ($D_7 = 4H$), establishing (44) and Table 3.

Now let $X \in \mathrm{Pic}(S_{-12,\overline{\mathbb{Q}}})^{H_0}$ be any H_0-invariant integral divisor class. By Lemma 5.3, the 48 lines span $\mathrm{Pic}(S_{-12,\overline{\mathbb{Q}}})$, so we can write $X = \sum_L c_L [L]$ with $c_L \in \mathbb{Z}$. Applying the Reynolds averaging operator over the 4-element group H_0 yields

$$X = \frac{1}{4} \sum_{\tau \in H_0} \tau(X) = \sum_L \frac{c_L}{4} \left(\sum_{\tau \in H_0} [\tau(L)] \right) = \left(\frac{1}{4} \sum_L c_L \right) H \in \mathbb{Q}H.$$

Hence $\mathrm{Pic}(S_{-12,\overline{\mathbb{Q}}})^{H_0} = \mathrm{Pic}(S_{-12,\overline{\mathbb{Q}}}) \cap \mathbb{Q}H$. If $H = mD_0$ for some integer $m \geq 1$ and $D_0 \in \mathrm{Pic}(S_{-12,\overline{\mathbb{Q}}})$, intersecting with any line L_0 gives $1 = H \cdot L_0 = m(D_0 \cdot L_0) \in m\mathbb{Z}$, forcing $m = \pm 1$. Thus H is primitive in $\mathrm{Pic}(S_{-12,\overline{\mathbb{Q}}})$, so $\mathrm{Pic}(S_{-12,\overline{\mathbb{Q}}}) \cap \mathbb{Q}H = \mathbb{Z}H$. Since $S_{-12}(\mathbb{Q}) \neq \emptyset$ (containing $Q_1 = (0:1:0:1) \in S_{-12}(\mathbb{Q})$), the Hochschild–Serre exact sequence (2)–(3) splits and gives $\mathrm{Pic}(S_{-12}/\mathbb{Q}) = \mathrm{Pic}(S_{-12,\overline{\mathbb{Q}}})^{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})} = \mathbb{Z}H$ of rank 1.

(3) By Lemma 3.1(1), every birational map $f : S_{-12} \dashrightarrow S_{-12}$ extends to a regular isomorphism $f \in \mathrm{Aut}(S_{-12}/\mathbb{Q})$. Because $\mathrm{Pic}(S_{-12}/\mathbb{Q}) = \mathbb{Z}H$ and H is the unique ample generator, $f^*(H) = H$. Hence f acts linearly on $H^0(S_{-12}, \mathcal{O}_S(H)) = \mathrm{span}(a, b, c, d)$, so f is induced by a projective linear transformation $A \in \mathrm{PGL}_4(\mathbb{Q})$ preserving the smooth quartic surface $F(a, b, c, d) = -12a^4 + b^4 + c^4 - d^4 = 0$, i.e., $F(A(x)) = \lambda F(x)$ for some $\lambda \in \mathbb{Q}^\times$. Taking the Hessian determinant

$$H(F) = \det\left(\frac{\partial^2 F}{\partial x_i \partial x_j}\right) = (-144a^2)(12b^2)(12c^2)(-12d^2) = +12^5\, a^2b^2c^2d^2 = 248832\, a^2b^2c^2d^2,$$

covariant transformation $H(F \circ A) = (\det A)^2 H(F) \circ A$ implies that A must preserve the zero locus $\{abcd = 0\}$ in $\mathbb{P}^3$. By unique factorization in $\mathbb{Q}[a, b, c, d]$, A must permute the four coordinate linear forms a, b, c, d up to non-zero scalars: $A(x)_i = \alpha_i x_{\pi(i)}$ for some permutation $\pi \in S_4$ and $\alpha_i \in \mathbb{Q}^\times$. Substituting into $-12a^4+b^4+c^4-d^4 = \lambda(-12a^4+b^4+c^4-d^4)$ requires $-12\alpha_a^4 x_{\pi(a)}^4 + \alpha_b^4 x_{\pi(b)}^4 + \alpha_c^4 x_{\pi(c)}^4 - \alpha_d^4 x_{\pi(d)}^4 = \lambda(-12a^4 + b^4 + c^4 - d^4)$. Comparing fourth-power ratios in $\mathbb{Q}^\times/\mathbb{Q}^{\times 4}$, since $-12, 1, -1$ are pairwise distinct modulo $\mathbb{Q}^{\times 4}$ (as $-1 \notin \mathbb{Q}^{\times 4}$ and $\pm 12 \notin \mathbb{Q}^{\times 4}$), the permutation π must fix a and d ($\pi(a) = a, \pi(d) = d$) and can only permute $\{b, c\}$ ($\pi \in \{\mathrm{id}, (b\ c)\}$). Normalize $\alpha_d = 1$; then $\lambda = 1$, forcing $\alpha_a^4 = \alpha_b^4 = \alpha_c^4 = 1$ in $\mathbb{Q}^\times$, so $\alpha_a, \alpha_b, \alpha_c \in \{\pm 1\}$. Thus $\mathrm{Aut}(S_{-12}/\mathbb{Q}) \cong (\mathbb{Z}/2\mathbb{Z})^3 \rtimes S_2$ has exact order $2^3 \times 2 = 16$. Moreover, any base-point-free genus-1 fibration over $\mathbb{Q}$ would have a non-zero fiber class $[F] \in \mathrm{Pic}(S_{-12}/\mathbb{Q}) = \mathbb{Z}H$ with $F^2 = 0$ [12, Proposition 2.23], which is impossible since $(mH)^2 = 4m^2 > 0$ for all $m \neq 0$.

(4) Any curve $C \subset S_{-12}$ defined over $\mathbb{Q}$ has class $[C] = mH$ ($m \geq 1$), degree $H \cdot (mH) = 4m$, and arithmetic genus $p_a(C) = \frac{1}{2}(mH)^2 + 1 = 2m^2 + 1$ by the adjunction formula on a K3 surface. For $m = 2$, $C_0 = \mathcal{Q}_6 \cap S_{-12} \in |2H|$ has $p_a(C_0) = 9$. Away from $rs = 0$, dividing $(a_0 : b_0 : c_0 : d_0)$ by $4rs^7$ gives $(1 : \frac{3r^8-6r^4s^4-4s^8}{4rs^7} : \frac{3}{2}(r/s)^6 : \frac{3r^8+6r^4s^4-4s^8}{4rs^7})$. Two distinct parameters $(r_1 : 1)$ and $(r_2 : 1)$ map to the same point in $\mathbb{P}^3$ if and only if $r_2^6 = r_1^6$, $r_2^3 = r_1^3$ (from $(d-b)/a = 3(r/s)^3$), and $(3r_1^8 - 4)/r_1 = (3r_2^8 - 4)/r_2$ (from $(d+b)/a$). The first two conditions give $r_2 = \omega r_1$ with $\omega^3 = 1, \omega \neq 1$, and substituting $r_2 = \omega r_1$ into the third condition yields $3r_1^8(1-\omega) = -4\omega^2(1-\omega) \iff r_1^8 = -\frac{4}{3}\omega^2$, which are the 16 simple roots of $(3r^8 + 4\omega^2 s^8)(3r^8 + 4\omega s^8) = 9r^{16} - 12r^8 s^8 + 16s^{16} = 0$ paired into 8 ordinary double points. Together with the node at $Q_1 = (0:1:0:1)$ where $(1:0)$ and $(0:1)$ meet with transverse tangents $\{a = 0, d = b\}$ and $\{c = 0, d = b\}$, C_0 has $1 + 8 = 9$ nodes, giving $p_g(C_0) = 9 - 9 = 0$. □

6 The Degree-17 Principal Asymptotic Tangent Self-Map on S_{-s^2}

When $n = -s^2$ ($s \in \mathbb{Q}^\times$, so $n = -1, -4, -9, -16, -25, -36, -49, \ldots$), the product of the four surface coefficients $ABCD = (-s^2)(1)(1)(-1) = s^2$ is a rational square. By introducing logarithmic derivative coordinates, we express the classical Euler–Richmond inflection-tangent construction [7, 6] via a 4-monomial tangent vector and a completely factored degree-17 rational formula with explicit polynomial syzygies and base-locus classification.

Theorem 6.1 (The 4-Monomial Asymptotic Tangent Vector & Factored Degree-17 Self-Map $\mathcal{A}_\pm$ on S_{-s^2})**.** *Let $P_1 = (a_1, b_1, c_1, d_1)$ be a generic point on $S_{-s^2} : -s^2a^4 + b^4 + c^4 = d^4$.*

1. ***4-Monomial Tangent Vector:*** *Setting $X = v_a/a_1$, $Y = v_b/b_1$, $Z = v_c/c_1$, and $W = v_d/d_1$, the two principal asymptotic tangent directions satisfying $\sum A_i X_i = \sum A_i X_i^2 = 0$ are given symmetrically by*

$$(X, Y, Z, W) = \left(\frac{c_1^2}{sa_1^2},\ \pm\frac{d_1^2}{b_1^2},\ \frac{sa_1^2}{c_1^2},\ \pm\frac{b_1^2}{d_1^2}\right).$$

Clearing the denominator $sa_1b_1c_1d_1$ yields the 4-monomial degree-5 asymptotic tangent vector

$$V_\pm(P_1) = (b_1c_1^3d_1,\ \pm sa_1c_1d_1^3,\ s^2a_1^3b_1d_1,\ \pm sa_1b_1^3c_1). \tag{47}$$

2. ***Factored Degree-17 Self-Map $\mathcal{A}_\pm : S_{-s^2} \dashrightarrow S_{-s^2}$ and Base Locus:*** *Along the asymptotic tangent line $P(t) = P_1 + tV_\pm(P_1)$, the t^0, t^1, and t^2 terms of $F(P_1 + tV_\pm(P_1))$ vanish identically on S_{-s^2}, and the common factor $s^2(s^2a_1^4 - c_1^4) = s^2(b_1^4 - d_1^4)$ divides both the t^3 and t^4 coefficients modulo $F(P_1)$, yielding the closed-form degree-17 rational self-map*

$$\mathcal{A}_\pm(P_1) = K_4(P_1)P_1 - K_3^\pm(P_1)V_\pm(P_1), \tag{48}$$

where the degree-16 and degree-12 invariants $K_4, K_3^\pm \in \mathbb{Z}[a_1, b_1, c_1, d_1, s]$ factor completely into binomials:

$$\begin{aligned} K_4(P_1) &= (s^2a_1^4b_1^4 - c_1^4d_1^4)(s^2a_1^4d_1^4 - b_1^4c_1^4), \\ K_3^\pm(P_1) &= 4a_1b_1c_1d_1\Big[b_1^2d_1^2(s^2a_1^4 + c_1^4) \mp sa_1^2c_1^2(b_1^4 + d_1^4)\Big]. \end{aligned} \tag{49}$$

On S_{-s^2}, the divisorial base locus of the degree-17 polynomial formula (48) consists of the 16 lines $L_{\mathrm{I}}(j,k)$ and $L_{\mathrm{II}}(j,k)$ in Pairings I and II satisfying $c_1^2/(sa_1^2) = \pm d_1^2/b_1^2$ (resp. $b_1^2/(sa_1^2) = \pm d_1^2/c_1^2$), i.e., $j \equiv k \pmod 2$ for $\mathcal{A}_+$ and $j \not\equiv k \pmod 2$ for $\mathcal{A}_-$ (along which $V_\pm(P_1)$ is tangent to the line itself), together with the codimension-2 Eckardt points and asymptotic inflection points where $K_4(P_1) = K_3^\pm(P_1) = 0$.

Proof. (1) Let $F(a,b,c,d) = -s^2a^4 + b^4 + c^4 - d^4$. By Taylor's formula, for $V = (a_1X, b_1Y, c_1Z, d_1W)$, the coefficients of t^1 and t^2 in $F(P_1 + tV)$ are $4(-s^2a_1^4X + b_1^4Y + c_1^4Z - d_1^4W)$ and $6(-s^2a_1^4X^2 + b_1^4Y^2 + c_1^4Z^2 - d_1^4W^2)$. Substituting $X = \frac{c_1^2}{sa_1^2}$, $Z = \frac{sa_1^2}{c_1^2}$ and $Y = \pm\frac{d_1^2}{b_1^2}$, $W = \pm\frac{b_1^2}{d_1^2}$ gives

$$-s^2a_1^4X + c_1^4Z = -sa_1^2c_1^2 + sa_1^2c_1^2 = 0, \qquad b_1^4Y - d_1^4W = \pm b_1^2d_1^2 \mp b_1^2d_1^2 = 0,$$

so the t^1 coefficient vanishes identically in $\mathbb{Z}[a_1, b_1, c_1, d_1, s]$ without even using the surface equation. Multiplying (a_1X, b_1Y, c_1Z, d_1W) by $sa_1b_1c_1d_1$ produces $V_\pm(P_1) = (v_a, v_b, v_c, v_d)$ in (47), for which the t^2 coefficient of $F(P_1 + tV_\pm(P_1))$ is

$$c_2(P_1) = 6(-s^2a_1^2v_a^2 + b_1^2v_b^2 + c_1^2v_c^2 - d_1^2v_d^2) = 6s^2a_1^2b_1^2c_1^2d_1^2(s^2a_1^4 - b_1^4 - c_1^4 + d_1^4) = -6s^2a_1^2b_1^2c_1^2d_1^2F(P_1),$$

which vanishes identically on S_{-s^2}.

(2) Let $c_3^\pm(P_1) = 4(-s^2a_1v_a^3 + b_1v_b^3 + c_1v_c^3 - d_1v_d^3)$ and $c_4(P_1) = -s^2v_a^4 + v_b^4 + v_c^4 - v_d^4$ denote the t^3 and t^4 coefficients of $F(P_1 + tV_\pm(P_1))$. Substituting (47) and (49), direct commutative-ring expansion in $\mathbb{Z}[a_1, b_1, c_1, d_1, s]$ (verified symbolically in Appendix A) gives the exact polynomial syzygies

$$\begin{aligned} c_3^\pm(P_1) - s^2(s^2a_1^4 - c_1^4)K_3^\pm(P_1) &= \mp 4s^3a_1^3b_1c_1^3d_1(b_1^4 + d_1^4)F(P_1), \\ c_4(P_1) - s^2(s^2a_1^4 - c_1^4)K_4(P_1) &= -s^4a_1^4c_1^4(b_1^8 + b_1^4d_1^4 + d_1^8)F(P_1). \end{aligned}$$

On S_{-s^2} (where $F(P_1) = 0$), both right-hand sides vanish identically, and dividing $c_4(P_1)t + c_3^\pm(P_1) = 0$ by $s^2(s^2a_1^4 - c_1^4)$ gives $t = -K_3^\pm(P_1)/K_4(P_1)$. Clearing $K_4(P_1)$ (of degree $8 + 8 = 16$ in (a_1, b_1, c_1, d_1)) in $P_1 + tV_\pm(P_1)$ yields the homogeneous degree-17 formula (48). Finally, on the 32 lines of Pairings I and II ($\{s^2a_1^4 = c_1^4, b_1^4 = d_1^4\}$ and $\{s^2a_1^4 = b_1^4, c_1^4 = d_1^4\}$), $K_4(P_1)$ vanishes identically, while $K_3^\pm(P_1)$ vanishes on the 16 lines where $b_1^2d_1^2(s^2a_1^4 + c_1^4) = \pm sa_1^2c_1^2(b_1^4 + d_1^4)$ (i.e., $j \equiv k \pmod 2$ for $\mathcal{A}_+$ and $j \not\equiv k \pmod 2$ for $\mathcal{A}_-$). Away from these lines and the coordinate planes $a_1b_1c_1d_1 = 0$, P_1 and $V_\pm(P_1)$ are linearly independent in $\mathbb{P}^3$, so $\mathcal{A}_\pm(P_1) = 0$ occurs only on the codimension-2 inflection locus $K_4(P_1) = K_3^\pm(P_1) = 0$. □

7 The Master Corner Identity and Shifted Linearization Engines

To study plane and conic sections where b, c, d share a common shift $M = b + c - d$, we introduce the corner coordinates $X = d - c$ and $Y = d - b$, so that $b = X + M$, $c = Y + M$, and $d = X + Y + M$.

Proposition 7.1 (The Master Corner Identity & Conic Reduction)**.** *For all X, Y, M in any commutative ring, we have the exact identity*

$$(X+M+Y)^4 - (X+M)^4 - (Y+M)^4 = 2XY\big(2X^2+3XY+2Y^2+6M(X+Y)+6M^2\big) - M^4. \quad (50)$$

Consequently, on $S_n : na^4 + b^4 + c^4 = d^4$, setting $b = X + M$, $c = Y + M$, $d = X + Y + M$ gives

$$2XY\big(2X^2+3XY+2Y^2+6M(X+Y)+6M^2\big) = M^4 + na^4. \quad (51)$$

In particular, choosing $X = u^2 + 2uv + v^2$, $Y = u^2 - 2uv + v^2$, and $M = u^2 - v^2$ gives $b = X + M = 2u^2 + 2uv$, $c = Y + M = 2u^2 - 2uv$, $d = X + Y + M = 3u^2 + v^2$, and yields the classical degree-2 quartic-square identity

$$(3u^2+v^2)^4 - (2u^2+2uv)^4 - (2u^2-2uv)^4 = \big((u^2-v^2)(7u^2+v^2)\big)^2. \quad (52)$$

Proof. Write $A = X + M$. Expanding $(A + Y)^4 - A^4 - (Y + M)^4$ by the binomial theorem gives

$$(A^4 + 4A^3Y + 6A^2Y^2 + 4AY^3 + Y^4) - A^4 - (Y^4 + 4Y^3M + 6Y^2M^2 + 4YM^3 + M^4).$$

Canceling A^4 and Y^4 and substituting $A = X + M$ into $4Y(A^3 - M^3) + 6Y^2(A^2 - M^2) + 4Y^3(A - M)$ factors out $2XY = 2(A - M)Y$:

$$\begin{aligned} &4YX(X^2+3XM+3M^2) + 6Y^2X(X+2M) + 4Y^3X - M^4 \\ &\quad = 2XY\big(2X^2+3XY+2Y^2+6M(X+Y)+6M^2\big) - M^4, \end{aligned}$$

proving (50) and (51). Next, setting $X = (u + v)^2 = u^2 + 2uv + v^2$, $Y = (u - v)^2 = u^2 - 2uv + v^2$, and $M = u^2 - v^2$ gives $b = X + M = 2u^2 + 2uv$, $c = Y + M = 2u^2 - 2uv$, and $d = X + Y + M = 3u^2 + v^2$. Substituting these into the right-hand side of (50) gives

$$2(u^2-v^2)^2(25u^4+14u^2v^2+v^4) - (u^2-v^2)^4 = (u^2-v^2)^2(49u^4+14u^2v^2+v^4) = \big((u^2-v^2)(7u^2+v^2)\big)^2,$$

proving (52). □

8 The Master u_4-Eckardt Identity, Double-Eisenstein Identity, and Bitangent Planes

Using the classical Eisenstein quartic identity $b^4 + c^4 + (b + c)^4 = 2(b^2 + bc + c^2)^2$ and the geometry of the four Eckardt points $Q_1, Q_2, Q_3, Q_4 \in S_n$, we obtain a 1-parameter master family of polynomial identities valid on S_n for all n.

Theorem 8.1 (The 1-Parameter Master u_4-Eckardt Identity)**.** *Define the quadratic forms $A = b^2 + bc + c^2 - d(b + c)$ and $B = (d - b)(d - c)$. Then for every $u_4 \in \mathbb{Q}$, the following polynomial identity holds on $S_n : na^4 + b^4 + c^4 = d^4$:*

$$(A + u_4B)^2 + na^4 = (d - b)(d - c)\, Q_{\mathrm{res}}(b, c, d; u_4), \quad (53)$$

where the residual ternary quadratic form is

$$Q_{\mathrm{res}}(b, c, d; u_4) = 2(u_4 + 1)(b^2 + c^2) + (u_4^2 + 2u_4 + 3)bc - (u_4^2 + 2u_4 - 1)d(b + c) + (u_4^2 + 1)d^2, \quad (54)$$

with symmetric matrix determinant $\det M(Q_{\mathrm{res}}) = -\frac{1}{2}(u_4 + 1)(u_4^2 - 2u_4 - 1)(u_4^2 + 2u_4 + 3)$*. Specializing u_4 yields three fundamental identities on S_n:*

1. ***At*** $u_4 = +1$ ***(The Double-Eisenstein Identity):*** *Since* $A + B = (b + c - d)^2$, (53) *becomes*

$$(b + c - d)^4 + na^4 = 2(d - b)(d - c)(2b^2 + 3bc + 2c^2 - d(b + c) + d^2). \tag{55}$$

2. ***At*** $u_4 = 0$ ***(The Zero-Shift Eckardt Identity):***

$$\big(b^2 + bc + c^2 - d(b + c)\big)^2 + na^4 = (d - b)(d - c)\big(2b^2 + 3bc + 2c^2 + d(b + c) + d^2\big). \tag{56}$$

3. ***At*** $u_4 = -1$ ***(The Four-Eckardt-Plane Identity):*** *Here* $\det M(Q_{\mathrm{res}}) = 0$, $A - B = b^2 + c^2 - d^2$, *and* $Q_{\mathrm{res}}(-1) = 2(d + b)(d + c)$ *splits into two linear planes, yielding*

$$(b^2 + c^2 - d^2)^2 + na^4 = 2(d - b)(d + b)(d - c)(d + c) = 2(d^2 - b^2)(d^2 - c^2). \tag{57}$$

Proof. On S_n, substitute $na^4 = d^4 - b^4 - c^4$. First, observe that at $u_4 = 0$, when $d = b$, we have $A|_{d=b} = b^2 + bc + c^2 - b(b + c) = c^2$, so $A^2 + (d^4 - b^4 - c^4)|_{d=b} = c^4 - c^4 = 0$. By $b \leftrightarrow c$ symmetry, $A^2 + d^4 - b^4 - c^4$ is divisible by $B = (d - b)(d - c)$ in $\mathbb{Z}[b, c, d]$; explicit division gives

$$A^2 + d^4 - b^4 - c^4 = B\big(2b^2 + 3bc + 2c^2 + d(b + c) + d^2\big) = B\, Q_{\mathrm{res}}(b, c, d; 0),$$

proving (56). Next, for general u_4, expanding $(A+u_4B)^2+d^4-b^4-c^4 = (A^2+d^4-b^4-c^4)+B(2u_4A+u_4^2B)$ shows that $B = (d - b)(d - c)$ divides the entire expression, with quotient

$$\begin{aligned} Q_{\mathrm{res}}(b, c, d; u_4) &= Q_{\mathrm{res}}(b, c, d; 0) + 2u_4\big(b^2 + bc + c^2 - d(b + c)\big) + u_4^2(d^2 - d(b + c) + bc) \\ &= 2(u_4 + 1)(b^2 + c^2) + (u_4^2 + 2u_4 + 3)bc - (u_4^2 + 2u_4 - 1)d(b + c) + (u_4^2 + 1)d^2, \end{aligned}$$

establishing (53)–(54). In the basis (b, c, d), the 3×3 symmetric matrix of Q_{res} has determinant $-\frac{1}{2}(u_4 + 1)(u_4^2 - 2u_4 - 1)(u_4^2 + 2u_4 + 3)$. Specializing to $u_4 = 1$ gives $A + B = (b + c - d)^2$ and (55); specializing to $u_4 = -1$ gives $A - B = b^2 + c^2 - d^2$ and $Q_{\mathrm{res}}(b, c, d; -1) = 2bc + 2d(b + c) + 2d^2 = 2(d + b)(d + c)$, proving (57). □

Corollary 8.2 (Reducible Plane Sections & Singular/2-Adic Analysis of Vepir's $\mathbb{Q}(\omega)$ Bitangent Plane)**.** *From the Double-Eisenstein Identity* (55) *and the single-Eisenstein form* $2(b^2 + bc + c^2)^2 = (b + c)^4 + d^4 - na^4$*:*

1. ***Reducible Plane Section Classification*** *(*$n = -\alpha^4 \equiv -1 \pmod{\mathbb{Q}^{\times 4}}$*):* *On the plane* $d = b + c + \alpha a$ *(*$\alpha \in \mathbb{Q}^\times$*), the left-hand side of* (55) *is* $(\alpha^4 + n)a^4$*. Hence* $S_n \cap \{d = b + c + \alpha a\}$ *splits over* $\mathbb{Q}$ *into two rational lines* $b + \alpha a = 0$, $c + \alpha a = 0$ *and a residual conic if and only if* $n = -\alpha^4 \equiv -1 \pmod{\mathbb{Q}^{\times 4}}$.

2. ***Eisenstein Factorization, Complete Singular-Locus Proof, & 2-Adic Obstruction on Vepir's*** $\mathbb{Q}(\omega)$ ***Bitangent Plane*** *(*$n = p_0^4 + q_0^4$*):* *Let* $p_0, q_0 \in \mathbb{Z} \setminus \{0\}$ *be coprime integers* $(\gcd(p_0, q_0) = 1)$*. On* S_n *with* $n = p_0^4 + q_0^4$*, the plane* $\Pi : na + p_0^3(b + c) = q_0^3 d$ *introduced in Vepir's online communication [11] is bitangent to* S_n *at the* $\mathbb{Q}(\omega)$*-conjugate nodes* $P_\omega = (1 : p_0\omega : p_0\omega^2 : q_0)$ *and* $\overline{P_\omega} = (1 : p_0\omega^2 : p_0\omega : q_0)$ *(*$\omega = e^{2\pi i/3}$*). Substituting* $d = (na + p_0^3 S)/q_0^3$ *(*$S = b + c$*) into* $2(b^2 + bc + c^2)^2 = S^4 + d^4 - na^4$ *factors out* $(S + p_0 a)^2$ *explicitly:*

$$2q_0^{12}(b^2 + bc + c^2)^2 = (p_0^4 + q_0^4)(S + p_0 a)^2 K(S, a), \tag{58}$$

where $K(S, a) = A_K S^2 + B_K Sa + C_K a^2$ *has* $A_K = p_0^8 - p_0^4 q_0^4 + q_0^8 > 0$, $B_K = 2p_0(p_0^8 + p_0^4 q_0^4 - q_0^8)$, $C_K = p_0^2(p_0^8 + 3p_0^4 q_0^4 + 3q_0^8)$*, and satisfies the exact completed-square and shifted identities*

$$\begin{aligned} 4A_K K(S, a) &= (2A_K S + B_K a)^2 + 8p_0^2 q_0^8(p_0^8 + p_0^4 q_0^4 + q_0^8)a^2, \\ K(S, a) &= A_K W^2 + 4(p_0^4 - q_0^4)WZ + 6Z^2 \qquad (W = S + p_0 a,\ Z = p_0 q_0^4 a). \end{aligned} \tag{59}$$

Over $\overline{\mathbb{Q}}$*, the plane quartic* $C = S_n \cap \Pi$ *has no singular points other than the two ordinary double points* $P_\omega, \overline{P_\omega}$*, so its geometric genus is strictly* $g(C) = 3 - 2 = 1$ *for all* $p_0, q_0 \in \mathbb{Z} \setminus \{0\}$*; and local solubility over* $\mathbb{Q}_2$ *requires* $p_0 \equiv q_0 \equiv 1 \pmod 2$ *(so* $n \equiv 2 \pmod{16}$*).*

Proof. (1) Substituting $b+c-d=-\alpha a$ into (55) gives $(\alpha^4+n)a^4 = 2(b+\alpha a)(c+\alpha a)(2b^2+3bc+2c^2-d(b+c)+d^2)$. Thus the two linear factors $b+\alpha a, c+\alpha a$ and the residual conic split off over $\mathbb{Q}$ if and only if $\alpha^4+n=0 \iff n=-\alpha^4$.

(2) At $P_\omega=(1:p_0\omega:p_0\omega^2:q_0)$, since $1+\omega+\omega^2=0$ and $\omega^3=1$, we have $\omega^4+\omega^8=\omega+\omega^2=-1$, so $n(1)^4+p_0^4(\omega^4+\omega^8)=(p_0^4+q_0^4)-p_0^4=q_0^4$, meaning $P_\omega, \overline{P_\omega}\in S_n$. The tangent plane at P_ω has normal $(n, p_0^3\omega^3, p_0^3\omega^6, -q_0^3)=(n,p_0^3,p_0^3,-q_0^3)$, which is independent of ω and equals $\Pi: na+p_0^3(b+c)=q_0^3 d$. Thus Π is bitangent to S_n at P_ω and $\overline{P_\omega}$, where $S=b+c=p_0(\omega+\omega^2)a=-p_0a$. Consequently, the line $S+p_0a=0$ passes through both nodes $P_\omega, \overline{P_\omega}$, so $(S+p_0a)^2$ divides $q_0^{12}(S^4-na^4)+(na+p_0^3S)^4$ in $\mathbb{Z}[p_0,q_0,S,a]$. Explicit division by $(S+p_0a)^2$ yields $(p_0^4+q_0^4)K(S,a)$ with A_K, B_K, C_K and (59) verified by direct expansion.

To prove rigorously that $C=S_n\cap\Pi$ has no other singular points over $\overline{\mathbb{Q}}$, change coordinates on Π from $(a:b:c)$ to $(a:S:D_{bc})$ where $S=b+c$ and $D_{bc}=b-c$. Since $4(b^2+bc+c^2)=3S^2+D_{bc}^2$, multiplying (58) by 8 writes the defining equation of C as

$$G(a,S,D_{bc}) = q_0^{12}(3S^2+D_{bc}^2)^2 - 8(p_0^4+q_0^4)(S+p_0a)^2K(S,a)=0.$$

Any singular point over $\overline{\mathbb{Q}}$ must satisfy $\frac{\partial G}{\partial D_{bc}} = 4q_0^{12}D_{bc}(3S^2+D_{bc}^2)=0$, giving two cases:

- **Case A ($3S^2+D_{bc}^2=0$):** Here $G=0$ forces $(S+p_0a)^2K(S,a)=0$, and since $\nabla(3S^2+D_{bc}^2)^2=0$ when $3S^2+D_{bc}^2=0$, $\frac{\partial G}{\partial S}=\frac{\partial G}{\partial a}=0$ holds if and only if $(S:a)$ is a multiple root of the binary quartic $(S+p_0a)^2K(S,a)=0$. Since $A_K>0$ (so $a=0, S=1$ is not a root), $4A_KC_K-B_K^2 = 8p_0^2q_0^8(p_0^8+p_0^4q_0^4+q_0^8)>0$, and $K(-p_0a,a)=6p_0^2q_0^8a^2\neq 0$, the unique multiple root is $S=-p_0a$, which with $D_{bc}=\pm i\sqrt{3}S$ yields precisely the two ordinary double points P_ω and $\overline{P_\omega}$ (with non-degenerate Hessian $\det H_{S,D_{bc}}\neq 0$).
- **Case B ($D_{bc}=0$, i.e. $b=c=S/2$):** Here $\frac{\partial G}{\partial S}=\frac{\partial G}{\partial a}=0$ holds if and only if $(S:a)$ is a multiple root of $G(a,S,0)=q_0^{12}(8na^4+9S^4)-8(na+p_0^3S)^4-8q_0^{12}S^4 = 8q_0^{12}(na^4+S^4/8)-8(na+p_0^3S)^4=0$. At $a=0$, the coefficient of S^4 in $G(0,S,0)$ is $q_0^{12}-8p_0^{12}\neq 0$ (since $2\notin\mathbb{Q}^{\times 4}$), so $a\neq 0$. Setting $a=1$, direct resultant computation (Appendix A) shows that the discriminant of $G(1,S,0)$ with respect to S factors in $\mathbb{Z}[p_0,q_0]$ as

$$\begin{aligned}\Delta_S(G(1,S,0)) = &-2^{17}3^5p_0^{12}q_0^{36}(p_0^4+q_0^4)^3(p_0^2-p_0q_0+q_0^2)(p_0^2+p_0q_0+q_0^2)\\ &\times(p_0^4-p_0^2q_0^2+q_0^4)(p_0^8-p_0^4q_0^4+q_0^8)(3p_0^8+3p_0^4q_0^4+q_0^8),\end{aligned}$$

  which is strictly non-zero for all $p_0,q_0\in\mathbb{Q}^\times$.

Thus P_ω and $\overline{P_\omega}$ are the only singularities of C over $\overline{\mathbb{Q}}$, and $g(C)=3-2=1$.

Finally, suppose $\gcd(p_0,q_0)=1$ and $p_0\not\equiv q_0 \pmod 2$, so $p_0^4+q_0^4\equiv 1 \pmod 8$ and $A_K\equiv 1 \pmod 8$ are odd. For any point $(a:b:c:d)\in C(\mathbb{Q}_2)$, if $(b,c)=(0,0)$ then $S=0$, and (58) gives $0=(p_0^4+q_0^4)p_0^2C_Ka^4$, forcing $a=d=0$, which is impossible in $\mathbb{P}^3$. Thus $(b,c)\in\mathbb{Q}_2^2\setminus\{(0,0)\}$; setting $m=\min(v_2(b),v_2(c))$ and writing $(b,c)=2^m(b',c')$ makes $(b')^2+b'c'+(c')^2\equiv 1 \pmod 2$ (as x^2+xy+y^2 has no non-trivial zeros over $\mathbb{F}_2$). Thus $v_2(b^2+bc+c^2)=2m$ is even, and the left-hand side of (58) is non-zero with 2-adic valuation $v_2(\text{LHS})=1+12v_2(q_0)+4m\equiv 1 \pmod 4$ and odd 2-adic unit part $\equiv 1 \pmod 4$ (being a square). On the right-hand side of (58), let $w=v_2(W)$ and $z=v_2(Z)$ in (59):

- If $w\le z$, then $v_2(A_KW^2)=2w<\min(2+w+z,1+2z)$, so $v_2(\text{RHS})=4w\equiv 0 \pmod 4 \neq v_2(\text{LHS})$.
- If $w=z+1$, then $v_2(6Z^2)=1+2z<\min(2w,2+w+z)$, so $v_2(\text{RHS})=2(z+1)+(1+2z)=4z+3\equiv 3 \pmod 4 \neq v_2(\text{LHS})$.
- If $w\ge z+2$, then $v_2(6Z^2)=1+2z\le\min(2w,2+w+z)-3$, so $K(S,a)/2^{1+2z}\equiv 3u_Z^2\equiv 3 \pmod 4$, forcing the odd 2-adic unit part of RHS to be $\equiv 3 \pmod 4$, contradicting LHS.

Hence $S_n\cap\Pi(\mathbb{Q}_2)=\emptyset$ whenever $p_0\not\equiv q_0 \pmod 2$. □

9 Universal Quadric Pencils, the 3-Parameter BCU Syzygy, and Conic-Square Splittings

On $S_n : na^4 + b^4 + c^4 = d^4$, we have two universal quadric-product identities valid for all n:

$$(b^2 + c^2 - d^2)(b^2 + c^2 + d^2) = (b^2 + c^2)^2 - d^4 = 2b^2c^2 - na^4, \tag{60}$$

$$(d^2 + b^2 - c^2)(d^2 - b^2 + c^2) = d^4 - (b^2 - c^2)^2 = 2b^2c^2 + na^4. \tag{61}$$

When $n = +2s^2$ (resp. $n = -2s^2$), the right-hand side of (60) (resp. (61)) factors over $\mathbb{Q}$ as $2(bc - sa^2)(bc + sa^2)$, cutting out the genus-1 quadric pencil of Bremner, Choudhry, and Ulas [1]. We now establish the two algebraic theorems that govern quadric-square splittings on diagonal quartic surfaces, including exact parameter-range and degenerate-locus classifications.

Theorem 9.1 (The 3-Parameter Quadratic-Square Syzygy & Complete-Intersection Parameter Classification)**.** *For any binary quadratic forms $Q_1(x, y) = rx^2 + sxy + ty^2$ and $Q_2(x, y) = rsx^2 + 2rtxy + sty^2$, all three cross-term coefficients (x^3y, x^2y^2, xy^3) of $Q_2^2 - 2rtQ_1^2$ vanish simultaneously:*

$$(rsx^2 + 2rtxy + sty^2)^2 - 2rt(rx^2 + sxy + ty^2)^2 = (s^2 - 2rt)(r^2x^4 + t^2y^4). \tag{62}$$

In normalized form ($r = -1, s = 2, t = -m$), (62) becomes

$$2(X^2 - mXY + mY^2)^2 - m(X^2 - 2XY + mY^2)^2 = (2 - m)(X^4 + m^2Y^4). \tag{63}$$

Consequently, for any $m \in \mathbb{Q} \setminus \{0, 1, 2, 4\}$ and $\alpha, \beta \in \mathbb{Q}^\times$, intersecting the two quadrics

$$\mathcal{F}_1 : \quad \alpha W^2 = X^2 - mXY + mY^2, \qquad\qquad \mathcal{F}_2 : \quad \beta Z^2 = X^2 - 2XY + mY^2 \tag{64}$$

in $\mathbb{P}^3$ embeds a geometrically smooth genus-1 complete-intersection curve $C = \mathcal{F}_1 \cap \mathcal{F}_2$ defined over $\mathbb{Q}$ into the diagonal quartic surface

$$\left(\frac{2\alpha^2}{2 - m}\right) W^4 - \left(\frac{m\beta^2}{2 - m}\right) Z^4 = X^4 + m^2Y^4.$$

Here geometric smoothness of $C = \mathcal{F}_1 \cap \mathcal{F}_2$ is governed by the four roots of the pencil determinant polynomial $\det(\lambda M_{\mathcal{F}_1} + \mu M_{\mathcal{F}_2}) = 0$ being distinct in $\mathbb{P}^1(\overline{\mathbb{Q}})$ over the algebraic closure $\overline{\mathbb{Q}}$ (not by requiring those roots to lie in $\mathbb{P}^1(\mathbb{Q})$; for instance, at $m = 3$, the residual quadratic factor $-3\lambda^2 - 12\lambda\mu - 8\mu^2 = 0$ has irrational but distinct roots $(\lambda : \mu) = (-6 \pm 2\sqrt{3} : 3) \in \mathbb{P}^1(\overline{\mathbb{Q}}) \setminus \mathbb{P}^1(\mathbb{Q})$, so C is still a smooth genus-1 curve over $\mathbb{Q}$). At the four excluded parameter values $m \in \{0, 1, 2, 4\}$, the intersection degenerates over $\overline{\mathbb{Q}}$ as follows: at $m = 0$ or $m = 2$, the quartic identity (63) collapses to $2X^4 = 2X^4$ (and the pencil determinant acquires a double root in $\mathbb{P}^1(\overline{\mathbb{Q}})$); at $m = 1$, $\mathcal{F}_2$ degenerates into the singular rank-3 quadric cone $\beta Z^2 = (X - Y)^2$ (splitting into two planes over $\mathbb{Q}(\sqrt{\beta})$); and at $m = 4$, $\mathcal{F}_1$ degenerates into the singular rank-3 quadric cone $\alpha W^2 = (X - 2Y)^2$ (splitting into two planes over $\mathbb{Q}(\sqrt{\alpha})$).

Proof. Write $U = rx^2 + ty^2$, so that $Q_1 = U + sxy$ and $Q_2 = sU + 2rtxy$. Expanding $Q_2^2 - 2rtQ_1^2$ in terms of U and xy gives

$$\begin{aligned}(sU + 2rtxy)^2 - 2rt(U + sxy)^2 &= (s^2U^2 + 4rstxyU + 4r^2t^2x^2y^2) - 2rt(U^2 + 2sxyU + s^2x^2y^2) \\ &= (s^2 - 2rt)U^2 - 2rt(s^2 - 2rt)x^2y^2 = (s^2 - 2rt)(U^2 - 2rtx^2y^2).\end{aligned}$$

Since $U^2 - 2rtx^2y^2 = (rx^2 + ty^2)^2 - 2rtx^2y^2 = r^2x^4 + t^2y^4$, this proves (62). Setting $r = -1, s = 2, t = -m$ and $(x, y) = (X, Y)$ gives $Q_1 = -(X^2 - 2XY + mY^2)$, $Q_2 = -2(X^2 - mXY + mY^2)$, $2rt = 2m$, and $s^2 - 2rt = 4 - 2m = 2(2 - m)$; dividing by 2 yields (63).

To analyze the geometric smoothness of $C = \mathcal{F}_1 \cap \mathcal{F}_2 \subset \mathbb{P}^3$, recall by the classical theorem on pencils of quadrics (Reid; Bright [12, Section 2.3]) that a pencil of quadrics $\lambda\mathcal{F}_1 + \mu\mathcal{F}_2$ in $\mathbb{P}^3$ cuts out a geometrically smooth curve C of degree 4 and genus 1 if and only if its degree-4 determinant polynomial $\det(\lambda M_{\mathcal{F}_1} + \mu M_{\mathcal{F}_2})$ has 4 distinct simple roots in $\mathbb{P}^1(\overline{\mathbb{Q}})$ over the algebraic closure $\overline{\mathbb{Q}}$. Since $\mathcal{F}_1$ has

Z-coefficient 0 and W-coefficient $-\alpha$, while $\mathcal{F}_2$ has W-coefficient 0 and Z-coefficient $-\beta$, the 4×4 matrix of $\lambda\mathcal{F}_1+\mu\mathcal{F}_2$ is block-diagonal with determinant

$$\det(\lambda M_{\mathcal{F}_1}+\mu M_{\mathcal{F}_2})=\frac{1}{4}\alpha\beta\,\lambda\mu\,\Delta_{\text{bin}}(\lambda,\mu),$$

where $\Delta_{\text{bin}}(\lambda,\mu)=(m\lambda+2\mu)^2-4m(\lambda+\mu)^2=m(m-4)\lambda^2-4m\lambda\mu+4(1-m)\mu^2$ is the discriminant of the binary quadratic $\lambda(X^2-mXY+mY^2)+\mu(X^2-2XY+mY^2)$. Thus the four roots of $\det(\lambda M_{\mathcal{F}_1}+\mu M_{\mathcal{F}_2})=0$ in $\mathbb{P}^1(\overline{\mathbb{Q}})$ occur at:

- $\mu=0$ ($\mathcal{F}_1$), which is a simple root (and a rank-3 cone) if and only if $\Delta_{\text{bin}}(1,0)=m(m-4)\neq 0$ (i.e. $m\notin\{0,4\}$);
- $\lambda=0$ ($\mathcal{F}_2$), which is a simple root (and a rank-3 cone) if and only if $\Delta_{\text{bin}}(0,1)=4(1-m)\neq 0$ (i.e. $m\neq 1$);
- the two roots $(\lambda:\mu)\in\mathbb{P}^1(\overline{\mathbb{Q}})$ of $\Delta_{\text{bin}}(\lambda,\mu)=0$, which are distinct in $\mathbb{P}^1(\overline{\mathbb{Q}})$ and distinct from $(1:0)$ and $(0:1)$ if and only if the discriminant of $\Delta_{\text{bin}}(\lambda,\mu)$ with respect to $(\lambda:\mu)$ is non-zero:

$$\Delta(\Delta_{\text{bin}})=(-4m)^2-16m(m-4)(1-m)=16m(m-2)^2\neq 0\iff m\notin\{0,2\}.$$

Therefore, $C=\mathcal{F}_1\cap\mathcal{F}_2$ is a geometrically smooth complete intersection of genus 1 over $\overline{\mathbb{Q}}$ (and hence a smooth genus-1 curve defined over $\mathbb{Q}$) if and only if $\alpha\beta\neq 0$ and $m\notin\{0,1,2,4\}$. (Notice that whether the two roots of $\Delta_{\text{bin}}(\lambda,\mu)=0$ lie in $\mathbb{P}^1(\mathbb{Q})$ or in $\mathbb{P}^1(\overline{\mathbb{Q}})\setminus\mathbb{P}^1(\mathbb{Q})$ affects only the field of definition of those two singular cones in the pencil, not the smoothness of the base curve $C=\mathcal{F}_1\cap\mathcal{F}_2$; for example, at $m=3$, $\Delta_{\text{bin}}(\lambda,\mu)=-3\lambda^2-12\lambda\mu-8\mu^2$ has discriminant $\Delta(\Delta_{\text{bin}})=16(3)(3-2)^2=144-96=48>0$ and two distinct roots $(\lambda:\mu)=(-6\pm 2\sqrt{3}:3)\in\mathbb{P}^1(\mathbb{Q}(\sqrt{3}))\subset\mathbb{P}^1(\overline{\mathbb{Q}})$, so C is smooth of genus 1.) At $m=1$ (resp. $m=4$), $\mathcal{F}_2:\beta Z^2=(X-Y)^2$ (resp. $\mathcal{F}_1:\alpha W^2=(X-2Y)^2$) is a rank-2 pair of planes whose singular line $Z=X-Y=0$ (resp. $W=X-2Y=0$) meets the other quadric in two singular points of C. □

Theorem 9.2 (The Symmetric Conic-Square Reduction Theorem Across All Parameter Regimes)**.** *Let $D(\lambda)=2\lambda^2-4\lambda+1$, and consider the symmetric quartic $x^4+y^4+c_3z^4$ modulo the symmetric conic $Q(x,y,z):xy=\lambda(x+y)^2+b(x+y)z+cz^2$.*

1. ***Complete Classification for** $D(\lambda)\neq 0$**:** Whenever $D(\lambda)=2\lambda^2-4\lambda+1\neq 0$ (which holds for all $\lambda\in\mathbb{Q}$), $x^4+y^4+c_3z^4$ reduces modulo $Q(x,y,z)$ to $D(\lambda)M(x,y,z)^2$ for a symmetric quadratic form $M(x,y,z)=(x+y)^2+\nu(x+y)z+\mu z^2$ if and only if both universal compatibility equations hold:*

$$b\Big[(\lambda-1)b^2-D(\lambda)c\Big]=0\qquad\text{and}\qquad D(\lambda)^3(2c^2+c_3)=\Big(b^2-2(\lambda-1)D(\lambda)c\Big)^2.\tag{65}$$

Specifically, (65) partitions into three exhaustive parameter regimes:

- ***Regime Ia (Non-Degenerate** $(\lambda-1)b\neq 0$**):** Here (65) is equivalent to*

$$(2\lambda^2-4\lambda+1)c=(\lambda-1)b^2\qquad\text{and}\qquad(\lambda-1)^2(2c^2+c_3)=(2\lambda^2-4\lambda+1)c^2.\tag{66}$$

Normalizing $b=D(\lambda)$ and $c=(\lambda-1)D(\lambda)$ gives $c_3=-D(\lambda)^2$ and yields the universal polynomial identity valid for all (x,y,z,λ):

$$x^4+y^4-D(\lambda)^2z^4=D(\lambda)\big((x+y)^2+2(\lambda-1)(x+y)z+D(\lambda)z^2\big)^2+Q_\lambda(x,y,z)R_\lambda(x,y,z),\tag{67}$$

where $Q_\lambda(x,y,z)=xy-\lambda(x+y)^2-D(\lambda)(x+y)z-(\lambda-1)D(\lambda)z^2$ and

$$R_\lambda(x,y,z)=2xy+2(\lambda-2)(x+y)^2+2D(\lambda)(x+y)z+2(\lambda-1)D(\lambda)z^2.\tag{68}$$

Moreover, choosing $\lambda=\frac{pq}{(p+q)^2}$ makes $D(\lambda)=\frac{p^4+q^4}{(p+q)^4}$ have a fourth-power denominator and forces $Q_\lambda(x,y,z)=0$ to contain the rational base points $(p:q:0)$ and $(q:p:0)$.

- ***Regime Ib (Degenerate $\lambda = 1$, where $D(1) = -1$):*** *At $\lambda = 1$, the first equation of (65) becomes $b(0-(-1)c) = bc = 0$ and the second becomes $-(2c^2 + c_3) = b^4$. Thus reduction to $-M(x,y,z)^2$ occurs if and only if $bc = 0$ and $2c^2 + c_3 = -b^4$ (i.e., either $c = 0, c_3 = -b^4$ giving $-((x+y)^2 - b^2z^2)^2$, or $b = 0, c_3 = -2c^2$ giving $-(x+y)^4$). In particular, if $bc \neq 0$ at $\lambda = 1$ (for instance, at the counterexample $(\lambda, b, c, c_3) = (1, 1, 1, -3)$, where $(\lambda-1)b = 0$ and $D(1)(2c^2 + c_3) = 1 = b^4$), the uncancelled cubic cross-term $4bc(x+y)z^3 = 4(x+y)z^3$ obstructs any symmetric-square reduction.*
- ***Regime Ic (Degenerate $b = 0$ with $D(\lambda) \neq 0$):*** *When $b = 0$, the first condition of (65) holds identically and the second becomes $D(\lambda)(2c^2 + c_3) = 4(\lambda-1)^2c^2$, yielding $D(\lambda)((x+y)^2 + \frac{2(\lambda-1)c}{D(\lambda)}z^2)^2$.*

2. ***Degenerate Locus $D(\lambda) = 0$ ($\lambda = 1 \pm \frac{\sqrt{2}}{2}$ over $\mathbb{Q}(\sqrt{2})$):*** *When $D(\lambda) = 0$, the $(x+y)^4$ term vanishes modulo $Q(x,y,z)$, leaving $4(\lambda-1)b(x+y)^3z + 2(b^2 + 2(\lambda-1)c)(x+y)^2z^2 + 4bc(x+y)z^3 + (2c^2+c_3)z^4$, which reduces to a scalar multiple of a square of a quadratic form in $(x+y, z)$ if and only if $b = 0$ and $c(2c^2 + c_3) = 0$.*

Proof. Let $\sigma_1 = x + y$ and $\sigma_2 = xy$. By Waring's formula, $x^4 + y^4 = \sigma_1^4 - 4\sigma_1^2\sigma_2 + 2\sigma_2^2$. Writing $\Sigma = \lambda\sigma_1^2 + b\sigma_1 z + cz^2$, we have $\sigma_2 = Q(x,y,z) + \Sigma$, so in $\mathbb{Z}[\lambda, b, c, c_3][x, y, z]$:

$$\begin{aligned} x^4 + y^4 + c_3z^4 &= (\sigma_1^4 - 4\sigma_1^2\Sigma + 2\Sigma^2 + c_3z^4) + Q(x,y,z)(2\sigma_2 + 2\Sigma - 4\sigma_1^2) \\ &= D\sigma_1^4 + 4(\lambda-1)b\sigma_1^3 z + (2b^2 + 4(\lambda-1)c)\sigma_1^2z^2 + 4bc\sigma_1 z^3 + (2c^2+c_3)z^4 \\ &\quad + Q(x,y,z)(2xy + 2(\lambda-2)(x+y)^2 + 2b(x+y)z + 2cz^2), \end{aligned} \tag{69}$$

where $D = D(\lambda) = 2\lambda^2 - 4\lambda + 1$.

(1) Suppose $D \neq 0$. Since $Q(x,y,z) = xy - \Sigma$, the quotient ring $k[x,y,z]/(Q)$ is a free module of rank 2 over the polynomial ring $k[\sigma_1, z]$ with basis $\{1, x\}$, so a symmetric polynomial in $k[\sigma_1, z]$ vanishes modulo $Q(x,y,z)$ if and only if it vanishes identically in $k[\sigma_1, z]$. For the quartic polynomial $P_{\mathrm{mod}}(\sigma_1, z) = D\sigma_1^4 + 4(\lambda-1)b\sigma_1^3z + (2b^2 + 4(\lambda-1)c)\sigma_1^2z^2 + 4bc\sigma_1z^3 + (2c^2+c_3)z^4$ to equal $D(\sigma_1^2 + \nu\sigma_1 z + \mu z^2)^2$ in $k[\sigma_1, z]$, matching the $\sigma_1^3 z$ coefficient uniquely determines $\nu = \frac{2(\lambda-1)b}{D}$, and matching the $\sigma_1^2z^2$ coefficient uniquely determines

$$D(\nu^2 + 2\mu) = 2b^2 + 4(\lambda-1)c \iff \mu = \frac{b^2(D - 2(\lambda-1)^2) + 2(\lambda-1)Dc}{D^2} = \frac{-b^2 + 2(\lambda-1)Dc}{D^2},$$

where we used the identity $D - 2(\lambda-1)^2 = (2\lambda^2 - 4\lambda + 1) - (2\lambda^2 - 4\lambda + 2) = -1$. With these unique values of ν and μ, direct symbolic subtraction gives the exact remainder

$$P_{\mathrm{mod}}(\sigma_1, z) - D(\sigma_1^2 + \nu\sigma_1 z + \mu z^2)^2 = \frac{4b[(\lambda-1)b^2 - Dc]}{D^2}\sigma_1 z^3 + \frac{D^3(2c^2+c_3) - (b^2 - 2(\lambda-1)Dc)^2}{D^3}z^4.$$

Thus $P_{\mathrm{mod}}(\sigma_1, z) = D(\sigma_1^2 + \nu\sigma_1 z + \mu z^2)^2$ identically in (σ_1, z) if and only if both coefficients on the right-hand side vanish, proving (65). In Regime Ia ($(\lambda-1)b \neq 0$), dividing $b[(\lambda-1)b^2 - Dc] = 0$ by $b \neq 0$ gives $Dc = (\lambda-1)b^2$. Substituting $b^2 = \frac{Dc}{\lambda-1}$ into μ gives $\mu = \frac{Dc(2(\lambda-1)^2-1)}{D^2(\lambda-1)} = \frac{c}{\lambda-1}$, so $D\mu^2 = \frac{Dc^2}{(\lambda-1)^2} = 2c^2 + c_3$, establishing (66). Normalizing $b = D(\lambda)$ and $c = (\lambda-1)D(\lambda)$ gives $\mu = D(\lambda)$ and $c_3 = -D(\lambda)^2$, establishing (67)–(68); and at $\lambda = \frac{pq}{(p+q)^2}$, $D(\lambda) = \frac{2p^2q^2 - 4pq(p+q)^2 + (p+q)^4}{(p+q)^4} = \frac{p^4+q^4}{(p+q)^4}$ and $Q_\lambda(p, q, 0) = pq - \frac{pq}{(p+q)^2}(p+q)^2 = 0$. Regimes Ib ($\lambda = 1, D = -1$) and Ic ($b = 0$) follow immediately by substituting $\lambda = 1$ and $b = 0$ into (65).

(2) When $D(\lambda) = 0$ (so $\lambda = 1 \pm \frac{\sqrt{2}}{2} \neq 1$), the σ_1^4 coefficient vanishes in (69). A polynomial of degree ≤ 3 in σ_1 can be a constant multiple of a square in $k[\sigma_1, z]$ only if its degree-3 coefficient $4(\lambda-1)b$ vanishes, forcing $b = 0$. Setting $b = 0$ leaves $4(\lambda-1)c\sigma_1^2z^2 + (2c^2+c_3)z^4$, which is a scalar multiple of a square in (σ_1, z) if and only if either $c = 0$ or $2c^2 + c_3 = 0$. □

10 The Tacnodal Tangent-Plane Method for Coordinate-Zero Points

While the tangent plane $T_P S_n$ at a generic point $P = (a_1 : b_1 : c_1 : d_1)$ ($a_1 b_1 c_1 d_1 \neq 0$) cuts out a plane quartic with a single ordinary double point (geometric genus $g = 3 - 1 = 2$), a genus drop to $g = 1$ occurs at any non-Eckardt point $P_0 \in S_n$ where one coordinate vanishes:

Theorem 10.1 (The Tacnodal Tangent-Plane Theorem for Coordinate-Zero Points)**.** *Let $P_0 = (a_0 : b_0 : c_0 : d_0) \in S_n(\mathbb{Q})$ be a non-Eckardt rational point on $S_n : na^4 + b^4 + c^4 = d^4$ ($n \in \mathbb{Q}^\times$) with one vanishing coordinate. (If $a_0 = 0$, then $b_0^4 + c_0^4 = d_0^4$ in $\mathbb{Q}$ forces $b_0 = 0$ or $c_0 = 0$ by Euler's exponent-4 Fermat theorem, which gives only the four Eckardt points $(0 : 1 : 0 : \pm 1)$ and $(0 : 0 : 1 : \pm 1)$; thus any non-Eckardt coordinate-zero rational point has $a_0 \neq 0$, normalized to $a_0 = 1$, and exactly one of b_0, c_0, d_0 equal to 0.) Because the third power of the vanishing coordinate is zero, that coordinate drops out of the linear tangent-plane equation $T_{P_0} S_n$, creating a tacnode (A_3-singularity, $\delta = 2$) at P_0 on the plane quartic $C_{P_0} = S_n \cap T_{P_0} S_n$ with no other singularities over $\overline{\mathbb{Q}}$ and dropping its geometric genus strictly from 3 to $g = 3 - 2 = 1$. Moreover, $C_{P_0}(\mathbb{R}) \setminus \{P_0\}$ is non-empty if and only if $n < 0$, in which case (up to $b \leftrightarrow c$ symmetry and $\mathbb{Q}^{\times 4}$-scaling) all such surfaces fall into two infinite families:*

1. ***Tacnodal Family A*** *($c_0 = 0$* ***with*** *$|b_0| > |d_0| \implies n = d_0^4 - b_0^4 < 0$,* ***base point*** *$P_0 = (1 : b_0 : 0 : d_0)$,* ***also covering*** *$b_0 = 0$* ***by*** *$b \leftrightarrow c$* ***symmetry):*** *On the tangent plane $(d_0^4 - b_0^4)a + b_0^3 b - d_0^3 d = 0$ parametrized by $a = 1$, $b = b_0 + d_0^3 T$, $d = d_0 + b_0^3 T$, the surface equation $c^4 = d^4 - b^4 - na^4$ factors as $c^2 = TU$, where (T, U) lies on the conic*

$$U^2 = (b_0^4 - d_0^4)\Big[(b_0^8 + b_0^4 d_0^4 + d_0^8)T^2 + 4b_0 d_0 (b_0^4 + d_0^4)T + 6b_0^2 d_0^2\Big]. \tag{70}$$

Completing the square reduces (70) to the diagonal ternary quadratic form

$$X^2 + 2M_- Z^2 = m_v M_+ Y^2, \qquad m_v = b_0^4 - d_0^4 > 0, \qquad M_\pm = b_0^8 \pm b_0^4 d_0^4 + d_0^8. \tag{71}$$

Whenever $6(b_0^4 - d_0^4) = (6w)^2$ is a rational square—for instance, at $(d_0, b_0) = (1, 7)$, where $n = 1^4 - 7^4 = -2400 = -150 \cdot 2^4 \equiv -150 \pmod{\mathbb{Q}^{\times 4}}$ and $w = 20$—the point $(T_0, U_0) = (0, 6w b_0 d_0)$ is a rational point on (70), and $c^2 = T(r,s)U(r,s)$ reduces directly to the $j = 1728$ Weierstrass elliptic curve

$$Y^2 = 2w\,\xi(\xi^2 + 2(b_0^8 - b_0^4 d_0^4 + d_0^8)). \tag{72}$$

(By contrast, when $c_0 = 0$ and $|d_0| > |b_0| \implies n = d_0^4 - b_0^4 > 0$, such as $P_0 = (1 : 1 : 0 : 2)$ on S_{15}, the bracketed quadratic in (70) is positive definite on $\mathbb{R}$ while $b_0^4 - d_0^4 = -n < 0$, so $U^2 < 0$ for all $T \in \mathbb{R}$ and P_0 is an isolated real point on $C_{P_0}(\mathbb{R})$.)

2. ***Tacnodal Family B*** *($d_0 = 0 \implies n = -(b_0^4 + c_0^4) < 0$,* ***base point*** *$P_0 = (1 : b_0 : c_0 : 0)$):* *On the tangent plane $(b_0^4 + c_0^4)a - b_0^3 b - c_0^3 c = 0$ parametrized by $a = 1$, $b = b_0 + c_0^3 T$, $c = c_0 - b_0^3 T$, the surface equation $d^4 = na^4 + b^4 + c^4$ factors as $d^2 = TU$, where (T, U) lies on the conic*

$$U^2 = (b_0^4 + c_0^4)\Big[(b_0^8 - b_0^4 c_0^4 + c_0^8)T^2 + 4b_0 c_0 (c_0^4 - b_0^4)T + 6b_0^2 c_0^2\Big], \tag{73}$$

which reduces to the diagonal ternary quadratic form $X^2 + 2M_+ Z^2 = (b_0^4 + c_0^4)M_- Y^2$.

Proof. (1) At $P_0 = (1 : b_0 : 0 : d_0)$ with $n = d_0^4 - b_0^4 \neq 0$ ($b_0 d_0 \neq 0$, since P_0 is not an Eckardt point), the gradient of $F(a, b, c, d) = na^4 + b^4 + c^4 - d^4$ is $4(n, b_0^3, 0, -d_0^3)$, so $T_{P_0} S_n$ is $(d_0^4 - b_0^4)a + b_0^3 b - d_0^3 d = 0$. In homogeneous coordinates $(a : T_h : c)$ on $T_{P_0} S_n$, setting $b = b_0 a + d_0^3 T_h$ and $d = d_0 a + b_0^3 T_h$ satisfies the tangent plane identically since $(d_0^4 - b_0^4)a + b_0^3(b_0 a + d_0^3 T_h) - d_0^3(d_0 a + b_0^3 T_h) = 0$. Substituting $b = b_0 a + d_0^3 T_h$ and $d = d_0 a + b_0^3 T_h$ into $c^4 = d^4 - b^4 - (d_0^4 - b_0^4)a^4$ cancels both the T_h^0 and T_h^1 terms identically:

$$\begin{aligned} H(a, T_h, c) = c^4 - \Big[&(4d_0^3(b_0^3 T_h)a^3 + 6d_0^2(b_0^3 T_h)^2 a^2 + 4d_0(b_0^3 T_h)^3 a + (b_0^3 T_h)^4) \\ &- (4b_0^3(d_0^3 T_h)a^3 + 6b_0^2(d_0^3 T_h)^2 a^2 + 4b_0(d_0^3 T_h)^3 a + (d_0^3 T_h)^4)\Big] \end{aligned}$$

$$= c^4 - T_h^2(b_0^4 - d_0^4)Q_A(T_h, a) = 0,$$

where $Q_A(T_h, a) = (b_0^8 + b_0^4 d_0^4 + d_0^8)T_h^2 + 4b_0 d_0(b_0^4 + d_0^4)T_h a + 6b_0^2 d_0^2 a^2$.

To find all singular points of $C_{P_0} = \{H(a, T_h, c) = 0\}$ over $\overline{\mathbb{Q}}$, observe that $\frac{\partial H}{\partial c} = 4c^3 = 0 \implies c = 0$. At $c = 0$, both $\frac{\partial H}{\partial T_h} = 0$ and $\frac{\partial H}{\partial a} = 0$ hold if and only if $(T_h : a)$ is a multiple root in $\mathbb{P}^1(\overline{\mathbb{Q}})$ of the binary quartic $T_h^2 Q_A(T_h, a) = 0$. Setting $U = c^2/T$ (with $T = T_h/a$ at $a = 1$) yields (70), and multiplying (70) by $M_+ = b_0^8 + b_0^4 d_0^4 + d_0^8$ and completing the square in $X = M_+ T + 2b_0 d_0(b_0^4 + d_0^4)Z$ (with $Y = 1, Z = b_0 d_0$) gives

$$6b_0^2 d_0^2 M_+ - 4b_0^2 d_0^2(b_0^4 + d_0^4)^2 = 2b_0^2 d_0^2(3(b_0^8 + b_0^4 d_0^4 + d_0^8) - 2(b_0^8 + 2b_0^4 d_0^4 + d_0^8)) = 2b_0^2 d_0^2 M_-,$$

proving (71). Consequently, the discriminant of $Q_A(T, 1)$ with respect to T is

$$\Delta_T(Q_A) = 16b_0^2 d_0^2(b_0^4 + d_0^4)^2 - 24b_0^2 d_0^2 M_+ = -8b_0^2 d_0^2 M_- = -8b_0^2 d_0^2(b_0^8 - b_0^4 d_0^4 + d_0^8) < 0.$$

Since $\Delta_T(Q_A) \neq 0$ and $Q_A(0, 1) = 6b_0^2 d_0^2 \neq 0$, the quadratic $Q_A(T_h, a)$ has two distinct non-zero complex-conjugate geometric roots in $\mathbb{P}^1(\overline{\mathbb{Q}}) \setminus \mathbb{P}^1(\mathbb{R})$ (and no real or rational roots), so the unique multiple root of $T_h^2 Q_A(T_h, a) = 0$ in $\mathbb{P}^1(\overline{\mathbb{Q}})$ is $T_h = 0$ (which gives the rational tacnode $P_0 = (1 : b_0 : 0 : d_0) \in C_{P_0}(\mathbb{Q})$). Blowing up the origin $(c, T) = (0, 0)$ via $c^2 = UT$ replaces $H(1, T, c) = c^4 - T^2(b_0^4 - d_0^4)Q_A(T, 1) = 0$ by its smooth normalization conic $U^2 = (b_0^4 - d_0^4)Q_A(T, 1)$, over which $c = \sqrt{TU}$ is a double cover branched over the four distinct simple geometric roots of $T \cdot Q_A(T, 1) = 0$ in $\mathbb{P}^1_T(\overline{\mathbb{Q}})$ (the two complex-conjugate geometric roots of $Q_A(T, 1) = 0$ in $\mathbb{P}^1(\overline{\mathbb{Q}})$ together with the two rational branch points $T = 0$ and $T = \infty$). By the Riemann–Hurwitz formula, a double cover of $\mathbb{P}^1$ branched at 4 simple geometric points over $\overline{\mathbb{Q}}$ is a smooth curve of geometric genus $g = 1$ (confirming that P_0 is an A_3-tacnode with delta-invariant $\delta_{P_0} = p_a(C_{P_0}) - g = 3 - 1 = 2$). Moreover, since $Q_A(T, 1) > 0$ for all $T \in \mathbb{R}$, the normalization conic $U^2 = (b_0^4 - d_0^4)Q_A(T, 1)$ has real points with $T \neq 0$ if and only if $b_0^4 - d_0^4 = -n > 0$ $(n < 0)$, and whenever it possesses a $\mathbb{Q}$-rational base point $(T_0, U_0) \in \mathbb{Q}^2$ (as verified for the seed pairs in Table 4), stereographic projection from (T_0, U_0) parametrizes $(T(r, s), U(r, s))$ over $\mathbb{Q}$. In particular, when $6(b_0^4 - d_0^4) = 36w^2$, parametrizing (70) from the rational point $(T_0, U_0) = (0, 6wb_0 d_0)$ gives $T(r, s)$ with a single root at the tangent slope where $U(r, s)$ has its double-root vertex, collapsing the degree-4 product $T(r, s)U(r, s)$ to the depressed cubic (72).

(2) In Family B, replacing $(b_0, d_0) \mapsto (b_0, ic_0)$ in the algebraic expansion of Family A transforms $b_0^4 - d_0^4 \mapsto b_0^4 + c_0^4 > 0$ and $M_+ \leftrightarrow M_-$, proving (73) with $\Delta_T(Q_B) = -8b_0^2 c_0^2 M_+ < 0$ and $Q_B(0, 1) = 6b_0^2 c_0^2 \neq 0$, so $P_0 = (1 : b_0 : c_0 : 0)$ is likewise the unique singularity (an A_3-tacnode with $\delta_{P_0} = 2$) of C_{P_0} over $\overline{\mathbb{Q}}$, and its normalization has geometric genus $g = 3 - 2 = 1$. □

Example 10.2 (Explicit Genus-1 Quartic Model for the Fourth-Power-Free Constant $n = -369$). For the Family A seed pair $(d_0, b_0) = (4, 5)$, we have $n = 4^4 - 5^4 = 256 - 625 = -369 = -3^2 \cdot 41$, which is fourth-power-free in $\mathbb{Z}$ (note that -369 is distinct modulo $\mathbb{Q}^{\times 4}$ from the $(d_0, b_0) = (1, 7)$ surface $n = -2400 = -150 \cdot 2^4 \equiv -150 \pmod{\mathbb{Q}^{\times 4}}$ discussed in Theorem 10.1(1), since $369/150 = 123/50 \notin \mathbb{Q}^{\times 4}$). On the conic (70) for $(d_0, b_0) = (4, 5)$, the rational base point $T_0 = -\frac{3}{14}$, $U_0 = \frac{4797}{2}$ parametrizes (70) over $\mathbb{Q}$ via

$$T(r, s) = -\frac{3(r^2 + 22386rs + 105996849s^2)}{14(r^2 - 227363409s^2)}, \qquad U(r, s) = -\frac{369(13r^2 + 387178rs + 2955724317s^2)}{2(r^2 - 227363409s^2)},$$

reducing $-369a^4 + b^4 + c^4 = d^4$ on the tangent plane $369a - 125b + 64d = 0$ directly to the explicit genus-1 quartic model

$$7C^2 = 1107(r^2 + 22386rs + 105996849s^2)(13r^2 + 387178rs + 2955724317s^2),$$
$$c = \frac{C}{2(r^2 - 227363409s^2)}.$$

Table 4: Globally Soluble Tacnodal Conics, Explicit Fourth-Power Scalings $n = n_{\mathrm{red}} \cdot s_0^4$, and Machine-Verified Rational Base Points (T_0, U_0) on $U^2 = Q(T)$.

Family	**Seed Pair**	**Raw** $n = n_{\mathrm{red}} \cdot s_0^4$	**Reduced** n_{red}	**Verified Rational Point** (T_0, U_0) **on Conic** $U^2 = Q(T)$
Family A	$(d_0, b_0) = (3, 5)$	$-544 = -34 \cdot 2^4$	$n_{\mathrm{red}} = -34$	$T_0 = -\frac{54770}{808647}, \quad U_0 = \frac{62334920}{115521}$
Family A	$(d_0, b_0) = (1, 7)$	$-2400 = -150 \cdot 2^4$	$n_{\mathrm{red}} = -150$	$T_0 = 0, \quad U_0 = 840 \quad$ (and $T_0 = -\frac{67256}{5767203}, \ U_0 = 840$)
Family A	$(d_0, b_0) = (4, 5)$	$-369 = -369 \cdot 1^4$	$n_{\mathrm{red}} = -369$	$T_0 = -\frac{3}{14}, \quad U_0 = \frac{4797}{2}$
Family A	$(d_0, b_0) = (5, 11)$	$-14016 = -876 \cdot 2^4$	$n_{\mathrm{red}} = -876$	$T_0 = -\frac{830577}{89784422}, \quad U_0 = \frac{68868492}{7363}$
Family A	$(d_0, b_0) = (5, 13)$	$-27936 = -1746 \cdot 2^4$	$n_{\mathrm{red}} = -1746$	$T_0 = -\frac{184761850}{41567543967}, \quad U_0 = \frac{4846404477720}{322229023}$
Family A	$(d_0, b_0) = (12, 13)$	$-7825 = -7825 \cdot 1^4$	$n_{\mathrm{red}} = -7825$	$T_0 = -\frac{9}{4757}, \quad U_0 = \frac{1929645}{71}$
Family A	$(d_0, b_0) = (15, 17)$	$-32896 = -2056 \cdot 2^4$	$n_{\mathrm{red}} = -2056$	$T_0 = -\frac{13795343090}{11772661159987}, \quad U_0 = \frac{24328298706160}{268960297}$
Family A	$(d_0, b_0) = (7, 15)$	$-48224 = -3014 \cdot 2^4$	$n_{\mathrm{red}} = -3014$	$T_0 = -\frac{232069797777184}{25047155810962675}, \quad U_0 = \frac{563826937749768}{11740536925}$
Family B	$(b_0, c_0) = (3, 16)$	$-65617 = -65617 \cdot 1^4$	$n_{\mathrm{red}} = -65617$	$T_0 = -\frac{30724}{7807559}, \quad U_0 = \frac{350976146620}{7807559}$
Family B	$(b_0, c_0) = (8, 21)$	$-198577 = -198577 \cdot 1^4$	$n_{\mathrm{red}} = -198577$	$T_0 = -\frac{118240031}{31644876992}, \quad U_0 = \frac{6438572885107657}{31644876992}$
Family B	$(b_0, c_0) = (11, 24)$	$-346417 = -346417 \cdot 1^4$	$n_{\mathrm{red}} = -346417$	$T_0 = -\frac{29655868}{12284127477}, \quad U_0 = \frac{3423153386099524}{12284127477}$

11 Prior Literature Comparison and Curve-Equivalence Classification

To place the results of Sections 3–10 in their exact historical and mathematical context, we compare each identity, parametrization family, and geometric construction against the existing literature and provide complete, independently reconstructible Picard-lattice invariants for the rational curves on S_4 and S_{-1}.

11.1 Picard-Lattice Classification on S_{-1} and Comparison with Swinnerton-Dyer (2000) and Bremner (2025)

On Euler's surface $S_{-1} : -a^4 + b^4 + c^4 = d^4$ (equivalently $x^4 + y^4 = z^4 + w^4$), Swinnerton-Dyer [14, 8] described the divisor classes of curves of *arithmetic genus zero* ($p_a(C) = 0$, i.e. smooth rational curves $C \subset S_{-1}$ with self-intersection $C^2 = -2$), and Bremner [2] recently made this description effective up to any prescribed degree bound, showing that there are 86 distinct non-trivial arithmetic-genus-0 parametrizations of degree < 50.

By Lemma 5.3 and Theorem 5.4, the rational Picard lattice $\mathrm{Pic}(S_{-1}/\mathbb{Q}) \cong \mathbb{Z}^9$ is spanned by the 24 $\mathrm{Gal}(\mathbb{Q}(\zeta_8)/\mathbb{Q})$-orbits of the 48 lines (listed explicitly in Appendix A). Inside $\mathrm{Pic}(S_{-1}/\mathbb{Q})$, the hyperplane class $H = [L_0] + [L_1] + [L_2] + [L_3]$ together with 7 of the 8 rational lines

$$L_{\mathrm{I}}(j,k)\ (j,k \in \{0,2\},\ \text{indices } 0,2,8,10) \qquad \text{and} \qquad L_{\mathrm{II}}(j,k)\ (j,k \in \{0,2\},\ \text{indices } 16,18,24,26)$$

span a rank-8 sublattice Λ_8 with ordered basis $\mathcal{B}_8^{(-1)} = (H, L_0, L_2, L_8, L_{10}, L_{16}, L_{18}, L_{24})$ and intersection determinant $\det(G_8^{(-1)}) = -192$, while adjoining the $\mathbb{Q}(i)$-conjugate line pair $\mathcal{O}_4 = L_{\mathrm{I}}(1,1) + L_{\mathrm{I}}(3,3) = L_5 + L_{15}$ gives a rank-9 basis $\mathcal{B}_9^{(-1)} = (H, L_0, L_2, L_8, L_{10}, L_{16}, L_{18}, L_{24}, \mathcal{O}_4)$ of $\mathrm{Pic}(S_{-1}/\mathbb{Q})$ with explicit 9×9 intersection matrix ($\det(G_9^{\mathrm{T5}}) = 512$):

$$G_9^{\mathrm{T5}} = \begin{pmatrix} 4 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & -2 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & -2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & -4 \end{pmatrix}. \tag{74}$$

For every curve $C \subset S_{-1}$ in Section 4, its exact intersection multiplicity $C \cdot L_m$ with each of the 8 rational lines $L_0 = \{d = b, c = a\}$, $L_2 = \{d = b, c = -a\}$, $L_8 = \{d = -b, c = a\}$, $L_{10} = \{d = -b, c = -a\}$, $L_{16} = \{d = c, b = a\}$, $L_{18} = \{d = c, b = -a\}$, $L_{24} = \{d = -c, b = a\}$, and $L_{26} = \{d = -c, b = -a\}$ is the degree of the homogeneous polynomial GCD of the two linear defining forms of L_m evaluated on

$C(p,q)$ (and $C \cdot \mathcal{O}_4$ is the degree of $\gcd_{\mathbb{Q}(i)[p,q]}(d(p,q) - ib(p,q), c(p,q) - ia(p,q))$ doubled). Solving the non-singular linear system $G_9^{\mathrm{T5}} c = v_9$, where $v_9 = (\deg C, C \cdot L_0, C \cdot L_2, C \cdot L_8, C \cdot L_{10}, C \cdot L_{16}, C \cdot L_{18}, C \cdot L_{24}, C \cdot \mathcal{O}_4)^T$, uniquely reconstructs the divisor coordinate vector $c \in \mathbb{Z}^9$, the self-intersection $C^2 = c^T G_9^{\mathrm{T5}} c$, and the arithmetic genus $p_a(C) = \frac{1}{2}C^2 + 1$ in Table 5.

Table 5: Exact Divisor Coordinates in $\mathcal{B}_9^{(-1)} = (H, L_0, L_2, L_8, L_{10}, L_{16}, L_{18}, L_{24}, \mathcal{O}_4)$, Self-Intersections C^2, Arithmetic Genera $p_a(C)$, and Literature Comparison for the Rational Curves on S_{-1}.

Curve	Deg	Intersections $(C \cdot L_{1,\mathbb{Q}}\,;\, C \cdot L_{\mathrm{II},\mathbb{Q}}\,;\, C \cdot \mathcal{O}_4)$	Coordinates in $\mathcal{B}_9^{(-1)}$	C^2	$p_a(C)$	Classification & Relation to Bremner (2025) [2]
C_{eul}	7	$(1,6,0,1\,;\,4,1,1,0\,;\,4)$	$(4,-1,-3,0,-1,-2,-1,-1,0)$	-2	0	Euler (1772); smooth $p_a = 0$ base curve
C_{13} (29)	13	$(1,0,6,1\,;\,12,1,1,4\,;\,0)$	$(4,0,2,-1,0,-4,0,0,0)$	-2	0	Smooth $p_a = 0$ curve in Bremner's deg < 50 list; single-generator form F_{13}
C_{19} (31)	19	$(1,6,16,1\,;\,12,1,1,0\,;\,12)$	$(12,-3,-3,-8,-3,-6,-3,-3,0)$	-2	0	Smooth $p_a = 0$ curve in Bremner's deg < 50 list; single-generator form F_{19}
C_{31} (33)	31	$(24,1,1,4\,;\,1,6,24,1\,;\,20)$	$(10,-7,0,0,3,0,2,-7,0)$	-2	0	Smooth $p_a = 0$ curve in Bremner's deg < 50 list; single-generator form F_{31}
$\Psi_{-1}^2(C_{\mathrm{eul}}^{(1)})$	37	$k=2$ iterate on Branch 1	$(\Psi_{-1}^*)^2([C_{\mathrm{eul}}^{(1)}])$	-2	0	Smooth $p_a = 0$ curve in Bremner's deg < 50 list
$\Psi_{-1}^2(C_{\mathrm{eul}}^{(2)})$	55	$k=2$ iterate on Branch 2	$(\Psi_{-1}^*)^2([C_{\mathrm{eul}}^{(2)}])$	-2	0	Smooth $p_a = 0$ curve **above** Bremner's deg < 50 threshold
$\Psi_{-1}^2(C_{\mathrm{eul}}^{(3)})$	91	$k=2$ iterate on Branch 3	$(\Psi_{-1}^*)^2([C_{\mathrm{eul}}^{(3)}])$	-2	0	Smooth $p_a = 0$ curve **above** Bremner's deg < 50 threshold
$C_{43} = \mathcal{A}_+(C_{\mathrm{eul}})$	43	$(1,0,6,1\,;\,16,19,19,24\,;\,4)$	$(-2,11,12,9,11,4,2,2,0)$	$+130$	66	Singular $p_g = 0, p_a = 66$ curve (**outside** $p_a = 0$ classification)
$C_{45} = \mathcal{A}_-(C_{\mathrm{eul}})$	45	$(21,18,24,21\,;\,0,3,3,0\,;\,44)$	$(22,-10,-9,-12,-10,0,-1,-1,0)$	$+114$	58	Singular $p_g = 0, p_a = 58$ curve (**outside** $p_a = 0$ classification)

Table 5 establishes two key conclusions regarding the $n = -1$ families:

1. Because $\Psi_{-1} \in \mathrm{Aut}(S_{-1}/\mathbb{Q})$ is a regular K3 isomorphism preserving the intersection pairing, every iterate $\Psi_{-1}^k(C_{\mathrm{eul}}^{(r)})$ has self-intersection -2 and arithmetic genus $p_a = 0$. Consequently, the image curves at degrees $13, 19, 31$, and 37 lie within Bremner's [2] comprehensive classification of all 86 smooth arithmetic-genus-0 curves of degree < 50 on S_{-1}. For these curves, the contribution of Corollary 4.2 is their compact *single-generator involutive polynomial representation* $F_m(p,q)$ in (28)–(33), while Theorem 4.3 provides closed-form quadratic degree formulas ($9k^2 - 3k + 7$, $12k^2 + 7$, $18k^2 + 6k + 7$) that immediately generate smooth rational curves of degrees $55, 79, 91, 115, 139, 187, \ldots$ beyond the degree-50 threshold.

2. By contrast, the degree-43 and degree-45 curves $C_{43} = \mathcal{A}_+(C_{\mathrm{eul}})$ and $C_{45} = \mathcal{A}_-(C_{\mathrm{eul}})$ are images of C_{eul} under the *non-invertible* degree-17 asymptotic inflection-tangent endomorphism $\mathcal{A}_\pm : S_{-1} \dashrightarrow S_{-1}$ of Theorem 6.1. As shown in Table 5, they have positive self-intersections $C_{43}^2 = 130$ and $C_{45}^2 = 114$ and positive arithmetic genera $p_a(C_{43}) = 66$ and $p_a(C_{45}) = 58$ (acquiring $\sum \delta_P = 66$ and 58 singular delta-invariants on S_{-1} to achieve geometric genus $p_g = 0$). Because Swinnerton-Dyer [8] and Bremner [2] classify only curves of arithmetic genus $p_a = 0$ ($C^2 = -2$), C_{43} and C_{45} lie strictly outside that classification.

11.2 Picard-Lattice Classification on S_4 and Summary Across All Sections

On $S_4 : 4a^4 + b^4 + c^4 = d^4$ (with $\eta_0 = 1 + i$), the 48 lines partition into 20 $\mathrm{Gal}(\mathbb{Q}(\zeta_8)/\mathbb{Q})$-orbits $\mathcal{O}_0, \ldots, \mathcal{O}_{19}$ spanning $\mathrm{Pic}(S_4/\mathbb{Q}) \cong \mathbb{Z}^8$. Choosing the explicit 8-orbit basis $\mathcal{B}_8^{(4)} = (\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3, \mathcal{O}_4, \mathcal{O}_8, \mathcal{O}_{10}, \mathcal{O}_{11})$ gives intersection determinant $\det(G_8^{(4)}) = -4096$, where the 8 basis orbits are defined explicitly by

$$\begin{aligned}
\mathcal{O}_0 &= L_0 + L_3 = \{d = b,\ c = (1 \pm i)a\}, & \mathcal{O}_1 &= L_1 + L_2 = \{d = b,\ c = (-1 \pm i)a\}, \\
\mathcal{O}_2 &= L_4 + L_{15} = \{d = \pm ib,\ c = (1 \pm i)a\}, & \mathcal{O}_3 &= L_5 + L_{14} = \{d = \pm ib,\ c = (-1 \pm i)a\}, \\
\mathcal{O}_4 &= L_6 + L_{13} = \{d = \pm ib,\ c = (-1 \mp i)a\}, & \mathcal{O}_8 &= L_{16} + L_{19} = \{d = c,\ b = (1 \pm i)a\}, \\
\mathcal{O}_{10} &= L_{20} + L_{31} = \{d = \pm ic,\ b = (1 \pm i)a\}, & \mathcal{O}_{11} &= L_{21} + L_{30} = \{d = \pm ic,\ b = (-1 \pm i)a\},
\end{aligned} \tag{75}$$

and where $H = \mathcal{O}_0 + \mathcal{O}_1$. For any rational curve $C(r,s) \subset S_4$, its intersection number with each 2-line conjugate orbit $\mathcal{O}_m = L + \overline{L}$ in (75) is $C \cdot \mathcal{O}_m = 2 \deg \gcd_{\mathbb{Q}(i)[r,s]}(\ell_1(C(r,s)), \ell_2(C(r,s)))$, and solving $c = (G_8^{(4)})^{-1} v_8$ uniquely determines its coordinate vector $c \in \mathbb{Z}^8$ and self-intersection $C^2 = c^T G_8^{(4)} c$. Table 6 gives the exact intersection vectors and basis coordinates of Carmichael's degree-4 curve C_0 and our four new families $C_{\mathrm{I}}, C_{\mathrm{II}}, C_{\mathrm{III}}$ (degree 8) and C_{IV} (degree 16) from Corollary 3.3: all five are smooth rational curves ($C^2 = -2, p_a = 0$) representing distinct divisor classes in $\mathrm{Pic}(S_4/\mathbb{Q})$. Table 7 summarizes the prior-literature status of all identities and theorems across Sections 3–10.

Table 6: Exact Divisor Coordinates in $\mathcal{B}_8^{(4)}$ ($\det G_8^{(4)} = -4096$) for the Smooth Rational Curves on S_4.

Curve	**Deg**	**Intersection Vector v_8 with $\mathcal{B}_8^{(4)}$**	**Coordinates in $\mathcal{B}_8^{(4)}$**	C^2	**Equivalence Status**
C_0 (12)	4	$(4,0,0,0,4,2,0,4)$	$(-1,-1,1,1,0,1,1,0)$	-2	Gerardin (1911), Carmichael (1913)
C_{I} (19)	8	$(2,6,12,4,0,4,0,0)$	$(2,1,-1,0,0,0,1,1)$	-2	New smooth $p_a = 0$ family (deg $= 8$)
C_{II} (20)	8	$(2,6,12,4,0,0,4,4)$	$(3,4,-2,0,1,0,-1,-1)$	-2	New $\mathbb{Q}$-rational class ($\mathbb{Q}(i)$-twist of C_{I})
C_{III} (21)	8	$(4,4,12,0,4,2,0,4)$	$(1,1,-1,1,0,1,1,0)$	-2	New smooth $p_a = 0$ family (deg $= 8$, distinct orbit)
C_{IV} (22)	16	$(2,14,24,0,4,4,8,8)$	$(4,5,-3,2,2,0,-1,-1)$	-2	New smooth $p_a = 0$ family (deg $= 16$)

Table 7: Summary of Prior Literature and New Contributions by Section.

Result / Identity	**Prior Literature**	**Contribution in This Work**
Master difference (4) & Table 2 ($n = 20p^8 - q^8$)	Classical identity (4) [3, 5]; 5-row table and $n = 20p^8 - q^8$ posted in Vepir's MSE online discussion (Oct 2026) [11]	Surveyed as foundational generating identity with full citation to Vepir [11]
Weighted (h,t)-binomial engine (8)	$k = 1, t = 0$ in Bremner–Choudhry–Ulas [1]; $k = 1$ (h,t) in Piezas & Vepir [11, Remark III]	Extended to arbitrary degree $4k$ alongside degree-15 trinomial and cancellation constructions
Difference-of-two-cubes collapse (Thm 2.1)	Implicit in Elkies' $n = -12$ construction [5]	Explicit universal ring identity $(X-Y+L)^4 - (X-Y-L)^4 = 8L(X^3 - Y^3)$ under $L^2 = 3XY$
$n = 4$ involution $\Phi_{L,\overline{L}}$, deg-8/16 families, tower $8k^2 - 4k + 4$ (Sec 3)	Only Gerardin–Carmichael deg-4 curve C_0 previously recorded [3, 11]	New explicit $\mathbb{Q}(i)$-skew-line involution ($M^2 = -I_4$), 4 new primitive families (Table 6), quadratic tower
$n = -1$ single-generator families F_{13}, F_{19}, F_{31}, towers, & C_{43}, C_{45} (Sec 4)	$p_a = 0$ curves of deg < 50 classified by Swinnerton-Dyer [8] and Bremner (2025) [2]	Single-generator forms $F_m(p,q)$; exact towers $9k^2 - 3k + 7, 12k^2 + 7, 18k^2 + 6k + 7$; singular $p_a > 0$ curves C_{43}, C_{45}
Galois skew-line, Cardano cone, & Picard rigidity for $n = -12$ (Sec 5)	48-line Picard framework in Pinch–Swinnerton-Dyer [13] and Bright [12]; C_0 in Elkies [5]	Explicit 20-line $\mathbb{Z}$-basis ($\det = -64$) and C_{48}, Reynolds-averaging proof of $\mathrm{Pic}(S_{-12}/\mathbb{Q}) = \mathbb{Z}H$, Hessian proof of $\lvert \mathrm{Bir} \rvert = 16$, 9-node proof for C_0
Factored degree-17 asymptotic tangent map $\mathcal{A}_\pm$ on S_{-s^2} (Sec 6)	Geometric tangent construction in Euler (1772), Richmond [7], Logan–McKinnon–van Luijk [6]	Explicit 4-monomial vector $V_\pm$, factored binomial decomposition $K_4 P_1 - K_3^{\pm} V_\pm$, syzygies, and base locus
Corner identity (Sec 7) & u_4-Eckardt / bitangent plane (Sec 8)	Classical Eisenstein identity; $\mathbb{Q}(\omega)$ bitangent plane for $n = p_0^4 + q_0^4$ in Vepir [11]	Master u_4-Eckardt identity; complete singular-locus proof ($g = 1$) and 2-adic obstruction proof ($p_0 \equiv q_0 \pmod 2$)
3-parameter BCU syzygy & symmetric conic-square identity (Sec 9)	Underlying quadric/conic methods in Bremner–Choudhry–Ulas (2014) [1]	Complete parameter-domain exclusion $m \notin \{0,1,2,4\}$ over $\mathbb{P}^1(\overline{\mathbb{Q}})$ and exact degenerate/non-degenerate compatibility criteria
Tacnodal tangent-plane method for coordinate-zero points (Sec 10)	Ramified tangent planes noted in Swinnerton-Dyer [8] and Bright [12]	Explicit conic/Weierstrass reduction for Families A and B, blow-up singular-locus proof over $\overline{\mathbb{Q}}$, and 11 verified base points (Table 4)

A Symbolic Verification Certificates and Reproducibility Package

All polynomial identities, exact greatest common divisors, primitivity assertions, 48×48 line-intersection matrices, 20×48 coordinate matrices, Galois-orbit ranks, and rational conic points in this paper have been machine-verified using exact symbolic arithmetic in Python 3.11 with SymPy 1.12. Below we record the complete numerical and algebraic certificates together with a self-contained executable verification script so that every computational claim can be reproduced independently.

A.1 Complete 20×48 Line-Coordinate Matrix C_{48} and Galois Orbit Partitions

Indexing the 48 lines on S_n by $m=16p+4j+k\in\{0,\dots,47\}$ ($p=0,1,2$ for Pairings I, II, III; $0\le j,k\le3$), the 20 pivot lines $\mathcal{B}_{20}=(L_0,\dots,L_6,L_8,L_9,L_{10},L_{16},L_{17},L_{18},L_{20},L_{21},L_{22},L_{32},L_{33},L_{34},L_{36})$ correspond to standard unit basis vectors $e_1,\dots,e_{20}\in\mathbb{Z}^{20}$ with intersection matrix G_{20} given in (43) ($\det(G_{20})=-64$). The remaining 28 lines are expressed as exact integer linear combinations of $\mathcal{B}_{20}$ by the rows of $C_{48}^T\in\mathbb{Z}^{48\times20}$ in Table 8, and the full 48×48 intersection matrix is reconstructed identically as $G_{48}=C_{48}^TG_{20}C_{48}$.

Table 8: Complete Integer Coordinate Matrix C_{48}^T of the 28 Non-Pivot Lines $L_m\notin\mathcal{B}_{20}$ in the Integral Basis $\mathcal{B}_{20}=(L_0,\dots,L_6,L_8,L_9,L_{10},L_{16},L_{17},L_{18},L_{20},L_{21},L_{22},L_{32},L_{33},L_{34},L_{36})$ (each pivot line $L_{b_r}\in\mathcal{B}_{20}$ has coordinate vector e_r).

Line	Coordinates in $\mathcal{B}_{20}$	Line	Coordinates in $\mathcal{B}_{20}$
L_{07}	$(1,1,1,1,-1,-1,-1,0,0,0,0,0,0,0,0,0,0,0,0,0)$	L_{29}	$(0,0,-1,0,1,1,0,0,1,0,0,-1,-1,1,0,0,0,0,0,0)$
L_{11}	$(1,1,1,1,0,0,0,-1,-1,-1,0,0,0,0,0,0,0,0,0,0)$	L_{30}	$(-1,-1,-1,-2,0,1,1,0,0,1,1,1,0,0,1,0,0,0,0,0)$
L_{12}	$(0,1,1,1,-1,0,0,-1,0,0,0,0,0,0,0,0,0,0,0,0)$	L_{31}	$(0,1,1,1,-1,-1,0,-1,-1,-1,0,1,1,0,0,1,0,0,0,0)$
L_{13}	$(1,0,1,1,0,-1,0,0,-1,0,0,0,0,0,0,0,0,0,0,0)$	L_{35}	$(1,1,1,1,0,0,0,0,0,0,0,0,0,0,0,0,-1,-1,-1,0)$
L_{14}	$(1,1,0,1,0,0,-1,0,0,-1,0,0,0,0,0,0,0,0,0,0)$	L_{37}	$(1,1,1,1,-1,-1,0,-1,-1,0,0,1,1,0,1,1,-1,-1,0,-1)$
L_{15}	$(-1,-1,-1,-2,1,1,1,1,1,1,0,0,0,0,0,0,0,0,0,0)$	L_{38}	$(-1,-1,-2,-2,2,2,0,1,1,0,1,0,-1,1,0,-1,1,0,-1,1)$
L_{19}	$(1,1,1,1,0,0,0,0,0,0,-1,-1,-1,0,0,0,0,0,0,0)$	L_{39}	$(1,1,2,2,-1,-1,0,0,0,0,0,-1,-1,0,-1,-1,0,0,1,1,-1)$
L_{23}	$(1,1,1,1,0,0,0,0,0,0,0,0,0,-1,-1,-1,0,0,0,0)$	L_{40}	$(2,1,1,1,1,0,-1,1,0,0,0,-1,-1,0,-1,-1,0,0,-1,0)$
L_{24}	$(-1,0,-1,-1,0,1,1,-1,0,0,0,1,0,0,1,1,0,0,0,0)$	L_{41}	$(0,1,1,1,0,0,0,0,0,-1,-1,-1,0,-1,-1,0,1,1,1,0)$
L_{25}	$(1,1,2,1,-1,-1,0,0,-1,0,0,0,1,-1,-1,0,0,0,0,0)$	L_{42}	$(-1,-1,0,-1,-1,0,1,0,0,1,0,1,1,0,1,1,-1,0,0,0)$
L_{26}	$(2,2,2,3,0,-1,-1,0,0,-1,-1,-1,-1,0,-1,-1,0,0,0,0)$	L_{43}	$(0,0,-1,0,0,0,0,-1,0,0,1,1,0,1,1,0,0,-1,0,0)$
L_{27}	$(-1,-2,-2,-2,1,1,0,1,1,1,1,0,0,1,1,0,0,0,0,0)$	L_{44}	$(-1,0,0,0,-1,0,1,-1,0,0,0,1,1,0,1,1,-1,0,1,-1)$
L_{28}	$(2,1,2,2,0,-1,-1,1,0,0,-1,-1,0,-1,-1,-1,0,0,0,0)$	L_{45}	$(0,-1,-1,-1,1,1,0,1,1,1,1,0,-1,1,0,-1,0,-1,-1,1)$
L_{46}	$(3,3,3,4,-1,-2,-1,-1,-1,-1,-1,-1,0,-1,-1,0,0,0,0,-1)$	L_{47}	$(-1,-1,-1,-2,1,1,0,1,0,0,0,0,0,0,0,0,1,1,0,1)$

- **Galois Orbits on S_{-1} ($\eta_0=1,\theta_0=\zeta_8$):** Under $\mathrm{Gal}(\mathbb{Q}(\zeta_8)/\mathbb{Q})=\{\sigma_u:\zeta_8\mapsto\zeta_8^u\}_{u\in\{1,3,5,7\}}$, the 48 lines partition into 24 orbits:

$$\begin{aligned}
&\mathcal{O}_0=\{0\},\ \mathcal{O}_1=\{1,3\},\ \mathcal{O}_2=\{2\},\ \mathcal{O}_3=\{4,12\},\ \mathcal{O}_4=\{5,15\},\ \mathcal{O}_5=\{6,14\},\\
&\mathcal{O}_6=\{7,13\},\ \mathcal{O}_7=\{8\},\ \mathcal{O}_8=\{9,11\},\ \mathcal{O}_9=\{10\},\ \mathcal{O}_{10}=\{16\},\ \mathcal{O}_{11}=\{17,19\},\\
&\mathcal{O}_{12}=\{18\},\ \mathcal{O}_{13}=\{20,28\},\ \mathcal{O}_{14}=\{21,31\},\ \mathcal{O}_{15}=\{22,30\},\ \mathcal{O}_{16}=\{23,29\},\ \mathcal{O}_{17}=\{24\},\\
&\mathcal{O}_{18}=\{25,27\},\ \mathcal{O}_{19}=\{26\},\ \mathcal{O}_{20}=\{32,37,42,47\},\ \mathcal{O}_{21}=\{33,36,43,46\},\\
&\mathcal{O}_{22}=\{34,39,40,45\},\ \mathcal{O}_{23}=\{35,38,41,44\}.
\end{aligned}$$

  Summing the columns of C_{48} over each orbit gives a 20×24 matrix of rank 9. The 9 independent Galois orbits with indices $(0,1,2,3,4,5,10,11,13)$ have the 9×9 intersection matrix G_9 of

determinant $\det(G_9) = 512$ (matching $\det(G_9^{\mathrm{T5}}) = 512$ in the basis $\mathcal{B}_9^{(-1)}$ of (74)):

$$G_9 = \begin{pmatrix} -2 & 2 & 1 & 2 & 0 & 0 & 1 & 0 & 0 \\ 2 & -4 & 2 & 0 & 2 & 0 & 0 & 2 & 2 \\ 1 & 2 & -2 & 0 & 0 & 2 & 0 & 0 & 0 \\ 2 & 0 & 0 & -4 & 2 & 2 & 0 & 2 & 2 \\ 0 & 2 & 0 & 2 & -4 & 2 & 2 & 0 & 0 \\ 0 & 0 & 2 & 2 & 2 & -4 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & 2 & 0 & -2 & 2 & 2 \\ 0 & 2 & 0 & 2 & 0 & 2 & 2 & -4 & 0 \\ 0 & 2 & 0 & 2 & 0 & 2 & 2 & 0 & -4 \end{pmatrix}.$$

- **Galois Orbits on S_4 ($\eta_0 = 1 + i, \theta_0 = \zeta_8(1 + i)$):** Under $\mathrm{Gal}(\mathbb{Q}(\zeta_8)/\mathbb{Q})$, the 48 lines partition into 20 orbits:

$$\begin{aligned} &\mathcal{O}_0 = \{0, 3\},\ \mathcal{O}_1 = \{1, 2\},\ \mathcal{O}_2 = \{4, 15\},\ \mathcal{O}_3 = \{5, 14\},\ \mathcal{O}_4 = \{6, 13\},\ \mathcal{O}_5 = \{7, 12\}, \\ &\mathcal{O}_6 = \{8, 11\},\ \mathcal{O}_7 = \{9, 10\},\ \mathcal{O}_8 = \{16, 19\},\ \mathcal{O}_9 = \{17, 18\},\ \mathcal{O}_{10} = \{20, 31\},\ \mathcal{O}_{11} = \{21, 30\}, \\ &\mathcal{O}_{12} = \{22, 29\},\ \mathcal{O}_{13} = \{23, 28\},\ \mathcal{O}_{14} = \{24, 27\},\ \mathcal{O}_{15} = \{25, 26\},\ \mathcal{O}_{16} = \{32, 33, 42, 43\}, \\ &\mathcal{O}_{17} = \{34, 35, 40, 41\},\ \mathcal{O}_{18} = \{36, 39, 45, 46\},\ \mathcal{O}_{19} = \{37, 38, 44, 47\}. \end{aligned}$$

  Summing the columns of C_{48} over each orbit gives a 20×20 matrix of rank 8, and the 8 independent orbits $\mathcal{B}_8^{(4)} = (\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3, \mathcal{O}_4, \mathcal{O}_8, \mathcal{O}_{10}, \mathcal{O}_{11})$ have the 8×8 intersection matrix

$$G_8^{(4)} = \begin{pmatrix} -2 & 4 & 2 & 0 & 0 & 2 & 2 & 2 \\ 4 & -2 & 0 & 2 & 2 & 0 & 0 & 0 \\ 2 & 0 & -4 & 2 & 2 & 2 & 2 & 2 \\ 0 & 2 & 2 & -4 & 4 & 2 & 2 & 2 \\ 0 & 2 & 2 & 4 & -4 & 0 & 0 & 0 \\ 2 & 0 & 2 & 2 & 0 & -2 & 2 & 0 \\ 2 & 0 & 2 & 2 & 0 & 2 & -4 & 2 \\ 2 & 0 & 2 & 2 & 0 & 0 & 2 & -4 \end{pmatrix}, \qquad \det(G_8^{(4)}) = -4096.$$

A.2 Certificates for $\mathcal{A}_{\pm}(C_{\mathrm{eul}})$ GCDs and Table 4 Rational Conic Points

1. **Certificate for Theorem 4.3 and Theorem 6.1 ($\mathcal{A}_+(C_{\mathrm{eul}})$ and $\mathcal{A}_-(C_{\mathrm{eul}})$ GCD Factorizations):** Substituting Euler's degree-7 curve $C_{\mathrm{eul}}(p, q)$ into the degree-17 map $\mathcal{A}_{\pm}$ on S_{-1} ($s = 1$) produces raw degree-119 polynomials whose exact greatest common divisors in $\mathbb{Z}[p, q]$ are:

$$\begin{aligned} G_{\mathcal{A}_+}(p, q) &= 24p^5q^5(p - q)^3(p + q)^3(p^2 + q^2)^4(p^4 + q^4)^3(p^2 - pq - q^2)^3(p^2 + pq - q^2)^3 \\ &\quad \times (p^4 - p^2q^2 + q^4)^3(p^4 + 3p^2q^2 + q^4)(p^4 + 6p^2q^2 + q^4) \\ &\quad \times (p^8 + 2p^6q^2 + 11p^4q^4 + 2p^2q^6 + q^8) \qquad (\deg = 76), \\ G_{\mathcal{A}_-}(p, q) &= 8p^5q^5(p^8 - p^4q^4 + q^8)(p^6 - 2p^4q^2 + p^2q^4 + q^6)(p^6 + p^4q^2 - 2p^2q^4 + q^6) \\ &\quad \times (p^6 + 4p^4q^2 - 6p^3q^3 + 4p^2q^4 + q^6)^3(p^6 + 4p^4q^2 + 6p^3q^3 + 4p^2q^4 + q^6)^3 \\ &\quad \times (p^8 - 4p^6q^2 + 8p^4q^4 - 4p^2q^6 + q^8) \qquad (\deg = 74), \end{aligned}$$

   leaving primitive polynomial families of degrees $119 - 76 = 43$ and $119 - 74 = 45$.

2. **Certificate for Table 4 (Tacnodal Conic Base Points):** Exact rational arithmetic confirms $U_0^2 - (b_0^4 - d_0^4)Q_A(T_0) = 0$ for all 8 Family A seed pairs (including both rational points $(T_0, U_0) = (0, 840)$ and $(T_0, U_0) = (-\frac{67256}{5767203}, 840)$ for $(d_0, b_0) = (1, 7)$) and $U_0^2 - (b_0^4 + c_0^4)Q_B(T_0) = 0$ for all 3 Family B seed pairs.

A.3 Executable Python / SymPy Verification Script

```
import sympy as sp
from sympy import (
    symbols, Poly, gcd, Matrix, Rational, sqrt, I, factor, expand, discriminant
)

# 1. Section 1 & Section 2: Table 2, Theorem 2.1, and Proposition 2.3
p, q, r, s, a, b, c, d, t, u, v, X, Y, Z, L, U, V, M, h, n = symbols(
    'p q r s a b c d t u v X Y Z L U V M h n'
)
assert expand((X + Y)**4 - (X - Y)**4 - 8*X*Y*(X**2 + Y**2)) == 0

X0 = p**7 + p**5*q**2 - 2*p**3*q**4 + p*q**6
Y0 = 3*p**2*q**5
beul = p**6*q + 3*p**5*q**2 - 2*p**4*q**3 + p**2*q**5 + q**7
aeul = p**6*q - 3*p**5*q**2 - 2*p**4*q**3 + p**2*q**5 + q**7
ceul, deul = X0 - Y0, X0 + Y0
assert expand(-aeul**4 + beul**4 + ceul**4 - deul**4) == 0

diff_21 = sp.rem(expand((X - Y + L)**4 - (X - Y - L)**4), L**2 - 3*X*Y, L)
assert expand(diff_21 - 8*L*(X**3 - Y**3)) == 0
assert expand(
    (3*U**2 + 6*U*V - 4*V**2)**4 - (3*U**2 - 6*U*V - 4*V**2)**4
    - (1296*U**7*V - 3072*U*V**7)
) == 0

assert expand(
    (h*p**4 + q**4)**4 - (h*p**4 - q**4)**4 - (t*p**3*q)**4
    - ((8*h**3 - t**4)/16 * p**8 + h/2 * q**8)*(2*p*q)**4
) == 0
assert expand(
    q**3*(p**33 + 4*p**3*q**30 - q**33)*(2*p**3*q**3)**4
    + (2*p**3*q**12)**4 + (p**15 - 2*q**15)**4 - (p**15 + 2*q**15)**4
) == 0

# 2. Section 3 (n = 4): Involution, 4 Families, and Tower Matrix
U4 = 3*(c**2 - 2*a*c + 2*a**2)**2
V4 = -2*(4*a**4 - 4*a**3*c + 2*a*c**3 - c**4 - b*d*(b**2 + d**2))
at, bt, ct, dt = a + t*(c - a), b + t*d, c + t*(c - 2*a), d - t*b
assert expand(
    4*at**4 + bt**4 + ct**4 - dt**4
    - ((1 - t**4)*(4*a**4 + b**4 + c**4 - d**4) + 2*t*(t**2 + 1)*(U4*t + V4))
) == 0

def phi_4(A, B, C, D):
    u4 = U4.subs({a: A, b: B, c: C, d: D})
    v4 = V4.subs({a: A, b: B, c: C, d: D})
    out = [
        expand(u4*A - v4*(C - A)), expand(u4*B - v4*D),
        expand(u4*C - v4*(C - 2*A)), expand(u4*D + v4*B)
    ]
    G = sp.gcd_list(out)
    return G, [expand(sp.quo(x, G, r, s)) for x in out]

branches_4 = [
    ([2*r*s**3, -(r**4 - 2*s**4), 2*r**3*s, r**4 + 2*s**4], 8),
    ([2*r*s**3, -(r**4 - 2*s**4), -2*r**3*s, r**4 + 2*s**4], 8),
    ([2*r*s**3, 2*r**3*s, -(r**4 - 2*s**4), r**4 + 2*s**4], 8),
    ([2*r*s**3, r**4 - 2*s**4, 2*r**3*s, r**4 + 2*s**4], 16),
]
for branch, deg_exp in branches_4:
    G, fam = phi_4(*branch)
    assert expand(4*fam[0]**4 + fam[1]**4 + fam[2]**4 - fam[3]**4) == 0
    assert Poly(fam[0], r, s).total_degree() == deg_exp
    assert sp.gcd_list([x.subs(s, 1) for x in fam]) == 1
    assert sp.gcd_list([x.subs(r, 1) for x in fam]) == 1

G_Lam = Matrix([[4, 2, 2], [2, -4, 4], [2, 4, -4]])
M_Phi4 = Matrix([[5, 8, 0], [-3, -5, 0], [-1, -2, 1]])
M_taub = Matrix([[1, 0, 0], [0, 0, 1], [0, 1, 0]])
M_Psi4 = M_Phi4 * M_taub
assert G_Lam.det() == 64 and M_Psi4.T * G_Lam * M_Psi4 == G_Lam
assert (M_Psi4 - sp.eye(3))**2 != sp.zeros(3, 3)
assert (M_Psi4 - sp.eye(3))**3 == sp.zeros(3, 3)
```

```
# 3. Section 4 (n = -1): Single-Generator Families F13, F19, F31 & Tower
F13 = (
    p**13 + p**12*q + 2*p**11*q**2 - 4*p**10*q**3 - 3*p**9*q**4
    + 3*p**8*q**5 + 7*p**7*q**6 + 4*p**6*q**7 - 12*p**5*q**8
    - 6*p**4*q**9 + 5*p**3*q**10 - p**2*q**11 + p*q**12 + q**13
)
a13 = F13
b13 = -F13.subs({p: q, q: p}, simultaneous=True)
c13 = F13.subs({q: -q})
d13 = -F13.subs({p: -q, q: p}, simultaneous=True)
assert expand(-a13**4 + b13**4 + c13**4 - d13**4) == 0
assert sp.gcd_list([x.subs(q, 1) for x in [a13, b13, c13, d13]]) == 1

F19 = (
    p**18*q + 6*p**17*q**2 - 6*p**16*q**3 + 12*p**15*q**4
    + 6*p**14*q**5 + 36*p**13*q**6 + 19*p**12*q**7 - 12*p**11*q**8
    - 24*p**10*q**9 + 36*p**9*q**10 - 12*p**8*q**11 + 12*p**7*q**12
    - 5*p**6*q**13 + 6*p**5*q**14 - 18*p**4*q**15 + 6*p**2*q**17 + q**19
)
a19 = F19
b19 = F19.subs({p: q, q: p}, simultaneous=True)
c19 = -F19.subs({q: -q})
d19 = F19.subs({p: q, q: -p}, simultaneous=True)
assert expand(-a19**4 + b19**4 + c19**4 - d19**4) == 0
assert sp.gcd_list([x.subs(q, 1) for x in [a19, b19, c19, d19]]) == 1

F31 = (
    p**31 + p**30*q + 11*p**29*q**2 - p**28*q**3 + 42*p**27*q**4
    - 24*p**26*q**5 - 19*p**25*q**6 + 32*p**24*q**7 - 154*p**23*q**8
    + 254*p**22*q**9 + 266*p**21*q**10 - 718*p**20*q**11
    + 126*p**19*q**12 + 303*p**18*q**13 - 478*p**17*q**14
    + 830*p**16*q**15 + 770*p**15*q**16 - 916*p**14*q**17
    - 738*p**13*q**18 - 21*p**12*q**19 + 350*p**11*q**20
    + 434*p**10*q**21 + 50*p**9*q**22 - 142*p**8*q**23
    - 91*p**7*q**24 - 76*p**6*q**25 + 15*p**5*q**26 + 3*p**4*q**27
    + 8*p**3*q**28 + 8*p**2*q**29 + p*q**30 + q**31
)
a31 = -F31.subs({p: -q, q: p}, simultaneous=True)
b31 = -F31.subs({p: q, q: p}, simultaneous=True)
c31 = F31.subs({q: -q})
d31 = F31
assert expand(-a31**4 + b31**4 + c31**4 - d31**4) == 0
assert sp.gcd_list([x.subs(q, 1) for x in [a31, b31, c31, d31]]) == 1

# 4. Section 5: Cardano Identity (40), 48-Line Basis B20, C48, & Orbits
assert expand(
    (3*Z**2 + 3*Z - 1)**4 - (3*Z**2 - 3*Z - 1)**4 - (6*Z*X)**4 - n*(6*Z*Y)**4
    - 24*Z*(27*Z**6 - 54*(X**4 + n*Y**4)*Z**3 - 1)
) == 0
assert expand((3*Z)**3 - 27*(3*Z)**(-3) - (27*Z**3 - Z**(-3))) == 0

F_12 = -12*a**4 + b**4 + c**4 - d**4
assert expand(sp.hessian(F_12, (a, b, c, d)).det() - 12**5*a**2*b**2*c**2*d**2) == 0

zeta8 = (1 + I)/sqrt(2)
planes = []
for p_idx in range(3):
    for j in range(4):
        for k in range(4):
            if p_idx == 0:
                planes.append(([0, -I**j, 0, 1], [-I**k, 0, 1, 0]))
            elif p_idx == 1:
                planes.append(([0, 0, -I**j, 1], [-I**k, 1, 0, 0]))
            else:
                planes.append(([-zeta8*(I**j), 0, 0, 1], [0, -zeta8*(I**k), 1, 0]))

G48 = sp.zeros(48, 48)
for m1 in range(48):
    G48[m1, m1] = -2
    for m2 in range(m1 + 1, 48):
        m_det = Matrix(
            [planes[m1][0], planes[m1][1], planes[m2][0], planes[m2][1]]
        ).det()
        if sp.simplify(m_det) == 0:
            G48[m1, m2] = G48[m2, m1] = 1
```

```
B20 = [0, 1, 2, 3, 4, 5, 6, 8, 9, 10, 16, 17, 18, 20, 21, 22, 32, 33, 34, 36]
G20 = G48.extract(B20, B20)
C48 = G20.inv() * G48.extract(B20, list(range(48)))
assert G20.det() == -64 and G48.rank() == 20 and C48.T * G20 * C48 == G48
assert all(val.is_integer for val in C48)

H_vec = C48[:, 0] + C48[:, 1] + C48[:, 2] + C48[:, 3]
orbits_12 = [
    ([0, 1, 2, 3], 1), ([4, 5, 6, 7, 12, 13, 14, 15], 2),
    ([8, 9, 10, 11], 1), ([16, 17, 18, 19], 1),
    ([20, 21, 22, 23, 28, 29, 30, 31], 2), ([24, 25, 26, 27], 1),
    (list(range(32, 48)), 4)
]
for orb, mult in orbits_12:
    assert sum((C48[:, idx] for idx in orb), sp.zeros(20, 1)) == mult * H_vec

# 5. Section 6: Degree-17 Map A_pm Syzygies & GCDs on Euler's Curve
F_s = -s**2*a**4 + b**4 + c**4 - d**4
K4 = (s**2*a**4*b**4 - c**4*d**4)*(s**2*a**4*d**4 - b**4*c**4)
for sgn in [+1, -1]:
    V_pm = [b*c**3*d, sgn*s*a*c*d**3, s**2*a**3*b*d, sgn*s*a*b**3*c]
    K3 = 4*a*b*c*d*(b**2*d**2*(s**2*a**4 + c**4) - sgn*s*a**2*c**2*(b**4 + d**4))
    c1 = 4*(-s**2*a**3*V_pm[0] + b**3*V_pm[1] + c**3*V_pm[2] - d**3*V_pm[3])
    c2 = 6*(-s**2*a**2*V_pm[0]**2 + b**2*V_pm[1]**2 + c**2*V_pm[2]**2 - d**2*V_pm[3]**2)
    c3 = 4*(-s**2*a*V_pm[0]**3 + b*V_pm[1]**3 + c*V_pm[2]**3 - d*V_pm[3]**3)
    c4 = -s**2*V_pm[0]**4 + V_pm[1]**4 + V_pm[2]**4 - V_pm[3]**4
    assert expand(c1) == 0 and expand(c2 + 6*s**2*a**2*b**2*c**2*d**2*F_s) == 0
    assert expand(
        c3 - s**2*(s**2*a**4 - c**4)*K3 + sgn*4*s**3*a**3*b*c**3*d*(b**4 + d**4)*F_s
    ) == 0
    assert expand(
        c4 - s**2*(s**2*a**4 - c**4)*K4 + s**4*a**4*c**4*(b**8 + b**4*d**4 + d**8)*F_s
    ) == 0

# 6. Sections 7, 8, 9, 10, and 11 Verification
assert expand(
    (X + M + Y)**4 - (X + M)**4 - (Y + M)**4
    - (2*X*Y*(2*X**2 + 3*X*Y + 2*Y**2 + 6*M*(X + Y) + 6*M**2) - M**4)
) == 0

u4_s = symbols('u4')
A_e = b**2 + b*c + c**2 - d*(b + c)
B_e = (d - b)*(d - c)
Q_r = (
    2*(u4_s + 1)*(b**2 + c**2) + (u4_s**2 + 2*u4_s + 3)*b*c
    - (u4_s**2 + 2*u4_s - 1)*d*(b + c) + (u4_s**2 + 1)*d**2
)
assert expand((A_e + u4_s*B_e)**2 + (d**4 - b**4 - c**4) - B_e*Q_r) == 0

m_s, lam, mu = symbols('m lam mu')
assert expand(
    2*(X**2 - m_s*X*Y + m_s*Y**2)**2 - m_s*(X**2 - 2*X*Y + m_s*Y**2)**2
    - (2 - m_s)*(X**4 + m_s**2*Y**4)
) == 0
assert expand((-4*m_s)**2 - 16*m_s*(m_s - 4)*(1 - m_s) - 16*m_s*(m_s - 2)**2) == 0

table4_A = [
    (3, 5, Rational(-54770, 808647), Rational(62334920, 115521)),
    (1, 7, Rational(0, 1), Rational(840, 1)),
    (1, 7, Rational(-67256, 5767203), Rational(840, 1)),
    (4, 5, Rational(-3, 14), Rational(4797, 2)),
    (5, 11, Rational(-830577, 89784422), Rational(68868492, 7363)),
    (5, 13, Rational(-184761850, 41567543967), Rational(4846404477720, 322229023)),
    (12, 13, Rational(-9, 4757), Rational(1929645, 71)),
    (15, 17, Rational(-13795343090, 11772661159987),
     Rational(24328298706160, 268960297)),
    (7, 15, Rational(-23206979777184, 25047155810962675),
     Rational(563826937749768, 11740536925)),
]
for d0_v, b0_v, T0_v, U0_v in table4_A:
    qa = (
        (b0_v**8 + b0_v**4*d0_v**4 + d0_v**8)*T0_v**2
        + 4*b0_v*d0_v*(b0_v**4 + d0_v**4)*T0_v + 6*b0_v**2*d0_v**2
    )
    assert U0_v**2 == (b0_v**4 - d0_v**4)*qa
```

```
table4_B = [
    (3, 16, Rational(-30724, 7807559), Rational(350976146620, 7807559)),
    (8, 21, Rational(-118240031, 31644876992), Rational(6438572885107657, 31644876992)),
    (11, 24, Rational(-29655868, 12284127477), Rational(3423153386099524, 12284127477)),
]
for b0_v, c0_v, T0_v, U0_v in table4_B:
    qb = (
        (b0_v**8 - b0_v**4*c0_v**4 + c0_v**8)*T0_v**2
        + 4*b0_v*c0_v*(c0_v**4 - b0_v**4)*T0_v + 6*b0_v**2*c0_v**2
    )
    assert U0_v**2 == (b0_v**4 + c0_v**4)*qb

B9_mat = Matrix.hstack(
    H_vec, C48[:, 0], C48[:, 2], C48[:, 8], C48[:, 10],
    C48[:, 16], C48[:, 18], C48[:, 24], C48[:, 5] + C48[:, 15]
)
G9_T5 = B9_mat.T * G20 * B9_mat
assert G9_T5.det() == 512

B8_mat = Matrix.hstack(*[
    C48[:, i1] + C48[:, i2] for i1, i2 in [
        (0, 3), (1, 2), (4, 15), (5, 14), (6, 13), (16, 19), (20, 31), (21, 30)
    ]
])
G8_S4 = B8_mat.T * G20 * B8_mat
assert G8_S4.det() == -4096
print("All symbolic verification checks passed!")
```

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